

Program overview

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Year 2023/2024
Organization Civil Engineering and Geosciences
Education Master Transport, Infrastructure & Logistics

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master TIL 2023			
Fundamentals (23 EC)			
CIEQ6002	Transport Modelling and Analysis	5	
ME44206	Quantitative Methods for Logistics	5	
ME44312	Machine Learning for Transport and Multi-Machine Systems	3	
SEN1221	Statistical Analysis of Choice Behaviour	5	
TIL4030-20	TIL Research and Design Methods	7	
TIL6022	TIL Python Programming	3	
Specialisations (27 EC) (1 specialisation)			
Specialization Transport Governance (27 EC)			
AR0168	People, Movement and Public Space	5	
CIEQ6212	Regions, transport and networks	4	
SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5	
SEN1721	Travel Behaviour Research	5	
TPM032A	Multi-criteria Decision Analysis	5	
WM1301TU	Ethics of Transportation	3	
Specialization Traffic Operations (27 EC)			
AE4423-20	Airline Planning and Optimisation	4	
CIEQ6003	Traffic Modelling and Management	6	
CIEQ6223	Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment	4	
ME44200	Operations and Maintenance	3	
ME44305	System Analysis and Simulation	5	
SEN9110	Simulation Masterclass	5	
Specialization Transport Networks (28 EC)			
AE4423-20	Airline Planning and Optimisation	4	
CIEM6301	Railway Traffic Management	5	
CIEQ6212	Regions, transport and networks	4	
CIEQ6232	Public Transport Demand and Network Planning and Operations	5	
CIEQ6233	Railway Operations and Control	5	
SEN1721	Travel Behaviour Research	5	
Specialization Logistics Systems (26 EC)			
CIEQ6213	Freight transport networks and systems	4	
ME44101	Dynamics and Interaction of Material and Equipment	4	
ME44300	Multi-Machine Coordination for Logistics	3	
ME44311	Advanced Operations and Production Management	5	
SEN9720	Logistics and Supply Chain Innovations	5	
TPM028A	Decision Making in Multimodal Transport Systems	5	
TIL interdisciplinary projects			
TIL4020-20	TIL Research Project	5	
TIL4020-20 P1	Research Proposal	4	
TIL4020-20 P2	Literature Review	1	
TIL5050-20	TIL Design Project	10	
TIL5050-20 P0	Preparation	0	
TIL5050-20 P1	Vision & Scope	2	
TIL5050-20 P2	Analysis & Design	8	
Thesis (30 EC)			
TIL5060	TIL Thesis	30	
Electives: between 19 and 21 (depending on the chosen specialization)			
Transport and Logistics (T&L)			
SEN115B	Law and Institutions	5	
SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5	
SEN1721	Travel Behaviour Research	5	
SEN1741	Innovations in Transport and Logistics	5	
SEN9110	Simulation Masterclass	5	
SEN9720	Logistics and Supply Chain Innovations	5	
SEN9725	Supply Chain Gaming	5	
TPM004A	Transport Safety	4	
TPM023B	Cost-Benefit Analysis: Theory and Application	4	
TPM028A	Decision Making in Multimodal Transport Systems	5	
TPM032A	Multi-criteria Decision Analysis	5	

Multi Machine Engineering (MME)			
ME44101	Dynamics and Interaction of Material and Equipment	4	
ME44106	Structural Design with FEM	4	
ME44110	Integrated Design Project for Multi-Machine Systems	5	
ME44115	Discrete Element Method (DEM) simulation	4	
ME44125	Reliability and Maintenance of Transport Equipment	3	
ME44200	Operations and Maintenance	3	
ME44210	Drive & Energy Systems	3	
ME44300	Multi-Machine Coordination for Logistics	3	
ME44305	System Analysis and Simulation	5	
ME44311	Advanced Operations and Production Management	5	
Traffic & Transport Engineering (TTE)			
CIEM6301	Railway Traffic Management	5	
CIEM6303	Transitions, Sustainability & Innovation	5	
CIEQ6003	Traffic Modelling and Management	6	
CIEQ6211	Transport Modelling: Traffic Assignment	3	
CIEQ6212	Regions, transport and networks	4	
CIEQ6213	Freight transport networks and systems	4	
CIEQ6221	Active Modes: Traffic and Transport	3	
CIEQ6222	Traffic Safety	3	
CIEQ6223	Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment	4	
CIEQ6224	Urban and motorway Traffic Flow Modelling and Control	5	
CIEQ6231	Public Transport System and Supply Planning and Operations	5	
CIEQ6232	Public Transport Demand and Network Planning and Operations	5	
CIEQ6233	Railway Operations and Control	5	
Control and Operations (C&O)			
AE4321-15	Air Traffic Management	4	
AE4423-20	Airline Planning and Optimisation	4	
AE4446	Airport and Cargo Operations	4	
Urbanism (U)			
AR0168	People, Movement and Public Space	5	
AR0228	Infrastructure and Environment Method Module	5	
AR3CS021	Seminar Cross Domain City of the Future	5	
AR8003TU	Legal and Governance	5	
Other TIL fields			
CME2300	Financial Engineering	4	
ME41106	Intelligent Vehicles 3ME	5	
MT44070	Shipping Management	5	
RO47016	Human Factors of Automated Driving	5	
TIL6020	TIL Scientific Assignment	7	
WI4062TU	Transport, Routing and Scheduling	3	
WM1301TU	Ethics of Transportation	3	
Convergence Program TIL (9 EC)			
IFEEMCS010500	Probability and Statistics	3	
IFEEMCS012100	Calculus for Engineering, part 1	3	
WI1807TH1-21	Linear Algebra	3	

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Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Master TIL 2023

Program Title	MSc Transport, Infrastructure & Logistics
Director of Education	Dr.ir. D.L. Schott
Program Coordinator	Dr. S. (Stefano) Fazi
EC Program	120 ECTS
Introduction 1	For more information, please check til.tudelft.nl

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Fundamentals (23 EC)

CIEQ6002	Transport Modelling and Analysis	5
Responsible Instructor	Dr. S. Sharif Azadeh	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	Dr. O. Cats	
Instructor	Dr. S. Sharif Azadeh	
Instructor	Ir. A.J. Pel	
Contact Hours / Week x/x/x/x	x/5/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Four-step transport models, output in terms of travel time, congestion, flows and modelling of impact on accessibility, traffic safety and environment. Demand modelling: trip generation, distribution and modal split using methods such as gravity models and OD estimation. Assignment problems part I (Uncongested networks and stochastic assignment, Congested networks, equilibrium assignment). Assignment problems part II (Public transport assignment System optimal assignment and dynamic assignment). Location planning part I (Facility location problem (service point). Application: location of charging stations for EVs.) Location planning part II (Facility location problem (transshipment hubs). Application: multimodal mobility service, DRT integrated in Fixed Line and Schedule systems).	
Study Goals	On completion of this Course, the student will be able to: TBU2-LO1: Describe and apply four-step transport models both for mobility and freight (TTE-TB-LO3, TTE-TB-LO4). TBU2-LO2: Describe and apply the key performance measurements of a transport system and their impacts (TTE-TB-LO2, TTE-TB-LO5). TBU2-LO3: Describe and estimate travel demand models at aggregated and disaggregated levels (TTE-TB-LO3, TTE-TB-LO4). 331 TBU2-LO4: Describe, compare and apply assignment models combining travel demand and transport supply models for road and public transport (TTE-TB-LO3, TTE-TB-LO4). TBU2-LO5: Formulate a location planning problem (service point and transshipment hubs) as a mixed integer linear model and solve using a branch and bound (cut) algorithm ((TTE-TB-LO3, TTE-TB-LO4).	
Education Method	Lectures, in class small groups practice problems, in-class discussions	
Assessment	Written exam (100%)	
Expected prior Knowledge	Technical Bc. especially with respect to calculus. statistics. the contents of mathematical modelling and algorithms Knowledge on transport systems in general is recommended.	
Academic Skills	Lecture notes and scientific articles provided by the teachers and are made available through Brightspace.	
Literature & Study Materials	Course slides and provided reading material	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Written exam (100%)	
Permitted Materials during Exam	Non-programmable calculator	
Collegerama	No	

ME44206	Quantitative Methods for Logistics	5
Responsible Instructor	Dr. B. Atasoy	
Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	The course contains of two parts; roughly 7 weeks each. In the first part of the course an introduction to basic operations research techniques will be presented: - linear programming (simplex, duality etc.) - integer programming (IP, BIP, MILP etc.) - basic probability and statistics; brief overview of models to deal with uncertainty This part of the course will be largely be based on the first chapters of the book by Hillier and Lieberman. During the first quarter students must complete an individual practical assignment where they develop a mathematical model and implement it using Python together with Gurobi as a solver. In the second part of the course several common quantitative methods for transportation and logistics are presented. - networks and routing: network flow problem, facility location problems, transportation problem, transshipment problem, vehicle routing problem and other standard problems; standard algorithms i.e. the branch and bound method. - queueing theory: overview of basic models and results. During the second quarter students must complete a practical assignment in groups on a case study with a transportation network again using Python.	
Study Goals	At the end of this course the students will be able to: - Explain the importance and use of operations research techniques to analyze and improve systems (i.e. queuing, simulation, forecasting, routing, scheduling) - Formulate mathematical models of real-life problems (e.g., resource allocation, production planning, transportation) with operations research techniques - Verify and validate mathematical models - Evaluate operation research techniques based on strengths and weaknesses (in terms of accuracy and efficiency) for improving real-life problems - Apply basic operation research techniques to small scale problems as well as realistic case studies	
Education Method	Flipped classroom approach: The students are expected to be prepared by studying the online material that is indicated every week. Lectures in two parts: First 45 min course content, Second 45 min assignment Course content includes a short intro and exercises related to the subject. Assignment part will be dedicated to introduction of the software (Python+Gurobi), introduction of the assignments and any further needed assistance and feedback. It is strongly recommended to bring a laptop to the lectures. There will be also additional reading material that will be provided during the course.	
Books	Hillier & Lieberman. Introduction to Operations Research. 10th Edition, McGraw-Hill, ISBN 9781259253188	
Assessment	Practical Q1 - 25% (individual: Mathematical Model, Python implementation and report) Practical Q2 - 25% (group: Mathematical Model, Python implementation and report) Exam Q2 - 50%. For all 3 parts a minimal grade of 5 is required. The exam will be given at the end of Q2, the resit for the exam in Q3 Both assignments are only given once a year; with a limited option to resubmit if the grade is insufficient. The highest grade for a resubmission is 6. In case of measures resulting from COVID-19, the prescribed assessment will be an online examination. If the opportunity arises for examinations on campus, the method of assessment will be on campus.	
Remarks	TIL-students: please be aware that there is an overlap in content between CIE4835 (elective T&P 3.1) and ME44206 (compulsory TIL). You are only allowed to choose one of the courses in your degree program	
Department	3mE Department Maritime & Transport Technology	

ME44312	Machine Learning for Transport and Multi-Machine Systems	3
Responsible Instructor	F. Schulte	
Instructor	Dr. B. Atasoy	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic programming	
Course Contents	The course gives an introduction to the basic concepts of machine learning (ML) (supervised, unsupervised, reinforcement learning) and puts them into relation to artificial intelligence and data analytics. Important fundamental machine learning algorithms including regression, classification, clustering, and neural networks are introduced and implemented in Python. The course also gives a brief overview of more advanced ML techniques that are used in the transportation and multi-machine domains such as reinforcement learning, graph-neural networks, and recommender systems. On this basis, machine learning methods are applied in group projects to transport and multi-machine systems using different types of data from different domains (e.g., AIS, GPS, New York Taxi, or machine failure data).	
Study Goals	-Understand the basic concepts of machine learning -Learn to implement widely-used machine learning algorithms -Apply machine learning methods to transport and multi-machine systems -Evaluate different machine learning methods for a problem setting -Conduct small ML projects	
Education Method	-Lectures -Group Work -Individual exercises for each machine learning algorithm	
Computer Use	Python on laptop	
Assessment	Group report	
Elective	Yes	
Tags	Abstract Algorithmics Artificial intelligence Broad Business Design Group work Industry Modelling Optimisation Programming Research Methods Stochastics Sustainability Technology Transport & Logistics Transport phenomena	
Percentage of Design	40%	
Design Content	Designing the right machine learning methodologies for different problem settings in the context of transport and multi-machine systems	
Department	3mE Department Maritime & Transport Technology	
Contact	f.schulte@tudelft.nl b.atasoy@tudelft.nl	

SEN1221	Statistical Analysis of Choice Behaviour	5
Module Manager	Dr. E.J.E. Molin	
Instructor	Ir. S. van Cranenburgh	
Instructor	Dr. E.J.E. Molin	
Responsible for assignments	Dr. E.J.E. Molin	
Co-responsible for assignments	Ir. S. van Cranenburgh	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	If one wants to design effective (transport) systems, one needs to understand the behavior of -future- users of these systems, e.g. the market share of a new public transport service. If one wants to design effective (transport) policies, one needs to know how people will respond to these policies, e.g. impact of a tax incentive on electric vehicle sales. This course sets out to provide students with in-depth knowledge and hands-on experience with behavioral theories, mathematical models and statistical methods that have proved very effective in understanding and predicting human choice behaviour. Importantly, this course takes a quantitative (i.e., mathematical and statistical) perspective on choice behaviour; this is quite different from the usual perspective in behavioral science-courses, which tends to be more qualitative. More specifically, we formalize behavioral theories into mathematical models, which we then estimate on empirical datasets using cutting edge statistical methods. In combination, the estimated econometric 'choice models' support rigorous analysis and forecasts of quantitative phenomena, thereby playing a crucial role in private and public sector policy development. The theories, models and methods that are taught in this course are applicable -and have been used- throughout a wide variety of domains, including but not limited to Transportation, Health, Marketing, Environment & Energy, and Political sciences. As such, our course optimally prepares students for a wide range of (quantitative, empirical) topics for their graduation projects. The course will cover both canonical (and Nobel Prize winning) methods which are based on the notion of rational decision making, as well as more recently proposed ones which are based on notions of bounded rationality (economic psychology) and moral decision-making. Both lecturers are highly active in the international research community, and their work forms part of the academic state of the art.	
Study Goals	Students will acquire an in-depth knowledge concerning the below mentioned theories, models and methods, as well as the skills to use them in their own research. Finally, they will acquire an understanding of the usefulness and limitations of these theories, models and methods; and an understanding of how they can and should be used in combination. Knowledge transfer will take the form of a series of lectures, whereas skills will be (further) developed in tailor made computer exercises. Specifically, during the computer exercises, students will learn how to design a statistically optimal choice experiment, and how to analyze data obtained from such experiments. In addition, they will learn how to interpret the outcomes of such analyses and how to use them to evaluate and design policies. - Choice behaviour modelling methods (van Cranenburgh) - o Discrete choice theory and the Logit model - o Advanced (Mixed) Logit models - o Choice models for bounded rationality and morality - Stated Choice data collection methods (Molin) - o Data-collection paradigms - o Stated choice experiments - o Efficient design approaches.	
Education Method	Each method will we covered by means of: - o Lecture series - o Exercises, in which concepts taught during the lectures will be used in hands on exercises. In case covid-regulations require this, a hybrid form of teaching will be adopted, where on campus lectures are livestreamed and computer exercises will be discussed online. IMPORTANT INFORMATION ON KNOWLEDGE REQUIREMENTS: We expect you to have substantial background in mathematics (calculus and statistics), including knowledge of mathematical modeling, and of statistical metrics and testing procedures such as variance, correlation, normal distribution, standard errors, t-test, and statistical testing of hypotheses (including the use of p-values). We also expect you to have knowledge of survey data collection concepts and methods (sample, population, sampling procedures, questionnaires). Note that students with insufficient background in mathematics and statistics will struggle to keep up with the pace of the course.	
Assessment	The final grade is determined by two partial exams which are held during the course and by a regular exam (in the exam week). The partial exams focus primarily on skills and practices, i.e. the analyses, modeling, and experimental design (which are practiced in the computer practice sessions). In contrast, the regular exam focuses mainly on theory and conceptual thinking. Note, however, that both exam types pose questions on theory and practical skills. Furthermore, Mobius or ANS examination software is used for both exams. Exams consist of multiple choice, true/false, and numerical questions and, to a lesser extent, open-ended (essay) questions. Since no re-sit will be offered for the partial exams, the way in which the final grade is determined depends on the period the exam is taken: regular exam period: Final grade = 7.5% * grade partial exam 1 + 7.5% * grade partial exam 2 + 85% * grade regular exam re-sit exam period: Final grade = 100% * grade re-sit exam Thus, the grades of the partial exams are only used in combination with the grade of the regular exam period. After this, the grades of the partial exams lose their validity. Also, the grades of the partial exams are not transferred to the next year and are not registered in Osiris.	
Elective	Yes	
Tags	Analysis Calculus Economics	

Involved
Modelling
Research Methods
Stochastics
Transport & Logistics

TIL4030-20	TIL Research and Design Methods	7
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Dr. J.A. Annema	
Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Dr.ir. S.C. Calvert	
Contact Hours / Week x/x/x/x	0/4 or 8/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The introductory course TIL4030 Research & Design introduces, discusses and practices research & design methodologies in the Transport, Infrastructure and Logistics domain on the basis of cases and problems along the four specialisations in the MSc TIL program. Through workshops, exercises, an integrated design exercise (in Q1) and a research project (Q2) students bring the knowledge and skills into practice.	
Study Goals	This introductory Msc TIL course, TIL4030 Research & Design, has three main learning objectives: 1. To train students in understanding and using systems design and engineering methods in TIL design problems. Students are able to: (a) use and discuss concepts and terminology related to TIL design; (b) choose the appropriate methods for addressing TIL design challenges; (c) formulate requirements for TIL systems that are unambiguous and SMART; (d) apply methods and tools to enhance creativity in design processes; (e) design and specify TIL systems engineering solutions through the use of requirements analysis, conceptual designs, and design architecting; (f) define KPIs (criteria) by which design alternatives can be compared in an effective manner; (g) use the appropriate methods and tools to evaluate and choose among design concepts. 2. To train students in critical thinking and in using scientific methods for solving TIL research problems. Students are able to: (a) explain what makes transport systems complex systems; (b) explain concepts like evolution, emergence, abstraction and reduction and give examples within the TIL domain; (c) perform literature search and review a scientific paper; (d) formulate research questions and choose appropriate (scientific) methods to answer these (e) collaboratively write a scientific paper; (f) Identify and rationally discuss fallacies (flaws in reasoning) and biases in their own thinking and work and that of others. 3. To train students in solving complex TIL design and research problems in (multi-disciplinary) group work. These (soft) skills are not explicitly assessed or graded, but are used throughout Q1 and Q2 in group work (design, research and TIL specialisations related). Students are able to: (a) Brainstorm content and scope of research and design problems in a team; (b) Discuss the validity of assumptions and evaluation results; (c) Organize project work in teams; (d) Use each others talents and expertise.	
Education Method	This course provides lectures, workshops and group work (design exercise in Q1 + a research project in Q2)	
Assessment	A. Integrated design exercise report (assignment, group work) B. Written exam on design methodology (individual), via digital examination C. Scientific Paper (assignment, group work) Parts A + B = 5 ECTS (in Q1) Part C = 2 ECTS (in Q2) A. The integrated design exercise is a group assignment and is thus graded as group work. To be able to pass, the grade of this assignment must be equal or higher than 5.0. The weight of this part is 2/3 of 5 ECTS (48% of the course result). B. The written exam on the design methodology is assessed individually. To be able to pass, the grade of this individual exam must be equal or higher than 6.0. The weight of this part is 1/3 of 5 ECTS (24% of the course result). C. The scientific Paper is again a group assignment, graded as group work. To be able to pass, the grade of this assignment must be equal or higher than 5.0. The weight of this part is 2 ECTS (28% of the course result). GIVEN $A \geq 5.0$, $B \geq 6.0$, $C \geq 5.0$, the overall score for the course is determined by: $0.48 * A + 0.24 * B + 0.28 * C$	
Expected prior Knowledge	none	
Academic Skills	We explicitly train some of the academic skills (see learning goals). What we expect of students is an academic attitude, both in individual work (exam) and in in group work: (a) We expect all students to study and work regularly and seriously --- we also expect that students expect the same from each other. Should problems arise in groups only as a very last resort will we step in to take measures (b) We expect students to engage in scientific debate, which is a very different thing than having a personal argument. Whos right about say proposition A is not determined by popular vote nor by strength of voice but by the amount of EVIDENCE for proposition A. (c) We will check all project reports for plagiarism---this is a very serious offence that can cost you dearly.	
Literature & Study Materials	Presentations (full slide decks) form the base material for the individual exam and can be used, alongside literature sources, to do the assignments. All needed materials are made available via BrightSpace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/.	
Permitted Materials during	The exam is 'closed book', but one-side written or printed A4 sheet of notes is allowed. No further materials or tools are allowed.	

Exam	
Collegerama	No

TIL6022	TIL Python Programming	3
Responsible Instructor	P.K. Krishnakumari	
Contact Hours / Week x/x/x/x	8 x 4 hours	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course is a hands-on course for students to learn the fundamentals of coding and Python programming language. With the course, we will cover the software installation, setting up the environment, and introduction to collaborative/versioning tools such as GitHub. Furthermore, this course will aid in developing programming skills from conceptualizing problems into schemes and data structures suitable for a computer to solve. Python, its tools and syntax, will be taught to develop these skills through hands-on learning activities and lab sessions. Assignments about traffic and logistics provide concrete examples of using Python to solve real-world problems. While the students will work with different datasets during their in-class learning activities, they will delve in-depth into a single dataset for their assignment.	
Study Goals	At the end of this course, the student should be able to: - familiarize themselves with python fundamentals - conceptualize and translate problems into modular programmable workflows - implement the data pipeline of a real-world problem - use visualization to convey the insights and findings - use versioning systems for efficient code management and to collaborate with peers	
Education Method	Lectures + Lab assignments + Projects Lectures: 8 x 2 hours - lectures with in-class coding exercises Lab assignments: 8 x 2 hours - scheduled lab sessions with supervision Projects: 7 x 2 hours - scheduled sessions with supervision Projects: 7 x 2 hours - group work with no supervision Self-study: 24 hours Total = 84 hours	
Assessment	The final grade is composed of three components: 1. Individual exam (30%) 2. Lab assignments (Pass or Fail) 3. Group projects (70%) You need a minimum of 5.8 for each of the graded components (1 & 3). You also need to successfully complete 5 out of 8 lab assignments to get a pass for component 2.	
Expected prior Knowledge	None. This is a coding course for beginners and those who want to learn about structured programming.	
Academic Skills	Developing fundamental programming skills for simulation, optimization and data analysis.	
Literature & Study Materials	Python manuals, Online resources, Coding cheatsheets	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	None	
Collegerama	No	

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Specialisations (27 EC) (1 specialisation)

Year	2023/2024
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Specialization Transport Governance (27 EC)
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AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'inteken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

CIEQ6212	Regions, transport and networks	4
Responsible Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr.ir. M. Saeednia	
Instructor	M. Rinaldi	
Instructor	Dr. C. Maat	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The concept of accessibility and its relevance for user behaviour and transport network and spatial design; Spatial and social external effects of transport infrastructures and use, and its relevance for transport network and spatial design; Characteristics, data and data analysis of population, economic activity and land use; Interactions in regional, land use and transport network developments and their dynamics in time; Urban and regional transport demand models; Transport networks and regional development optimisation and design problem; Base introduction and application of Transport Economics, System Dynamics, and Agent-Based Modelling.	
Study Goals	On completion of this course, the student will be able to: LO1. To explain the concept of accessibility and its indicators. LO2. To analyse population, economic activity and land-use data and to 342 evaluate the accessibility and spatial, social and economic impact of existing or new land use transport plans. LO3: To explain the conceptual, dynamic relationships between land use, transport system quality and transport behaviour, both of passengers and freight transport. LO4: To apply location theory and network design optimization methods on land use and transport challenges. LO5: To create transport network land use design on basis of formulated societal requirements. LO6: To model the complex, dynamic relationship between land use and transportation development.	
Education Method	Lectures, practical lecture/workshop	
Assessment	Written exam (100%)	
Expected prior Knowledge	Transport Demand Modeling and optimization techniques.	
Academic Skills	Mathematical modeling	
Literature & Study Materials	- Lecture slides; - Book: Jean-Paul Rodrigue (2020) 'The geography of Transport Systems'. Fifth Edition; New York: Routledge, 456 pages. ISBN 978-0-367-36463-2	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Closed book.	
Collegerama	No	

SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5
Module Manager	N. Mouter	
Instructor	Dr. J.A. Annema	
Instructor	J. Rezaei	
Instructor	N. Mouter	
Instructor	Dr. R.M. Shortall	
Co-responsible for assignments	J. Rezaei	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Transport policies are almost always controversial. Several methods are applied worldwide to support decision-making in transport. Nowadays, societal cost-benefit analysis (CBA) and Multi Criteria Decision Methods (MCDM) are the most prominent and well-known transport policy appraisal methods. In this course you will learn about the newest scientific developments in these appraisal methods. Additionally, you will learn about new appraisal methods such as Participatory Value Evaluation.	
Study Goals	Describe and explain traditional evaluation methods (CBA and MCDM) and non-traditional evaluation methods. Explain the pros and cons of different methods for different evaluation circumstances Describe the ethical critiques for different types of evaluation methods Define/pick a research gap (to be used in the paper) Write an academic paper, based on completed case study exercise, where: A research gap that the student can fill/contribute to is identified ; Pros and cons of ONE chosen method are critically reflected upon; The suitability of the method is convincingly argued; The results are analysed and interpreted, criticised with particular attention to ethical issues; Policy recommendations are made on the use of the results	
Education Method	In formal lectures the methods and theories will be explained. By two exercises students will apply the methods and theories. Finally, by analysing a pre-defined real-world controversial transport policy case and at least ten scientific papers (which they have selected themselves) students will learn to critically reflect on appraisal outcomes. They have to write a concise essay to report their findings and reflections.	
Assessment	The exam is an individual paper of 3500 words (excluding appendices and list of references). (70%) A CBA assignment (15%) A MCDM assignment (15%)	

SEN1721	Travel Behaviour Research	5
Module Manager	M. Kroesen	
Instructor	B. Pudane	
Instructor	Dr. E.J.E. Molin	
Instructor	M. Kroesen	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. E.J.E. Molin	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Knowledge about basic statistics and data-analysis (probability density functions, hypothesis testing, regression) is absolutely required as prior knowledge. Without this knowledge it will be very difficult to follow this course.	
Course Contents	In this course students learn how to perform travel behaviour research and how to translate the findings from such research into recommendations for transport policy design. The students will acquire knowledge about six modelling paradigms in travel behaviour research, namely the econometric, psychological, geographical, biographical, mobility style and sociological paradigm. The strengths and weaknesses of each paradigm will be discussed. In addition, three state-of-the-art methods to analyse travel behaviour (structural equation models, latent class choice models and latent class clusters models) will be treated. These methods figure prominently within the six modelling paradigms. For each method there will be a group assignment (so 3 in total). These group assignments will count for 50% of the grade.	
Study Goals	After this course the students will be able: (1) to carry out travel behavior research using three state-of-the-art statistical methods, namely structural equation modeling, latent class cluster analysis and latent class choice models. (2) to translate the study findings to recommendations for transport policy design.	
Education Method	Lectures Computer practicals Group project (4 persons)	
Computer Use	In the course the following statistical softwarepackages will be used: SPSS, Latent Gold, Apollo and AMOS	
Assessment	Knowledge about the modelling paradigms and three methods will be tested in an individual written exam in week 10, which accounts for 50% of the final grade. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be an online "open book" Mobius exam. The group project will consist of three assignments which together account for 50% of the final grade. Both grades should be >5.75 to pass the course. Both parts can be separately passed.	

TPM032A	Multi-criteria Decision Analysis	5
Module Manager	J. Rezaei	
Instructor	J. Rezaei	
Responsible for assignments	J. Rezaei	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic mathematics and programming	
Course Contents	During this course, students will learn how a multi-criteria decision problem can be structured, solved, and analyzed. A multi-criteria decision problem is a problem where a number of alternatives (options) need to be evaluated with respect to a number of criteria (attributes) in order to (i) select the best alternative, (ii) rank all the alternatives, or (iii) sort the alternatives into a number of classes. A multi-criteria decision analysis (MCDA) method is a method to find a solution for (one or more parts of) an MCDA problem, where the main phases are: - Formulating the problem (identifying the goal(s); alternatives; and criteria), - Evaluating the alternatives with respect to the criteria, - Finding the relative importance of (or tradeoff among) the criteria, - Synthesizing the data collected in the previous phases to find a solution, - Checking the reliability and validity of the outcomes. These phases are executed with close interaction between decision-maker(s)(DMs) and analysts. This course covers the main phases of MCDA problems and some of the main MCDA methods. An overview of the contents of the course is as follows. - Introduction - Structuring an MCDA problem - Multi-Attribute Value methods - Multi-Attribute Value Theory (MAVT) - Analytic Hierarchy Process (AHP) - Best Worst Method (BWM) - Outranking Methods - ELECTRE methods - PROMETHEE methods - Group Decision-Making - Behavioral considerations	
Study Goals	On successful completion of this course, the students should be able to: - Structure multi-criteria decision problems; - Analyze decision problems using different MCDA methods, including multi-attribute value methods and outranking methods; - Structure and analyze decision problems considering different views of a group of decision-makers/stakeholders; - Explain and analyze behavioral aspects of decision problems (e.g., cognitive biases); - Apply an MCDA method to a real-world decision problem and analyze the outcomes.	
Education Method	Lectures Assignments Experimental analysis Real-world applications	
Assessment	The course has two partial assessments as follows: - Assignments (students in groups of size 2 apply MCDA methods to different decision-making problems): 40% - Application (students in groups of size 2 apply an MCDA method to a real-world decision-making problem and write a report of 8000 words): 60% The review of the Assignments and Application is done following a rubric. In order to pass the course, the grade of each part (Assignments, Application) should be at least 5.0, and the average of the parts $(0.4 * \text{Assignments} + 0.6 * \text{Application})$ should be at least 5.8. Resit: Students who fail the course have a possibility to repair the Assignments and Application to a maximum of 6.0. If other group members have already passed the course, the students may do this individually. The deadline for the resit is the end of the fifth week of Q2. If a student fails the course, they take separate partial grades the following year (partial grades are registered separately in Osiris).	
Elective	Yes	
Category	MSc level	

WM1301TU	Ethics of Transportation	3
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems - can see these problems within the broader background of moral, legal and political philosophy - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way	
Education Method	A series of 6 lectures and some working group sessions, focused on the discussion and application of the theories presented in the course	
Assessment	Written exam and assessment participation in working groups	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Specialization Traffic Operations (27 EC)
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AE4423-20	Airline Planning and Optimisation	4
Responsible Instructor	Dr.ir. B.F. Lopes Dos Santos	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	It is assumed that students are familiar with modelling and optimization techniques. It is assumed that students have experience in code development in Python or MatLab (or any other programming language).	
Course Contents	This course provides students with knowledge to analyse planning problems related to airline operations and to develop modeling approaches to solve these problems. The focus is on the relationship between planning models and their operational implications. It starts with a general overview of the airline demand analysis, followed by the study of the planning framework which airlines operate in. This planning framework includes strategic decisions, namely fleet planning and network development, tactical decisions, such as scheduling and maintenance planning.	
Study Goals	At the end of the course, the students should be able to: Obj1: identify the main strategic, tactical, and operational problems of an airline; Obj2: get familiarized with the development and implementation of some modeling techniques: (a) mix-integer linear programming, (b) multi-commodity flow networks, (c) time-space networks, (d) Markov decision programming. Obj3: get familiarized with the development and implementation of some solution techniques: (a) branch-and-bound, (b) column generation, (c) dynamic programming. Obj4: identify, formulate and solve airline strategic and tactical planning problems: (a) airline networks development, (b) fleet planning, (c) frequency planning, (d) aircraft assignment and routing planning, (e) crew scheduling, and (f) maintenance planning. Obj5: identify an airline operations problem, analyse and solve it; Obj6: explain the implications of airline planning decisions and report them in an academic manner.	
Education Method	The course will work according to a blended learning approach. Online lectures will be prepared for 9 content sessions. Discussion and Q&A sessions of 1 hour will be planned per week and there will be tutorial lectures to discuss the assignment(s). The students are assessed according to the grades of a group assignment (60%) and an individual online exam (40%).	
Literature and Study Materials	References literature: - Belobaba, P., Odoni, A., Barnhart, C. (2009), "The Global Airline Industry", John Wiley & Sons Ltd, West Sussex, UK. [ISBN: 978-0-470-74077-4] - Lectures notes and additional material posted on Brightspace during the course. Additional Literature: - Doganis, R. (2010), "Flying Off Course: Airline economics and Marketing", 4th Edition, Routledge: London, UK. [ISBN: 0-415-44737-2] - Bazargan, M. (2010), "Airline Operations and Scheduling", 2nd Edition, Ashgate Publishing Company, Surrey, UK. [ISBN: 978-0-7546-7900-4] - Clark, P. (2007), "Buying the Big Jets: Fleet Planning for Airlines", Ashgate Publishing Company, Hampshire, UK. [ISBN: 978-0-7546-7090-2]	
Prerequisites	The students should have completed the AE4441 course, or the ME44206 course, or an equivalent operations research course. The students should also be familiar with programming. For python courses, students can follow TIL6010, AE1205, or TI3105TU.	
Assessment	Group assignments - divided into two parts (60%) Final exam (40%) The students need to have a grade equal to or higher than 5.5 in both examination elements. The group assignments are done by groups of (maximum) 3 students.	
Permitted Materials during Tests	None	

CIEQ6003	Traffic Modelling and Management	6
Responsible Instructor	Dr. V.L. Knoop	
Instructor	Dr. M. Wang	
Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Prof.dr.ir. S.P. Hoogendoorn	
Contact Hours / Week x/x/x/x	x/x/6/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Roadway capacity and cumulative curves Intersection control Car-following behaviour, micro-macro relations, trajectories Railway capacity Timetable stability and max-plus algebra Speed contour plots & capacity drop Fundamental diagram & shock waves Traffic flow properties for buses, trams and metro, incl. macroscopic fundamental diagrams and dwell time functions. Transit signal priority. Pedestrians: expand traffic flow theory to two dimensional planes Safety, environment; control objectives Systems and control Network management - hierarchical and multi-level (Application of) traffic simulation tools (reliability, seeds, ...)	
Study Goals	On completion of this course, the student will be able to: TBU3-LO1 Describe traffic flows in space and time at microscopic and macroscopic level (TTE-TB-LO1) TBU3-LO2 Discuss and apply theories and models at microscopic and macroscopic level for different transport modes (TTE-TB-LO3). TBU3-ILO3 Develop and assess control strategies at local and network level (TTE-TB-LO2). TBU3-ILO4 Set-up, perform, and report a traffic flow simulation experiment (TTE-TB-LO4).	
Education Method	Lectures, feedback on group work	
Assessment	Written exam (100%)	
Tags	Analysis Modelling Transport & Logistics Transport phenomena	
Expected prior Knowledge	Calculus; basic insights into modelling and insights into simulation setup.	
Academic Skills	Critical thinking	
Literature & Study Materials	Lecture notes provided by the teachers	
Judgement	100% exam	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	Yes	

CIEQ6223	Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment	4
Responsible Instructor	M. Rinaldi	
Instructor	Dr. M.P. Hagenzieker	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	Dr.ir. W.J. Schakel	
Instructor	Dr. V.L. Knoop	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Introduction, classification and functional description of intelligent vehicles; System architecture, decision and control methods for intelligent vehicle systems; Technologies for intelligent vehicles, sensors, communication, and actuators; Behavioural adaptation to intelligent vehicles; Impact assessment of intelligent vehicles on traffic safety; Impact assessment of intelligent vehicles on traffic efficiency; Simulation experimental research of impacts of intelligent vehicles on traffic flow using traffic flow/driving simulation	
Study Goals	On completion of this course, the student will be able to: B2U3-LO1. describe and analyse the essential components of intelligent vehicle systems; B2U3-LO2. analyse and design the tactical and/or operational level decision-making systems of your intelligent vehicle system; B2U3-LO3. compare the characteristics of different sensing and communication technologies for intelligent vehicles; B2U3-LO4. identify and assess behavioural adaptation effects of intelligent vehicles; B2U3-LO5. (a) analyse the main methods of traffic safety impact assessment for intelligent vehicle systems; (b) assess the traffic safety impacts of intelligent vehicle systems; B2U3-LO6. (a) analyse the main methods of traffic flow impact assessment for intelligent vehicle systems; (b) assess the traffic flow impact of intelligent vehicle systems; B2U3-LO7. apply traffic simulation to assess the traffic flow impacts of a selected intelligent vehicle system.	
Education Method	Lectures (2x 2 hours per week)	
Assessment	100% group assignment.	
Expected prior Knowledge	The course is closely related to CIEM6220-U4 Traffic Flow Modelling and Control and CIEM6220-U1 Traffic Safety.	
Academic Skills	In addition to the technical materials, students will also practice scientific writing, literature research, presenting, giving feedback and working in teams	
Literature & Study Materials	Reader, recommended papers and slides	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	None	
Collegerama	No	

ME44200	Operations and Maintenance	3
Responsible Instructor	Dr.ir. Y. Pang	
Instructor	Dr. V. Reppa	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course focuses on intelligent control to optimize the operational performance and to maintain the reliability of multi-machine systems in large-scale transport and logistics domain. Along with the understanding of diverse transport systems and logistic processes, including the dynamic interactions of the equipment and facilities involved, the control for multi-machine systems contains two folds, the integrated control for efficient operations and the intelligent decision-making for reliability control. To ensure efficient and reliable operations at both equipment level and system level, technologies and methodologies of automated data acquisition, remote communication, intelligent data analysis and integrated operational decision-making will be studied. Condition-based maintenance of systems and equipment will be analysed and discussed in combination with on-line techniques for monitoring the health of the sensors used for maintenance. During the course industrial case studies will be given to present the state-of-the-art development and foreseen challenges. The course is concluded by practical assignments where such intelligent control systems will be researched and conceptually designed.	
Study Goals	After successfully completing this course, you will be able to: - explain the concepts of intelligent control for operational support systems and advanced maintenance strategies; - identify the properties that determine efficient operation and reliable performance; - analyse the complexity of the operations and interactions of multi-machine systems; - derive models to represent the interactions and operational processes of multiple equipment and facilities; - design intelligent operational control systems in term of mathematical models and integrated control methods; - apply the methodologies and technologies to improve the efficiency and reliability for both individual machine and overall multi-machine system; - describe the effects of sensor faults in operations and maintenance of transport systems; - explain the design of model-based sensor fault detection methods; - explain the design of multiple sensor fault isolation methods; - design a sensor fault detection and isolation method for the operation and/or maintenance of a transport system.	
Education Method	Lectures (2 hours per week), case study, practical assignments	
Literature and Study Materials	Handouts and references to relevant literature and media. Additional teaching materials to be provided during lectures and available via Brightspace . Expected prior knowledge: ME44101 (not obligatory). Prior knowledge in system engineering (dynamical systems) is recommended.	
Assessment	Group assignment: Assignment Reports & Oral Exam Final grade is calculated by weighted average of Assignment Part I (75%) and Assignment Part II (25%). In case of a final grade less than 6, the group can have one opportunity to resubmit or repair the insufficient work via one of two modalities: 1. Repair: to improve the work of the same assignment towards a maximum retake grade of 6; 2. Resubmit: to choose a new assignment to be assessed by general assessment criteria towards a retake grade higher than 6. After successful delivery of the group assignment reports, the Oral Exam takes place in the exam week of the education period.	
Department	3mE Department Maritime & Transport Technology	

ME44305	System Analysis and Simulation	5
Responsible Instructor	Ir. M.B. Duinkerken	
Instructor	F. Schulte	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Required for	Multi-Machine Engineering	
Expected prior knowledge	Basic programming	
Parts	Systems and Data Analysis Discrete-Event and Agent-Based Simulation	
Course Contents	The course is an introduction to "Problem Engineering"; how to formulate a well-defined problem in a complex situation. The course introduces a systems approach to define the elements and structure of a problem situation and the process approach of discrete simulation to understand and quantify the time-dependent relations (behavior)	
Study Goals	Learning objectives: - Describe all relevant elements and their relations for at least one complex industrial or transportation system (e.g., cyber-physical production systems) using a combination of at least two conceptual models: process models, system models, steady-state models; - Collect data for the described system and analyse the system using the collected data and introduced data analysis methods; - Design industrial and transportation systems and processes applying an established innovation model; - Transform the developed conceptual models into a process-oriented simulation model, using a suitable language (e.g., Delphi/Tomas/Lazarus or Python/Salabim/Simpy).	
Education Method	lectures + group assignments	
Computer Use	Windows laptop with Python	
Literature and Study Materials	"The Delft Systems Approach, Analysis and Design of Industrial Systems", 2008, H.P.M.Veeke, J.A.Ottjes, G. Lodewijks, Springer, ISBN 978-1-84800-176-3 Lecture notes	
Assessment	Report of group assignment	
Exam Hours	80 for assignment	
Tags	Abstract Broad Challenging Design Industry Logics Modelling Process Programming Programming concepts Prototyping Stochastics Transport & Logistics	
Percentage of Design	50%	
Design Content	Logistic systems	
Department	3mE Department Maritime & Transport Technology	

SEN9110	Simulation Masterclass	5
Module Manager	Prof.dr.ir. A. Verbraeck	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Experience with a simulation environment such as Arena, Simio, Tomas, NetLogo, GAMA or MESA. Basic knowledge about probability theory and statistics.	
Course Contents	System Theory, Object Orientation, Discrete Event System Specification, Multi-Formalism Simulation, Distributed Simulation, Real-Time Simulation, Multi-Paradigm Simulation, Multi-Resolution Simulation, and Simulation Language selection will be the core topics of the course. After an introduction to system theory, the inner working of simulation environments will be illustrated on the basis of the DEVS, DESS, and DTSS formalisms. Then, possible integrating of the different formalisms will be shown. Several special topics will be taught, based on the latest research in simulation. This material will be illustrated in intensive and interactive courses. In addition to the lecture topics, several other simulation topics will be studied by groups of students, who will write a scientific paper, and present their findings in class. These topics can be focused on the MSc program that the students participate in; special topics to study are available for TIL, CoSEM, EPA, Computer Science, and other students. Finally, groups of students will study a simulation package in-depth and discuss the commonalities and differences with other packages. Again, the package chosen can be targeted at the MSc program of the students. TIL students can, e.g. study a package that is more aimed at logistics and transport, CoSEM students can focus on a package that is used in systems design, whereas EPA students can focus on a package used in policy analysis. Most packages can be made available for use from home. This course requires active participation and contains group work.	
Study Goals	After taking this course the student will have knowledge about: - systemic principles behind simulation, and clear understanding of key terms, such as state, time, transition, event, process, and activity; - internal working of different kinds of discrete event simulation languages and environments, with a focus on state transitions over time; - underlying theories and formalisms of discrete event simulation, such as DEVS, DTSS and DESS; - the DEVS paradigm, including state trajectories and transition functions; quantized DEVS (QDEVS); discretized time; - Dynamic Structure DEVS (DSDEVS), and other DEVS variants; - transformation between simulation formalisms; the formalism-transformation graph; - object-oriented simulation environments; service-based simulation; - structure and abilities of distributed simulation; the concept of HLA to manage time and state in distributed simulations; classes of functions in a distributed simulation environment; - latest research activities in the field of simulation, with research topics like web-based simulation, real-time control using simulation, agent based simulation, interactive simulation and gaming, and simulation in special domains; - examples of successful and less successful simulation studies and the learning experiences of those studies; - important similarities and differences between simulation languages, and selection methods for simulation packages. Practical skills will have been developed with respect to: - usage of at least one simulation package that was previously unknown; - writing of a scientific paper on the topic of simulation. Insight will have been gained on: - comparing and contrasting simulation languages; - the importance of time and state relationships for any type of simulation modeling; - the formal foundations of the simulation domain.	
Education Method	Lectures about state-of-the-art simulation theory Computer use for studying one discrete-event or agent-based simulation package in-depth per group Paper writing by student groups on a special topic Lectures by student groups on a special topic.	
Computer Use	Each group of students will study and evaluate a discrete-event or agent-based simulation package. Examples are Simio, Arena, Enterprise Dynamics, Plant Simulation, AnyLogic, Extendsim, Flexsim, Jaamsim, DSOL, Salabim, MESA, GAMA, Netlogo, and Repast Symphony. Also here, the simulation package to study can be specifically adapted to the group of students. Most packages can be made available for use from home.	
Course Relations	The course contains very little overlap with other simulation courses such as EPA1324, EPA1352, TB231/TB233, etc.	
Literature and Study Materials	Set of scientific papers and book chapters that will be made available through Brightspace. Typically, every class will have on average 2 papers to study for the exam. The set of papers is also available as a Reader that can be obtained on paper.	
Reader	A paper-based or PDF version of the Reader is available. The content of the Reader is also electronically available in Brightspace.	
Assessment	The mark for this course will be based on the result of two summative group assignments (paper with presentation, lab study of simulation package consisting of 4 sub-assignments that all count equally heavy) and on a summative individual written exam. The individual written exam will count for 50%, the group paper (including the class presentation) for 30% and the study of the simulation package for 20%. Each of the three marks has to be 5.75 or higher to be included in the final mark. Partial marks will stay valid for the study year in which they were obtained and the study year thereafter. The written exam takes place at the end of Q1 with a retake at the end of Q2. When a group assignment is insufficient or missing, the group will be given the opportunity to repair the insufficient or missing assignment.	
Permitted Materials during Tests	All written materials are allowed during the exam: papers, class notes, slides.	
Elective	Yes	
Tags	Challenging Intensive Modelling Programming concepts Software Stochastics	
Targetgroup	Students in EPA, CoSEM, TIL, TEL, PEL, Computer Science, and any other MSc student with some prior knowledge about	

Extra Skills Trained	computer simulation. Writing a high-quality scientific paper; exploring a new simulation language / package.
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Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Specialization Transport Networks (28 EC)
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AE4423-20	Airline Planning and Optimisation	4
Responsible Instructor	Dr.ir. B.F. Lopes Dos Santos	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	It is assumed that students are familiar with modelling and optimization techniques. It is assumed that students have experience in code development in Python or MatLab (or any other programming language).	
Course Contents	This course provides students with knowledge to analyse planning problems related to airline operations and to develop modeling approaches to solve these problems. The focus is on the relationship between planning models and their operational implications. It starts with a general overview of the airline demand analysis, followed by the study of the planning framework which airlines operate in. This planning framework includes strategic decisions, namely fleet planning and network development, tactical decisions, such as scheduling and maintenance planning.	
Study Goals	At the end of the course, the students should be able to: Obj1: identify the main strategic, tactical, and operational problems of an airline; Obj2: get familiarized with the development and implementation of some modeling techniques: (a) mix-integer linear programming, (b) multi-commodity flow networks, (c) time-space networks, (d) Markov decision programming. Obj3: get familiarized with the development and implementation of some solution techniques: (a) branch-and-bound, (b) column generation, (c) dynamic programming. Obj4: identify, formulate and solve airline strategic and tactical planning problems: (a) airline networks development, (b) fleet planning, (c) frequency planning, (d) aircraft assignment and routing planning, (e) crew scheduling, and (f) maintenance planning. Obj5: identify an airline operations problem, analyse and solve it; Obj6: explain the implications of airline planning decisions and report them in an academic manner.	
Education Method	The course will work according to a blended learning approach. Online lectures will be prepared for 9 content sessions. Discussion and Q&A sessions of 1 hour will be planned per week and there will be tutorial lectures to discuss the assignment(s). The students are assessed according to the grades of a group assignment (60%) and an individual online exam (40%).	
Literature and Study Materials	References literature: - Belobaba, P., Odoni, A., Barnhart, C. (2009), "The Global Airline Industry", John Wiley & Sons Ltd, West Sussex, UK. [ISBN: 978-0-470-74077-4] - Lectures notes and additional material posted on Brightspace during the course. Additional Literature: - Doganis, R. (2010), "Flying Off Course: Airline economics and Marketing", 4th Edition, Routhledge: London, UK. [ISBN: 0-415-44737-2] - Bazargan, M. (2010), "Airline Operations and Scheduling", 2nd Edition, Ashgate Publishing Company, Surrey, UK. [ISBN: 978-0-7546-7900-4] - Clark, P. (2007), "Buying the Big Jets: Fleet Planning for Airlines", Ashgate Publishing Company, Hampshire, UK. [ISBN: 978-0-7546-7090-2]	
Prerequisites	The students should have completed the AE4441 course, or the ME44206 course, or an equivalent operations research course. The students should also be familiar with programming. For python courses, students can follow TIL6010, AE1205, or TI3105TU.	
Assessment	Group assignments - divided into two parts (60%) Final exam (40%) The students need to have a grade equal to or higher than 5.5 in both examination elements. The group assignments are done by groups of (maximum) 3 students.	
Permitted Materials during Tests	None	

CIEM6301	Railway Traffic Management	5
Responsible Instructor	E. Quaglietta	
Module Manager	E. Quaglietta	
Instructor	Prof.dr. R.M.P. Goverde	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Railway timetabling: Microscopic and macroscopic rail traffic simulation models Timetable design techniques and optimization models Timetable assessment using microscopic simulation Energy-efficient train operation Real-time railway traffic management: Driver Advisory Systems Disruption management Train delay prediction and analysis using a statistical train delay analysis tool Railway traffic management systems Conflict detection and resolution Simulation-based assessment of traffic management strategies This elective will include discussions about societal impacts of advanced tools and techniques for train service planning and management. Students will be aware of responsibilities and ethical implications in using automated decision support systems in rail traffic planning/management practice. Roles and responsibilities of railway stakeholders in planning / managing rail traffic will be tackled together with interaction processes during nominal and degraded operational conditions.	
Study Goals	On completion of this module, the student will be able to: LO1: Design railway timetables (CE ILO1) LO2: Assess rail traffic management methods (CE ILO4) LO3: Analyse train delays (CE ILO2) LO4: Apply rail traffic simulation (CE ILO2) LO5: Evaluate energy-efficient train operation (CE ILO3)	
Education Method	Blended learning mixing on-campus lectures, reading, videos, guest lectures and formative assignments	
Assessment	Formative group assignments on rail traffic simulation, scheduling and traffic data analysis with supervised practicum sessions. Formative assessment also includes pre- and/or post-lecture tests, on-line self-assessment, training with model exams with answers. Final digital written exam. The exam will include questions on lectures and formative group assignments. A minimum grade of 5.8 is required to successfully pass the exam.	
Expected prior Knowledge	Prior knowledge about planning and design of public transport systems is recommended, such as units of Module B3 of the MSc programme in Transport and Traffic Engineering (i.e. Public Transport Planning and Operations, Public Transport networks and modelling, Railway operations and control). In particular, blocking time theory is assumed working knowledge. Additional reading (one chapter, 3 hours) is required for those not familiar with blocking time theory. In addition, an introduction course on discrete optimization is recommended.	
Academic Skills	Problem formulation and problem solving Critical and analytical thinking Interpretation Writing technical reports Applying professionally-relevant tools and methods	
Literature & Study Materials	I.A. Hansen & J. Pachl (eds.), Railway Timetabling & Operations, Eurailpress, Hamburg, 2nd edition, 2014. Lecture slides of the course available on Brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Open book - digital exam	
Collegerama	No	

CIEQ6212	Regions, transport and networks	4
Responsible Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr.ir. M. Saeednia	
Instructor	M. Rinaldi	
Instructor	Dr. C. Maat	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The concept of accessibility and its relevance for user behaviour and transport network and spatial design; Spatial and social external effects of transport infrastructures and use, and its relevance for transport network and spatial design; Characteristics, data and data analysis of population, economic activity and land use; Interactions in regional, land use and transport network developments and their dynamics in time; Urban and regional transport demand models; Transport networks and regional development optimisation and design problem; Base introduction and application of Transport Economics, System Dynamics, and Agent-Based Modelling.	
Study Goals	On completion of this course, the student will be able to: LO1. To explain the concept of accessibility and its indicators. LO2. To analyse population, economic activity and land-use data and to 342 evaluate the accessibility and spatial, social and economic impact of existing or new land use transport plans. LO3: To explain the conceptual, dynamic relationships between land use, transport system quality and transport behaviour, both of passengers and freight transport. LO4: To apply location theory and network design optimization methods on land use and transport challenges. LO5: To create transport network land use design on basis of formulated societal requirements. LO6: To model the complex, dynamic relationship between land use and transportation development.	
Education Method	Lectures, practical lecture/workshop	
Assessment	Written exam (100%)	
Expected prior Knowledge	Transport Demand Modeling and optimization techniques.	
Academic Skills	Mathematical modeling	
Literature & Study Materials	- Lecture slides; - Book: Jean-Paul Rodrigue (2020) 'The geography of Transport Systems'. Fifth Edition; New York: Routledge, 456 pages. ISBN 978-0-367-36463-2	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	Closed book.	
	No	

CIEQ6232	Public Transport Demand and Network Planning and Operations	5
Responsible Instructor	Dr. O. Cats	
Instructor	N. van Oort	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Modelling PT planning problems. The course consists of three blocks: network and demand analysis, network assignment and design, and service design. The course covers the following topics: - Planning process - Network analysis - Demand and behavioral modelling - Network assignment - Network design - System resilience - Cost-benefit analysis - Data and modelling in planning practices - Demand and supply real-time management - Mobility-on-Demand - Service optimisation in operations and planning practices - Long-distance travel The course includes lectures, lab tutorials and talks highlighting insights from practice. Societal aspects of the entire public transport planning process are discussed. The unit will discuss the relations between public transport service design and societal/urban development goals such as accessibility, coverage, sustainability, equity and resilience. Attention is also given to communicating modelling and analysis results to decision makers and implications of mobility service developments for different user groups. This unit is the official substitute of the course CIE4811 "Planning and Operations of Public Transport Systems".	
Study Goals	LO1. Analyse key dependencies among design variables, parameters, and criteria in public transport service planning and management and their dynamics. LO2. Design and examine public transport networks while considering coverage, accessibility, travel and operational costs. LO3. Characterize the structure of public transport networks by applying graph theory techniques. LO4. Compare alternative network models, simulation, optimization and choice modelling techniques in the realm of public transport, and their key notions, strengths, and weaknesses. LO5. Select the modelling approach adequate for a given application concerning network modelling, optimization and assignment models, and interpret the results of these models	
Education Method	The course involves lectures, in-class discussions, feedback sessions, pre and post-lecture material such as reading scientific articles and watching videos.	
Assessment	Group assignment including an individual component: 100%.	
Tags	Algorithmics Analysis Business Modelling Programming Sustainability Transport & Logistics Transport phenomena	
Expected prior Knowledge	This courses provides knowledge on planning and operations of public transport systems and does not have specific prior knowledge requirements. Notions related to network science, transport fundamentals and mathematical programming will be used in several classes.	
Academic Skills	The main academic transferable skills that will be developed in this unit are critical thinking, analytical thinking and problem solving, specifically applied to the planning and operations of public transport networks and demand analysis. Students are expected to exercise the following academic skills: - Problem formulation - Data analysis - Statistical interpretation - Comparing alternatives quantitatively and qualitatively - Working in diverse groups - Critically reflecting on their own work - Report structuring and writing - Reading and summarizing scientific papers The main personal skills that the student will develop are organizational, team-building, and interpersonal and presentation abilities by means of concrete cooperation with colleagues in assignments having the scope of emulating projects and interactions in a professional environment.	
Literature & Study Materials	Lecture slides and pre- and post-lecture material including 3-5 reference papers are made available through Brightspace.	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable (no exam)	

Collegerama	No
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CIEQ6233	Railway Operations and Control	5
Responsible Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Dr. D.M. van de Velde	
Contact Hours / Week x/x/x/x	x/x/x/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Railway safety and signalling Railway safety principles Train separation, Interlocking, Automatic train protection Blocking time theory Railway capacity assessment Next generation signalling (ERTMS, CBTC, Virtual Coupling) Railway automation Automatic Train Operation Grades of Automation ATO-over-ETCS Railway traffic Control - Metro control Migration to ERTMS and ATO Coordination in the railway value chain Railway governance Railway policy Railway value chains Short-term, Mid-term and Long-term coordination This unit will underline the impacts of railway signalling and automation on railway performance and public safety, making the students aware of the responsibilities and ethical implications related to the introduction of fully automated railway operations. Roles and responsibilities of the different railway system components and stakeholders are introduced, alongside with potential conflicts of interest in the railway business. Regulations and policies over the entire railway value chain are discussed in line with international objectives on sustainable mobility.	
Study Goals	LO1: Assess railway safety and signalling systems LO2: Assess railway automation LO3: Analyse railway value chains and coordination misalignments LO4: Evaluate railway governance and main policy options LO5: Evaluate migration issues, including ERTMS and ATO	
Education Method	This course uses a mixture of lectures, reading, guest lectures, and formative exercises.	
Assessment	Written exam (100%)	
Expected prior Knowledge	None	
Academic Skills	Modelling, design, critical thinking, reasoning	
Literature & Study Materials	J. Pachl, Railway Signalling Principles, 2nd ed., 2021. And a set of papers or book chapters made available on BrightSpace. Each lecture will require reading one paper/book chapter.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Simple calculator	
Collegerama	No	

SEN1721	Travel Behaviour Research	5
Module Manager	M. Kroesen	
Instructor	B. Pudane	
Instructor	Dr. E.J.E. Molin	
Instructor	M. Kroesen	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. E.J.E. Molin	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Knowledge about basic statistics and data-analysis (probability density functions, hypothesis testing, regression) is absolutely required as prior knowledge. Without this knowledge it will be very difficult to follow this course.	
Course Contents	In this course students learn how to perform travel behaviour research and how to translate the findings from such research into recommendations for transport policy design. The students will acquire knowledge about six modelling paradigms in travel behaviour research, namely the econometric, psychological, geographical, biographical, mobility style and sociological paradigm. The strengths and weaknesses of each paradigm will be discussed. In addition, three state-of-the-art methods to analyse travel behaviour (structural equation models, latent class choice models and latent class clusters models) will be treated. These methods figure prominently within the six modelling paradigms. For each method there will be a group assignment (so 3 in total). These group assignments will count for 50% of the grade.	
Study Goals	After this course the students will be able: (1) to carry out travel behavior research using three state-of-the-art statistical methods, namely structural equation modeling, latent class cluster analysis and latent class choice models. (2) to translate the study findings to recommendations for transport policy design.	
Education Method	Lectures Computer practicals Group project (4 persons)	
Computer Use	In the course the following statistical softwarepackages will be used: SPSS, Latent Gold, Apollo and AMOS	
Assessment	Knowledge about the modelling paradigms and three methods will be tested in an individual written exam in week 10, which accounts for 50% of the final grade. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be an online "open book" Mobius exam. The group project will consist of three assignments which together account for 50% of the final grade. Both grades should be >5.75 to pass the course. Both parts can be separately passed.	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Specialization Logistics Systems (26 EC)

CIEQ6213	Freight transport networks and systems	4
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Prof.dr.ir. L.A. Tavasszy	
Instructor	Dr. S. Fazi	
Instructor	Dr.ir. A.J. van Binsbergen	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Economic activities, trade, logistics and freight flows; Logistics networks with consolidation and transshipment points; Modelling of freight transport demand and supply; Unimodal and intermodal freight transport networks; Freight transport and logistics optimisation from the user perspective; Terminal and port operations; Urban freight transport, urban distribution; Sustainability effects of freight transport, sustainability policy consequences for freight transport Transport Economics (applied)	
Study Goals	On completion of this course, the student will be able to: LO1. To explain the current state and future developments of freight transportation. 343 LO2. To explain logistic processes. LO3. To design, model and analyse freight flows. LO4. To explain and analyse the interaction between freight demand and supply. LO5. To design, model and analyse freight traffic flows for multimodal networks in global, national and urban settings.	
Education Method	Lectures, guest lectures, formative group assignment (multiple parts) with instruction	
Assessment	Written exam (100%)	
Expected prior Knowledge	1. Basic insights performance, impacts, ethical and social implications and their indicators - of traffic and transport networks; 2. Basic insights in transport economics (choice preferences, business perspectives); 3. Basic insights in Transport Modelling and Analysis, including aggregated, disaggregated models, gravity models and discrete choice models for freight networks; 4. Basic knowledge on distribution (OD-estimation), assignment and equilibrium models, Basic knowledge on mixed-integer linear optimisation; 5. Insight in characteristics of unimodal and intermodal networks, physical and service networks for freight transport; 6. Insight in data-driven modelling approaches for verification, calibration and validation of models, input and parameters in the context of transport infrastructure, network, operations and behaviour.	
Academic Skills	analytic skills; data analysis; academic writing (for the exam)	
Literature & Study Materials	- Lecture slides; - Book: Tavasszy, L., G. de Jong (eds) (2014), Modelling Freight Transport, Elsevier (digital edition) - Report: CE Delft (2020) 'STREAM Freight Transport' (digital edition; annual update) - Paper: Jin Y. Yen (2015) Finding the K shortest Loopless Paths in a Network, Management Science 17(11)	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	handheld calculator	
Collegerama	Yes	

ME44101	Dynamics and Interaction of Material and Equipment	4
Responsible Instructor	Prof. Dr.ir. D.L. Schott	
Instructor	Dr.ir. Y. Pang	
Co-responsible for assignments	H. Shi	
Co-responsible for assignments	E.F.L. Stok	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ME44115	
Course Contents	This course focuses on modelling and designing machine-cargo interactions and using those interactions to conceptually design equipment taking into account the transport demands and logistics processes. The course starts with the logistic context with emphasis on (bulk and container) terminal level, for which the functionality of equipment types will be addressed with regard to the cargo type (dry bulk, container, piece goods) and its characteristics. Specific equipment types will be discussed, analysed and designed, such as a belt conveying systems used for bulk materials (e.g. iron ore and coal) as well as for baggage and parcel handling. The interaction between cargo and equipment will be explicitly taken into account. Specifically for the bulk solid cargo type, an experimental assignment to determine the properties of a particular bulk solid material is part of the course. Bulk solid materials include granular materials such as coal, sand, limestone, iron ore, grain. These materials can be free flowing through bunkers and chutes as well as stored in silos, handled by stackers and reclaimers or transported by conveyors. With the experimentally determined properties conceptual design of a silo and belt conveyor will be undertaken. State-of-the-art particle based simulation with Discrete Element Method (DEM) will be introduced. The parameters, algorithms, and applications will be addressed, as well as the procedure for validation of DEM simulations to obtain realistic bulk material behaviour in a virtual environment. It will be shown how DEM has led to new breakthroughs and developments in the field of port handling equipment.	
Study Goals	The student will be able to: - Differentiate among various types of cargo terminals based on the specific characteristics of cargo and their corresponding logistical requirements. - Conduct experimental analysis to determine key properties of bulk solid materials, and utilize the findings to inform effective equipment design. - Design equipment solutions that are tailored to specific logistical requirements, taking the performance of the equipment within the logistics context into account, supported by thorough calculations. - Assess the effectiveness of equipment designs by considering the relationship between material characteristics, material behavior, and material equipment interaction parameters on equipment performance. - Evaluate the role of Discrete Element Method (DEM) in the design of equipment, including understanding the theory and algorithm used for DEM calculations and analyzing the impact of input parameters (particle level, simulation level) on the output, specifically on bulk behavior and erratic behavior.	
Education Method	Lectures, including guest lectures from industry experts. Practical for testing and characterising bulk solid materials and its material equipment interaction properties by experiments.	
Course Relations	ME44115 DEM simulations	
Literature and Study Materials	Schulze, D., Powders and Bulk Solids - Behavior, Characterization, Storage and Flow, Springer, 2008 ISBN 978-3-540-73767-4 (print), 978-3-540-73768-1 (online), DOI: 10.1007/978-3-540-73768-1 http://www.springerlink.com/content/155416/?p=fbeb6748815f4e4c92f56519a15f8837&pi=0 Rhodes, M., Introduction to Particle Technology by Martin Rhodes, John Wiley & Sons, 2008. ISBN 978-0-470-01427-1 (print), 9780470727102 (online), DOI: 10.1002/9780470727102 http://www3.interscience.wiley.com/cgi-bin/bookhome/117932420?CRETRY=1&SRETRY=0	
Assessment	Slides and supplementary material provided on Brightspace and/or during lectures Because of measures resulting from COVID-19, the prescribed assessment will be as follows: 1. practical assignment - remote (25% of the mark) 2. written exam - remote open book with several fraud prevention measures (75% of the mark) If the opportunity arises for examinations on campus, the method of assessment will be: 1. practical assignment (25% of the mark) 2. written exam - closed book (75% of the mark).	
Enrolment / Application	Please enroll yourself as soon as possible. It helps us to schedule and setup enrollment for practical (in pairs or threesomes)	
Remarks	MSc ME track Multi Machine Engineering MSc TIL Engineering specialisation	
Percentage of Design	20%	
Design Content	Conceptual design of material handling equipment considering cargo characteristics and interaction parameters between cargo and equipment	
Department	Dynamics of Material and Equipment Interaction 3mE Department Maritime & Transport Technology	

ME44300	Multi-Machine Coordination for Logistics	3
Responsible Instructor	Dr. V. Reppa	
Instructor	Prof.dr. R.R. Negenborn	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this course students will get familiar with automatic control techniques and their benefits and applications for real-time coordination of actions of large-scale transport systems. Theoretical concepts mostly related to model predictive control are discussed, as well as their application to real-world transport systems. Illustrative examples of large-scale systems include among others intermodal / synchromodal (i.e. road, waterborne, rail) transport networks. The operation of these networks relies on the interoperability of various machines such as cargo ships, trains, road tracks, automated guided vehicles, quay/yard cranes and many more.	
Study Goals	Students are able to: LO1. apply system theory to describe large-scale machines and transport systems LO2. explain the design of an automatically controlled system and of the automatic coordination of several controlled systems LO3. explain the design of control architectures for large-scale interconnected machines and transport systems (centralized, distributed, single-agent, multi-agent, single-level, multi-level) LO4. evaluate the pros and cons of different control architectures and their impact on multi-machine coordination for transport LO5. design an optimal control strategy for a single machine used in the transport network (truck, ship, crane, etc) LO6. develop a multi-agent framework based on model predictive control for real-time logistics of large-scale transport systems	
Education Method	Study material will be provided to support the lectures during which new concepts will be discussed and explained. Small groups of students will be created to conduct a project that will help them to develop deeper understanding of the theoretical (optimal) control and coordination concepts applied to a specific machine that is part of a larger transport network. Throughout the course illustrative examples will be used for showing the application of new concepts. Feedback sessions will be organized to discuss questions about the project.	
Course Relations	The course is related to the exit qualifications 1.b, 1.c, 3.b, 3.c and 7.a of the track MME of the Master Programme Mechanical Engineering, and to the specialization Engineering of the Master Programme Transport, Infrastructure and Logistics. Prior knowledge in automation of transport systems, dynamical systems, optimization and simulation is recommended.	
Literature and Study Materials	We will discuss chapters from the book "Intelligent Infrastructures", Negenborn et al., Springer, Dordrecht, The Netherlands, 2010. ISBN: 978-90-481-3597-4. Additional handouts and references to relevant scientific literature and media will also be provided.	
Assessment	Assessment will be based on a written group project report (50%) and on a written individual (closed-book) exam (50%). In principle, the written exam will take place on campus. Resit options are offered for both the project and the exam. Note that the grade of the exam and the grade of the project are only valid for the academic year 2023-2024.	
Tags	Information & Communication Modelling Optimalisation Project Transport & Logistics	
Department	3mE Department Maritime & Transport Technology	
Contact	Dr. Vasso Reppa (v.reppa@tudelft.nl)	

ME44311	Advanced Operations and Production Management	5
Responsible Instructor	A. Napoleone	
Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	To enroll in the course, students must have a BSc in Engineering. The course is especially for students following TIL, MME and SDPO.	
Course Contents	The course requires high engagement and commitment, and includes both self-study and group work. Students are expected to be willing to work in teams, to collaborate and use social and project management skills. The course revolves around different topics of operations and production management, including: system analysis and process mapping, modularity in product design and process design, smart manufacturing and Industry 4.0, reconfigurability and adaptability in operations management.	
Study Goals	With this course, the student is expected to develop a structured approach to address operational problems using the methods provided in the course and/or methods identified with individual study. After successfully completing this course, students will be able to: - Describe the fundamental concepts of operations and production management - Search for and use methods to solve operational problems in different applications - Have a structured approach to manage familiar as well as unknown operational problems	
Education Method	Problem-based and interest-driven approach. PART ONE in Q3: (i) Plenary lectures / Cases by industrial speakers AND (ii) Individual study and deepening of one/more operational topics of interest. Attending lectures is required to acquire the fundamentals of production and operations management. Active participation during plenary lectures is strongly advised. At the end of Q3, students submit individual reports (one report per student). PART TWO in Q4: Group projects (up to 6 students per group) with case companies AND Group supervision (on demand). In Q4, students work in groups at different projects with case companies. The course leader can assist students in finding a company. Groups work autonomously and ask for supervision when needed. At the end of Q4, students submit group reports that include multiple analyses of operational problem at the case company and the solution to the problem. Q5 is available groups if preferable for time constraints.	
Assessment	The assessment is based on the evaluation of the individual report (Q3) and group report (Q4).	
Department	3mE Department Maritime & Transport Technology	
Judgement	The report is judged by Dr A. Napoleone The final grading considers the individual report (50%) and the group report (50%)The grading is based on a Rubric. Results can be discussed in feedback sessions (on demand)	
Contact	a.napoleone@tudelft.nl	

SEN9720	Logistics and Supply Chain Innovations	5
Module Manager	Ir. M.W. Ludema	
Instructor	Ir. M.W. Ludema	
Responsible for assignments	Ir. M.W. Ludema	
Co-responsible for assignments	Dr. J.H.R. van Duin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	Supply chain management is about the exchange of product, information and money between companies or organizational entities that are part of a supply chains or demand/supply network. Supply chain management contributes to the competitive strength of companies in the market place. The supply chain management function co-ordinates and controls operational business value adding processes in and between companies. For a large extent the supply chain management function determines the flexibility of companies to respond to market changes and determines the efficiency of the internal processes within companies, as well as the inter-company processes. Logistical and supply chain decision-making frequently relates to situations in which conflicting wishes and desires from various parts of companies or organizations have to be brought together cohesively. Gaining insight into the performance indicators of a company/organization, its logistical costs, customer expectations and the opportunities to improve these aspects is of vital importance for survival of companies/organizations. All engineers will encounter to a larger or lesser extent logistics and supply chain management in the business and institutional environment in where they will be working in, or in where they will be carrying out their assignments.	
Course Contents	This course contains logistical, supply chain and innovations in logistics and supply chain topics. Logistics Topics: - The significance of the logistics system of companies in supply chains; - Methods for analyzing logistics systems of companies; - The role of inventories in logistics systems of companies; - The role of transport between logistics systems of companies; - Connecting purchasing and customer service within companies; - Value added logistics, outsourcing and third party logistics; - Performance Measurement in logistics systems of companies; - Logistics control and information systems; - Design of logistics networks and locations for distribution centers. Supply Chain Topics: - Global trade lanes, industrial clusters and supply chains; - Strategy issues and the supply chain; - Organizational issues and supply chains; - Supply Chain Mapping, Benchmarking and the SCOR-reference model; - Physical and Virtual Value Systems (MRP-I, MRP-II, ERP and APS); - Supply Chain Value Adding Processes; - Lean and Agile Supply Chains and the strategic role of decoupling points; - Supply Chain Collaboration (Supply Chain Partnerships & Alliances); - Supply Chain Analysis; - Supply Chain Design. Innovations in Logistics and Supply Chain Management: - The Last-Mile of Supply Chains; - Multi-modal and Synchro-modal Logistics and Supply Chains; - Customs and Trade Compliance; - Supply Chain Integration and the use of Cross Chain Control Centers (4C); - Service Logistics (Asset Management, Maintenance Management and spare parts logistics); - Sustainable, Green and Closed-Loop Supply Chains within a sustainable economy, - Smart Supply Chains and the Internet of Things; - Supply Chain Finance; - Smart Contracts & Block Chains Technology and the Supply Chain.	
Study Goals	This course covers the supply chain management function in and between companies. This course will provide the student insight in the theoretical background of the supply chain, its current use and expected developments and a wide set of technical and informational innovations that companies consider to implement. The student learns and puts into practice the basic theoretical skills of a supply chain analyst and engineer and can show the contribution of relevant logistics and supply chain innovations. Theoretical knowledge and practical skills will be explained and exercised by means of a set of relevant cases.	
Education Method	The course contains 17-24 lectures (2 or 3 time 2 hours each week), in-class discussions about presented topics and distributed articles and cases. During the course assignment groups of 4-5 students will work on a logistics and supply chain analysis/engineering assignment. In the final weeks of the course the assignment groups will compile their results in a final report. Furthermore the course contains self-study and some group assignment feedback sessions. The draft report of the course assignment groups will discussed with the groups prior to final hand-in.	
Literature and Study Materials	This course works with a virtual reader. Suggested Course Materials is continuously under construction (see manual on Bright Space), relevant articles will be related to weekly lectures.	
Practical Guide	See Bright Space for the updated course and assignment manual and course assessment issues.	
Prerequisites	This course builds on the knowledge gained during courses like operations management, production logistics, logistics management or equivalent courses. Students without (sufficient) knowledge and background are advised to spend additional study time during the first two to three weeks of this course. Please consult the module manager / lecturer during the first lecture or earlier. During the first two lectures an overview of relevant operational and logistics management issues will be provided by means of intensive lectures and a few in-class assignments.	
Assessment	Final deliverable and group assessment: Students will work in groups on a large group assignment in one of the following example domains:	

Transportation Equipment Supply Chain;
 Construction Supply Chain;
 After-sales service /spare parts supply chain for the maintenance of durable goods;
 Supply chain of conditioned goods for the retail market;
 Supply chain of the inter city last mile;
 Care supply chain of the Medical Equipment;
 Other issues in respect to the interest of the student.

A final report including analysis of the current supply chain, impact assessment of a chosen logistics and supply chain innovation and a design of a feasible future supply chain.

TPM028A	Decision Making in Multimodal Transport Systems	5
Module Manager	M.Y. Maknoon	
Instructor	M.Y. Maknoon	
Responsible for assignments	M. Farhani	
Co-responsible for assignments	J. Rezaei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Knowledge in operations research (mathematical modelling, integer programming) - Basic knowledge of optimization solver (e.g. CPLEX) and programming	
Course Contents	In case of deficits, students need to invest additional time to acquire these knowledges. Coordinating physical, information and decision flow is a focal point in managing multi-modal freight transportation systems. By having a global view on the entire supply chain - from raw material acquisition to the end customer delivery multimodal systems aim to provide a rapid, efficient and low-cost solution for freight transportation. On the other hand, social and environmental concerns have a central role in political and economic decision making. These concerns result in laws and regulations that affect the production of goods and services. Therefore, in addition to provide efficient, reliable and green services multimodal systems should be flexible enough to respond to short and long-term environmental, social and political changes. To address these challenges, effective, powerful and intelligent methods and techniques are required to analyse, plan and manage multi-modal systems. Operations research is at the heart of this intelligence. Operational research models and methods, integrated with enterprise information management systems, analyse and transform data into useful information for planning and management. This course focuses on the multimodal freight transportation. In particular design, planning and operation of multimodal network. The models and techniques taught in this course deal with a wide range of problems. From the long-term physical design and operating structure to the transport route selection, service mode and timetabling.	
Study Goals	The course focuses on multimodal transportation network. It presents the main models and methods to analyse and solve problems arises in this context. The course aims at knowledge acquisition and development of problem solving skills. The goal is to enhance students analytical and modeling capacities in research and practice.	
Education Method	The course is based on active participation. In case of unforeseen circumstances or measures resulting from COVID-19, student participation will be facilitated by on-line platforms (e.g. Microsoft Teams, Skype).	
Assessment	Readings will be given for each session and presentations will be supplemented by discussions with the students. The course is assessed based on two mid-term take-home exams (15% each adding up to 30% of the total grade) and one closed-book final exam (70% of the total grade). In case of unforeseen circumstances or measures resulting from COVID-19, the final exam will be take-home and open-book.	
Elective	To pass this course, students need to get more than 5.5/10 for the final exam. The results of mid-term exams are valid only for the current year and are not transferable to the next academic year. In addition, re-take is not allowed for the mid-term exams.	
Tags	Yes Design Mathematics Modelling Optimalisation Programming Transport & Logistics	
Category	MSc level	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

TIL interdisciplinary projects

TIL4020-20	TIL Research Project	5
Responsible Instructor	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	Starting academic year 2020-2021	
Education Period	1 3	
Start Education	1 3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This course teaches TIL students how to design and manage scientific research in the TIL domain. TIL students will become Master of Science. This means that they have to be able to do and manage scientific research. However, what is scientific research actually? What is the difference between scientific research and consultancy research? What is the role of scientific theory (or theories) in scientific research? What is the role of scientific literature in scientific research? In this course these kinds of questions are addressed using literature and insights from teachers experienced in doing TIL related research and from fellow TIL-students who just finished their master thesis. In this course also writing a good quality conclusion and discussion section is taught.	
Study Goals	After this course students are able: - To design a scientific research in the TIL domain; - To write a coherent research proposal in which scientific gaps are identified; research question (s) is/are formulated which clearly address the gaps; scientific literature is properly reviewed; method(s) is/are proposed which fit to the research question(s); - To scope the research in such a way that it is doable in the time specified (e.g., for TIL master thesis six month is specified); - To identify and find good quality scientific literature in the TIL domain; - To recognize and apply clear scientific storylines; - To identify and manage pitfalls in doing scientific research based on experiences from TIL master students who just finished their master thesis and TIL scientists. - To write a good quality conclusions and discussion section.	
Education Method	Workshops of four hours will be given. Students will be taught briefly (in lectures) some background and theory. The main purpose of the workshops, however, is that students (in groups) will practice in finding an interpreting scientific literature, theories and methods. They will also practice with defining research questions, scoping, storyline building and managing their own research.	
Assessment	Students have to write individually a scientific research proposal of around ten pages including scientific gap identified, research question(s), methods, literature review, scoping and planning. They may choose their own research related to the TIL domain. They have to use at least ten peer reviewed scientific papers. They have to write a conclusion and discussion section related to their literature review part. This conclusion and discussion section will also be assessed with a grade.	
Expected prior Knowledge	None.	
Academic Skills	Students should be able to write in a scientific style and have good English language skills.	
Literature & Study Materials	Sheets and practice material will be made available to the students. Two books are recommended: - Piet Verschuren Hans Doorewaard (2013), Designing a research project, 2nd Revised edition, Boom Lemma Uitgevers. - David E. Gray (2013), Doing Research in the Real World, 3rd Revised edition, Sage Publications Ltd.	
Judgement	Two grades will be given. The full research proposal is graded (80%), and the conclusion/discussion section of the literature review also (20%).The requirements for the marks will be made clear to the students beforehand based on a rubric. The two parts (80/20%) should each be at least 5.0 and the weighted average of both parts >5.75 to pass.	
Permitted Materials during Exam	Not applicable.	
Collegerama	No	

TIL4020-20 P1	Research Proposal	4
Responsible Instructor	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	Check TIL4020-20	
Education Period	1 3	
Start Education	1 3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Check TIL4020-20	
Study Goals	Check TIL4020-20	
Education Method	Check TIL4020-20	
Assessment	Check TIL4020-20	
Expected prior Knowledge	Check TIL4020-20	
Academic Skills	Check TIL4020-20	
Literature & Study Materials	Check TIL4020-20	
Judgement	Check TIL4020-20	
Permitted Materials during Exam	Check TIL4020-20	
Collegerama	No	

TIL4020-20 P2	Literature Review	1
Responsible Instructor	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	Check TIL4020-20	
Education Period	1 3	
Start Education	1 3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Check TIL4020-20	
Study Goals	Check TIL4020-20	
Education Method	Check TIL4020-20	
Assessment	Check TIL4020-20	
Expected prior Knowledge	Check TIL4020-20	
Academic Skills	Check TIL4020-20	
Literature & Study Materials	Check TIL4020-20	
Judgement	Check TIL4020-20	
Permitted Materials during Exam	Check TIL4020-20	
Collegerama	No	

TIL5050-20	TIL Design Project	10
Responsible Instructor	Dr. J.M. Vleugel	
Instructor	Dr.ir. J.H. Baggen	
Contact Hours / Week x/x/x/x	n/a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc TIL students late in their final year.	
Summary	Who, what, how, when?? Week x.1-x.5 -> P0 info (preparation) Week x.6-x.10 -> P1 info (pvs) Week x.11-x.20 -> P2 info (project) Advanced project course of 20 weeks in systems/design engineering. Process information: Project by interdisciplinary teams: 4-5 students with different BSc's, at least 2 different TIL specialisations, preferably mixed m/f, cultures. This corresponds with the student mix and the (international) labour market for engineers. Course coordinator checks these criteria before accepting a proposed team. Mandatory enrolment: Form on Brightspace. You have to fulfill the 2 entry requirements. Arrangements: Q1+Q2, Q2+Q3, Q3+Q4 and Q4+Q1. Three modules with entry and exit requirements: P0 Preparation (3 lectures, build team, find commissioner + topic, start PVS); until end of week x.5. P1 Vision & Scope (PVS document); until end of week x.9. P2 Analysis & Design (project work, presentations and reporting): 10* weeks (flexible; exams), P0 + P1 = 2 EC. P2 = 8 EC. Supervision: Selected by course coordinator. Commissioner appoints own supervisor(s). Weekly supervision meetings arranged by the team with the two TUD supervisors (if possible together, online or at the TUD) and between the team and the commissioner supervisor(s). 4 joint mandatory meetings: Course coordinator, TUD- + commissioner supervisors meet with team in Kick-off, mid-term, green light, final presentation + defence meetings. Course coordinator chairs these meetings (he attends the mid-term meeting only if he is a weekly supervisor or on special request). All meetings start with (20 min.) presentations. Mid-term interviews (5*15 min.) with an independent person (if available). Use: Course manual, advised project timeline, tools (flyer, company contact list, list with finished projects); on Brightspace. Contact: Course coordinator is available in weeks 1.1-5.1, but not in weeks 5.2-5.6. For a full course overview without login: https://www.tudelft.nl/en/student/ceg-student-portal/education/master/transport-infrastructure-logistics/programme-2023-2024/4-projects *ADVICE: IF YOU CAN, START AS EARLY AS POSSIBLE, TO PREVENT LAST MINUTE RUSH AND STRESS.	
Course Contents	Coursebase, Brightspace or MSc TIL website section 4.	
Study Goals	After the course all team members have reached the level to start a MSc thesis. This includes 1. Project acquisition skills; 2. Research capabilities; generic as well as topic specific. 3. Design and evaluation capabilities; generic as well as topic specific. 4. Reporting and presentation skills; 5. Team work and project management skills.	
Education Method	Start-up lectures. Self-managed projects with weekly supervision by commissioner and TUD. Planning: TIL5050-20 takes 5 + 5 + 10 = 20 weeks, as follows: P0 Preparation Week 1.1 + 1.2 (resp. week 3.1 + 3.2) - 3 mandatory lectures for Q1 + Q2 (resp. Q3 + Q4) teams. Week n.1 - Enrolment forms TIL5050-20 received and accepted. Week n.1-n.5 - Team, commissioner, topic. Week n.5 - Course coordinator checks if enrolled students have 50 credits and communicates this to students. P1 Vision & Scope Week n.6-n.9 - Write a PVS (proposal). Week n.9 - V/O grade for PVS (OSIRIS), go/no go decision for the project (P2). Week n.10 - Kick-off meeting with TUD + commissioner supervisors. P2 Analysis & Design Week m.1-m.5 - Execution. Week m.5 - Mid-term meeting + mid-term interviews. Week m.6-m.10 - Execution. Week m.9 - Green light meeting. Week m.10 - Final presentation and defence meeting. Grade in OSIRIS. In case of P0/P1 in Q4 and P2 in Q1 next year:	

	Week 3.1 + 3.2 - 3 mandatory lectures. Week 4.1-4.5 for P0 + P1. Week 1.1-1.10 for P3. The P0/P1 planning spreads the time spent on preparation over a long period. In practice, some teams may study faster and their members may already have received the 50 EC at the beginning of P0. If a commissioner and topic are swiftly found, then they have a longer period to finalize their PVS. Other teams may be in a different position. P2 cannot start earlier than the next period, because of supervision planning. If Q4-Q1 is chosen, then the summer break extends the gross lead time of the project.
Course Relations	TIL5050-12 P0, P1, P2; TIL4030-16; TIL4020-20.
Practical Guide	Digest - Coursebase TIL5050-20 main, P0, P1 and P2 pages. - Brightspace: Course manual, acceptable topics, finished and current projects, open project calls, company contacts, project flyer. - TIL5050-20 lecture sheets + notes. - TIL4030-16 lectures notes + literature about the theory and practice of design. - TIL4020 lecture material when writing your PVS and project report(s). Then contact the Course coordinator, who will assist you where-ever needed. He is also one of the 2 TUD supervisors in many of the projects.
Prerequisites	2nd year course. Specific knowledge related with the content of your project is advisable. There is no time to learn a new subject or tool.
Assessment	P1: Grading scheme. P2: Grading rubric.
Enrolment / Application	Mandatory course for all 2nd year MSc TIL students. Not offered as an elective. No acceptance of exchange students or students from other MSc programmes or universities. Enrolment conditions (CiTG MSc TIL2022 TER & Annex 2022-2023): Sub 2: TIL4020-20 TIL5050-20 Sub 3: TIL4030-20 course TIL research and design methods: passed Sub 4: "The student is allowed to continue and thus complete the second project mentioned in Section 2 only, once he has successfully completed courses amounting to a total of at least 50 credits at the first day of the fifth week of the educational period in which the project starts." These entry conditions are a baseline. Exceptions can be granted for individual students in special cases. This refers to a received, discussed and accepted positive advice from a TIL study advisor. Regular study delays due to failing to pass exams are not one of these cases. Enrol on Brightspace in the period you want to start preparing your project (contact the course coordinator if enrolment is not open yet). Use enrolment form TIL5050-20. This should be received in week x.1. The Course coordinator checks your form and may accept or reject your application based on fulfilment of the enrolment conditions. Team composition: The course coordinator may add an additional student to a proposed team of 4 or in exceptional cases change an unbalanced team. The course coordinator also helps in finding additional team members, if requested. All proposed team members should fulfil the enrolment form requirements. The course coordinator can check OSIRIS (SVO).
Special Information	This course is closed during Q5. Students can start with P0 each quarter. P0 > P1 > P2. P0/P1/P2 are about the same project carried out in the same study year. Force major: If a granted (by the commissioner) and accepted (by the course coordinator) project would be cancelled by the commissioner during or just after P1, then the course coordinator makes a decision on how to proceed. A back-up plan may be handy. This means that you should always have at least 1 serious commissioner as backup. The course modules can start and finish with some flexibility, taking care of exams and holidays. Discuss flexibility with your supervisors or the course coordinator. *Project work and meetings can be held partially or fully online. Team activities are organized according to the needs of the team. Make your own arrangements in communication with the commissioner, your TUD supervisors and the course coordinator. Online work reduces the need and cost of traveling and softens the burden of meeting planning. The way eventual health regulations are applied at the commissioner's office may differ from the way the TUD applies them.
Tags	Analysis Building & Spatial Development Design Drawing Economics Energy Fieldwork Industry Modelling Policy Analysis Project planning / management Rail & Road Engineering / Planning Sustainability Technology Telecommunication Transport & Logistics
Contact	Course Coordinator and supervisor: Dr. Jaap Vleugel. Available for online meetings (Teams/Zoom).

	Data + meetings: Email meeting requests to j.m.vleugel@tudelft.nl. For in-between questions regarding P1 and P2 some pre-information by email eases my work considerably. For P2 supervision: Regular weekly + on demand. Prepare any meeting by sending an email with agenda and documents with marked changes to all attendees. Supervision meetings: 1 day before. Mid-term, green light and final presentation: 2 days before or on Friday afternoon if the meeting is on the Monday after the weekend. A checklist for each meeting can be found in the course manual and on Brightspace.
Expected prior Knowledge	Read: Literature & Study Materials on the P0 page in coursebase.
Academic Skills	This is a 2nd year course. Earlier experience with the field of application is advisable.
Literature & Study Materials	MSc 2nd year. See Practical Guide.
Judgement	https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/onderwijsinformatie/educational-rules-and-regulations . For more information on grading, see article 14 in the Rules and Guidelines MSc TIL R&G.
Permitted Materials during Exam	-
Collegerama	No

TIL5050-20 P0	Preparation	0
Responsible Instructor	Dr. J.M. Vleugel	
Instructor	Dr.ir. J.H. Baggen	
Contact Hours / Week x/x/x/x	n/a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	See Study Goals.	
Study Goals	1. To compose an interdisciplinary team. 2. To find a(n external) commissioner. 3. To acquire a project from this commissioner that is scientifically acceptable and practically feasible in terms of content, planning and composition of the team. 4. To prepare your PVS document. 5. To manage your team work effectively and efficiently. 6. To communicate effectively and efficiently with the commissioner and the course coordinator and your supervisors at the TUD (if already available).	
Education Method	September & January (repetition): 3 mandatory lectures: Team & Process, Problem Definition, Design + Evaluation. P0/P1: Start-up meeting and follow-up meetings with the course coordinator. Meetings between the team and the commissioner supervisor(s). Meetings with the weekly supervisors (selected & appointed by the course coordinator).	
Course Relations	TIL5050-12 Main, P1, P2. TIL4030-16, TIL4020-20.	
Assessment	None.	
Enrolment / Application	Week x.1 email enrolment form TIL5050-20 (on Brightspace) to course coordinator for approval. All enrolment requirements should be fulfilled by the end of week x.5.	
Expected prior Knowledge	Read - Coursebase TIL5050-20 main, P0, P1 and P2 pages. Literature & Study Materials (below). - Brightspace: Examples of earlier PVS, course manual, acceptable topics, finished and current projects, open project calls, company/agency contacts, project flyer. - TIL5050-20 lecture sheets + notes. - TIL4030-16 lectures notes + literature about the theory and practice of design. - TIL4020 lectures when writing your PVS and project report(s). Then contact the Course coordinator, who will assist you in the process. He is also one of the 2 TUD supervisors in many of the projects.	
Academic Skills	MSc 2nd year.	
Literature & Study Materials	Clive L Dym, Alice M Agogino, Ozgur Eris, Daniel D Frey, Larry J Leifer, Engineering Design Thinking, Teaching, and Learning, Journal of Engineering Education v94 n1 (January 2005): 103-120. TUD Library code: download. Clive L. Dym, Patrick Little, and Elizabeth J. Orwin, Engineering design : a project-based introduction, Harvey Mudd College, 4th edition, Wiley, 2014. TUD Library code: ATA214. Andrew P. Sage, James E. Armstrong, Jr., Introduction to systems engineering, Wiley, 2000. TUD Library code: MBA200. Courses Systems engineering: TIL4030-16. Material from other design courses (master or bachelor) if you followed these. Mandatory TIL5050-20 lectures and all other information on Brightspace. Mandatory TIL4020-20 lectures and all other information on Brightspace. All relevant mandatory and elective courses of the MSc TIL program. ----- P1: PVS template and accepted PVS can be found on Brightspace. The template is also practical in your discussions with the commissioner (e.g. needs statement, data acquisition and risk management). The grading scheme for P1 is based on the template.	
Judgement	https://www.tudel.ft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations For more information on grading, see article 14 in the Rules and Guidelines MSc TIL R&G.	
Permitted Materials during Exam	No exam, but Final Presentation & Defense at the end of P2.	
Collegerama	No	

TIL5050-20 P1	Vision & Scope	2
Responsible Instructor	Dr. J.M. Vleugel	
Instructor	Dr.ir. J.H. Baggen	
Contact Hours / Week x/x/x/x	na	
Education Period	1	
	2	
	3	
	4	
Start Education	1	
	2	
	3	
	4	
Exam Period	none	
Course Language	English	
Course Contents	In P1 you write your PVS. It is based on a small literature study and discussions with your commissioner, the course coordinator and if time-wise possible, also your TUD supervisors. Ideally, you start working on it as soon as the course coordinator approves your commissioner and topic; week x.6 or earlier. It should be ready by the end of week x.9. Baseline: You choose a design/systems engineering method to structure your project and build your story-line. On top of it you use other scientific methods and tools in support of the Problem analysis, Literature overview, Stakeholder analysis, Design requirements analysis, Designs, Evaluation. [Problem analysis] The problem as defined by the commissioner is a starting point for your actual definition of the problem: "Problem analysis [...] involves identifying the overriding problem and establishing the causes and effects related to that problem. A key element of this analysis will ensure that root causes, not just the symptoms of the problem, are identified and subsequently addressed in the project design. Projects that only address the effects of the problem, and not its underlying causes, are unlikely to produce sustainable benefits." (1) Problem analysis involves - A short literature review to see if there are theories that deal with the stated problem, to choose appropriate research and design tools, to elaborate relevant topics and to define the research gap. - Study of the commissioner's organisation and its context. This allows you to collect qualitative (interviews, brainstorm) and quantitative data (data collected by the commissioner and your own measurements). - Stakeholder analysis with a choice of (a coalition of) main stakeholders. This material helps you to scope your project, modify your research questions, specify requirements and constraint for your designs. - Requirements analysis: What are key requirements, constraints and (if applicable) kpi to be fulfilled in the designs? These are not given by the commissioner, but established by your team. [Design] The design phase is meant to develop and evaluate possible solutions (alternatives) to reduce or solve the problem(s) as identified in the problem analysis, to choose the one(s) best fitting with the requirements analysis. You then evaluate the design(s) based on the same (or subset of these) criteria. [Technical perspective] The project suits a technical university, hence it should have some technical perspective/application/component, which should be elaborated to a sufficient extent. Examples are RFID or barcodes, sorting or transport equipment in a warehouse or the use of driver assist tools in a car, fuel technology. A model is regarded as a practical tool, not as technical perspective. [Project choice: Balance science-practice] No consultancy-style projects = do exactly what the commissioner wants. As an academic you should be critical towards any information you receive and follow your own line when discussing and advising the commissioner. The course format does not allow you to learn new theories, methods or apply unfamiliar software. In case of software, you will not develop a real software tool, but you can help the commissioner by building a preliminary design. [Available topics] Not all fields and topics can be chosen. This depends on available commissioners and options for supervision at the TUD. The company list on Brightspace may change. The course coordinator will also assess the risk level of each project based on previous experiences. (1) https://www.elo.iastate.edu/engl-309/problem-analysis/ 1. To write a Project Vision and Scope document (PVS) that strikes a balance between - The needs expressed by the commissioner; - The team's interdisciplinary knowledge and ideas; - The requirements and ideas of the TUD supervisors. 2. To achieve a quality level for the PVS that leads to a sufficient grade (V). 3. To manage your team work effectively and efficiently. 4. To communicate effectively and efficiently with the commissioner and the course coordinator and your TIL supervisors. Research on-site and off-site (TUD or home). Writing draft PVS. Discussions with commissioner supervisor. Discussions with the course coordinator and TUD supervisors. TIL5050-12 Main, P0, P2; TIL4030-16; TIL4020-20. Read - Coursebase TIL5050-20 main, P0, P1 and P2 pages. - Brightspace: Examples of earlier PVS, course manual, acceptable topics, finished and current projects, open project calls, company contacts, project flyer. - TIL5050-20 lecture sheets + notes. - TIL4030-16 lectures notes + literature about the theory and practice of design. - TIL4020 lectures when writing your PVS and project report(s). Then contact the Course coordinator, who will assist you where-ever needed. He is also one of the 2 TUD supervisors of many of the projects. The company supervisor(s), the course coordinator and (to be appointed) TIL supervisors usually have a few comments on the draft PVS versions. Once you sufficiently responded to them a grading will follow in week x.10.	
Study Goals		
Education Method		
Course Relations		
Practical Guide		
Assessment		

	The final, graded, PVS is also the kick-off document. Grading scheme on Brightspace. PVS should get a sufficient (Voldoende) to acquire 2 EC and pass the course. Passing P1 is also an entry requirement for P2.
Contact	See P0.
Expected prior Knowledge	For process: Earlier projects about working in teams. For content: Literature & Study Materials of P0.
Academic Skills	MSc 2nd year.
Literature & Study Materials	-
Judgement	https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations For more information on grading, see article 14 in the Rules and Guidelines MSc TIL R&G.
Permitted Materials during Exam	No exam.
Collegerama	No

TIL5050-20 P2	Analysis & Design	8
Responsible Instructor	Dr. J.M. Vleugel	
Instructor	Dr.ir. J.H. Baggen	
Contact Hours / Week x/x/x/x	n/a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Process: Earlier team projects. Content: Literature & Study Materials of P0, P1/PVS.	
Course Contents	This is an advanced course in applied systems/design engineering. The PVS was written in P1. It is amended in the kick-off meeting. This amended PVS is your initial guidance for the content and planning of the project. You do not rewrite your PVS after the kick-off meeting, but use the comments in your draft project report. A project contains the following 3 building blocks: 1) A current state analysis: -an extensive research problem analysis of the organisation and its context -literature analysis with research gap -stakeholder analysis to study relationships among stakeholders and search for common ground for your later designs -requirements and constraints analysis as a basis for your design alternatives and for the filtering and evaluation of your designs 2) A future state analysis: -description of ideal future situation -one or more conceptual designs taking care of the results of the requirements analysis -one or more preliminary designs which are usually filtered (by a morphological analysis or other tool) -choosing the best final design(s) -detailing of the best final design(s) 3) Evaluation, conclusions and recommendations: -final evaluation of developed (detailed) designs -conclusions, recommendations for science and practice -[implementation advice] In appendix -technical details (statistics, PDL/simulation, unused cases or scenarios, ..) -team reflection about final results, project experience, experts and literature, supervision and the course itself -logbook (detailed weekly activities by each team member) -PVS (kick-off version) By email - individual reflections The goal of each project is to provide the commissioner with a dedicated advice on how to improve its business or policy process or product. A project is finished with a complete project report, which is defended in a final presentation and defense meeting. Read the extensive Course Manual on Brightspace. A complete overview of the course without login can be found here: https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/master/transport-infrastructure-logistics/programme-2022-2023/4-projects/til-design-project/.	
Study Goals	1: To carry out a full-scale problem analysis using relevant scientific tools and methods on the foundation of a systems engineering approach. The problem analysis will be finished with a set of design requirements, criteria, constraints and where applicable KPI's. 2: To develop and evaluate one or more design(s) (a set of complementary measures, solutions or tools) to deal with the problem as scoped earlier. The design(s) should be well-argued, if data allow also quantified and (to some extent) practically feasible. The evaluation of the design(s) should be carried out with a transparent evaluation method. Implementation is not part of the design, but an implementation advice may be added. 3: To write a (final) report with appendices. This may also contain an implementation advice. You present and defend (final) project results during P2 and in particular in the Final Presentation and Defense meeting. 4: As a team, you cooperate to reach the project goals in an effective and pleasant manner. You organise weekly supervision meeting with your commissioner supervisors and the TUD supervisors (independently). You keep your TUD supervisors informed about planning and team dynamics. This demands sound project management. You inform the course coordinator in case of more structural team issues.	
Education Method	[Set-up] Team work with active weekly supervision by the commissioner and separately with the two TIL supervisors. [Planning] Read the TIL5050-20 main page; the timeline is also in the enrolment form. [Start] Kick-off meeting - document + presentation You contact the course coordinator to plan the date, time and place of the meeting (reserve via secretariats) and prepare a kick-off meeting with an agenda. The invitees include all supervisors and the course coordinator. This meeting will be held at the TUD. Kick-off document is the TIL5050-20 P1 PVS. You send this to all attendees of kick-off meeting together with an agenda at least 1 day before this meeting. [Process planning] Execution takes 10* weeks from the kick-off meeting. The actual finish depends on your exam and holiday planning. [Supervision TUD - document + presentation]	

	TUD: Weekly progress meetings with both daily supervisors or at least alternating are strongly advised. Don't skip these meetings. Contact the course coordinator if you don't get enough support from them (e.g., illness or other reasons). Discuss content, organization, team. A short presentation is common in each meeting and mandatory at the kick-off, mid-term, green light and final presentation and defense meeting. Email docs to supervisors 1 day before progress meetings, 2 (4 in case of a weekend in between) days before a Mid-term / Greenlight/ Final Presentation. Take care of weekends and holidays. [Supervision by commissioner (- document + presentation)] At least weekly meetings with the commissioner supervisor(s). If any serious problem occurs with supervision (e.g. no dedicated help, not reading documents, late with contacts and meetings, access to employees or data), always inform the TUD supervisors and the course coordinator. [Joint meetings - document + presentation] Kick-off, mid-term, green light, final presentation & defense meetings with all supervisors. The course coordinator will not attend the mid-term meeting, unless he is also one of your supervisors or if you request this specifically. The date of the mid-term meeting can be reserved in the kick-off meeting. [Mid-term interviews - prepare topics - invitation schedule on Brightspace] In the same week as the mid-term meeting, there are also 1:1 mid-term interviews (15 min each) between the course coordinator (or his replacement - see Brightspace - if he is one of the weekly supervisors) and each student. They are about the process, team, issues, questions, the course itself etc. No documents are needed. The conversation is open, constructive and confidential. [Green light meeting & Final presentation and defense meeting - document + presentation] These dates cannot be scheduled at the start of your project, because of uncertainty about the progress. So wait with planning these two until you passed the mid-term meeting. TIL5050-12 Main, P0, P1; TIL4030-16; TIL4020-20. Course manual and other material on Brightspace and the TIL website of the course: https://www.tudelft.nl/en/student/ceg-student-portal/education/master/transport-infrastructure-logistics/programme-2023-2024 A TIL5050 P2 project consists of several phases and products: 1 -Mid-term report (meeting). About problem analysis until requirements analysis, with an outlook into the design phase. This report will be assessed and suggestions for improvements will be given by all supervisors and next steps will be discussed and planning adapted. Team dynamics can also be on the agenda if necessary. 2 - Green light report (meeting). 90% ready final report (incl. preliminary conclusions). The green light report will be assessed by all supervisors. Assessment may lead to a green, yellow or red light. Green means continue, yellow means major improvements are necessary (updated report, assessment email), while red means stop. Suggestions for improvement and next steps will be discussed. 4 - Final report (Final presentation & defence meeting). Here the complete final report with appendices is discussed. Report appendices are meant for details like model structure (not code), logbook (weekly per team member), group reflection, original PVS. Logbook, group reflection and PVS do not count for the number of pages. Individual reflections (1-2 A4) are sent by email to all supervisors and the course coordinator. The final assessment takes into consideration the following: - Research quality (Content) - Design quality (Design method + execution) - Written and oral Communication - Team work (Process). The grading rubric with assessment criteria and their weights can be found on Brightspace. The team is responsible for regular contacts with all supervisors and the course coordinator. Each of the 4 mandatory meetings will contain a (short) presentation (20 minutes). A major challenge in each project is to find a balance between the requirements by the university, the external customer (if available) and your own perspectives, ideas, intentions and wishes. You are not a consultant, so give enough attention to the scientific methods and reasoning. The balance between theory and practice depends on the topic. Convince a very practical commissioner of the benefits of a more scientific approach. Offer different perspectives on the problem; from the literature and other applications outside the narrow scope of the commissioner. Each project team has sufficient freedom for exploration and expression. As soon as it happens, discuss 'how to find the right balance' with your TUD supervisors and your customer supervisor or the course coordinator. Do not struggle for weeks with such a problem. [Final mark] The rubric group result will also be the individual final result unless the main supervisor has sound reasons to deviate. Reasons for a lower mark for a team member can be behaviour, communication and/or quality and/or quantity of individual input. They force other team members to correct the work and work more hours. They can also feel additional stress. Normally these problems should be addressed earlier in the project (mid-term meeting or earlier). The final mark will be rounded off with a precision of 0.5. In order to pass, the minimum mark should be a 6.0. The final mark is a joint and motivated decision by the multidisciplinary assessment committee consisting of all supervisors.
Course Relations	
Practical Guide	
Assessment	
Enrolment / Application	All (proposed) team members should fulfil these entry requirements: 1. P0: Have at least 50* EC in week x.5. 2. P1: Passed with a (V) grade in week x.10. *Exceptions for non-regular study delays. See the planning examples on the TIL5050-20 coursebase main page.
Special Information	[Meetings] You can organize any number of internal team sessions. You are free to contact other (external) experts in the Netherlands before and during the project. Individual team members do not have to attend all organised meetings, provided you have an acceptable excuse, in particular exams, sickness or planned holidays. However, communicate this early and clearly with your team members and your weekly supervisors. Compensate the hours during the project. Deliver what you agreed at the planned time. This helps maintain trust and lowers the risk of piggy-backing,
Remarks	The projects, carried out in one course year, are to be evaluated by one of the concerned professors, who will report to the Director of Education.
Expected prior Knowledge	See P0 page.

Academic Skills	MSc 2nd year.
Literature & Study Materials	See P0 page.
Judgement	https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations For more information on grading, see article 14 in the Rules and Guidelines MSc TIL R&G.
Permitted Materials during Exam	No exam. All meetings without restrictions. P2 has a Final presentation and defence meeting with sheet presentation. You are free to use any document, show functioning models, simulations, hand-out materials or show miniatures of your designs.
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Thesis (30 EC)

TIL5060	TIL Thesis	30
Responsible Instructor	Dr. S. Fazi	
Contact Hours / Week x/x/x/x	n/a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Summary	The final part of the study is an individual interdisciplinary master's thesis project. It contains elements from at least two fields offered by the faculties participating in the MSc-TIL program. This project may be carried out as an internal- or an external project (with a company, public or private institution) in the Netherlands or abroad. It takes half a year (30 EC). The student is supervised by a supervisory team composed of two supervisors, for cohorts 2022/2023 onwards, and three members (1 chair and 2 supervisors) for older cohorts. The composition of the committee is according to the rules and guidelines from the Board of Examiners. The student should refer to the guidelines of her/his year of enrollment. The project shall lead to: - A thesis report; - An academic research paper of 8-12 pages. It is mandatory for students who started their thesis project on or after September 1, 2017. It presents the main results of your research project to others in an orderly, logical manner. It is more than a simple summary, ask yourself: What does really stand out in my research? A content list + examples are provided in the TIL Graduation Portal. Include it as an appendix in your thesis report. The content and process of a thesis project are completely described here: https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/master/transport-infrastructure-logistics/programme-2022-2023/5-thesis-and-graduation	
Course Contents	Individual research (or design) project at MSc level.	
Study Goals	1) Identify the research problem and the research gap. These follow from a literature study in combination with a preview of the current state (problem-oriented) and potential ways to contribute to a solution of initially identified issues and problems (also known as future state in design-oriented studies); 2) Make a motivated choice of and apply relevant theories and methods learned during your studies in your project; 3) Synthesise the results of the project in a concise and academic (critical) way and to prepare a set of relevant conclusions and recommendations; 4) Reflect on the results, show how these compare to the state-of-the-art; discuss the pros and cons for society and for the commissioner; 5) Independently plan and execute the research project; 6) Act in coach-able manner. Use remarks and advice on your work to improve it; 7) Be communicative, on paper as well as in personal contacts with your supervisors; 8) Develop an open mind towards remarks and suggestions by supervisors; 9) Defend the results of your study in a satisfactory way during the final defense meeting.	
Education Method	The student plans, executes, reports and defends the thesis project independently, but supervised and supported by the members the thesis committee. This implies that the student is the primary process manager: he or she arranges all meetings, distributes (draft) reports and other relevant discussion material, such as minutes, (information about) datasets and tools. The daily supervisors from the TUD and the commissioner coach the student during the project. They meet on a regular basis. For the TUD regular means at least once a month. Meetings between the student and the commissioner should take place every week. The actual frequency of meetings depends on the progress and needs of the student. The supervising professor sees to it that the draft and final reports full the academic requirements (see grading rubric). Four meetings are mandatory: a kick-off meeting, a mid-term meeting, a green-light meeting and the final thesis defense, which consists of an open, public defense and a closed defense with the entire assessment committee (chair, daily supervisors, commissioner supervisor(s)). *The final meeting set-up differs per chair. The preparation of a thesis project starts several months before the kick-off meeting. During this period the student has to - find a suitable problem owner (commissioner) and an academically challenging and socially relevant research topic; - compose a supervising team (= assessment committee); - write a project proposal that identifies the research gap and the contribution of the student to reduce this gap while meeting the needs of the commissioner. The best way to do this is by working for about a month at the commissioner to get to know the problem and the customer organization and relevant staff. While being there, the thesis proposal can be written. The student can contact the TIL Graduation coordinators and lecturers to discuss the thesis proposal and other content and process questions. Finally, the student can organize the kick-off meeting.	
Literature and Study Materials	A student has freedom when collecting and processing relevant literature within the agreed scope of the thesis project. Any literature, data, methods, study material from BSc and MSc courses concerning the topic of the thesis could be applied next to new material from academic and professional sources. The supervisors will give advice regarding the content and direction of the research. If they have strong arguments to change the direction of the project or the way the student carries out the process, then the student must follow their advice. The MSc TIL graduation portal contains many writing and presentation tips.	
Practical Guide	https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/master/transport-infrastructure-logistics/programme-2022-2023/5-thesis-and-graduation	
Prerequisites	See the TER, which includes the allowed exceptions. Contact the program coordinator if you are in doubt about how to fill out your TIL-2 form.	
Assessment	Your thesis project is assessed at regular moments. The following 4 meetings with the thesis assessment committee are mandatory in order to guarantee the quality of each project. *Ask the chair if a presentation should be prepared in case of the kick-off, mid-term and green light meeting. *The formal start is the Kick-off meeting. Here the Kick-off document (= project proposal) is assessed. A key element in the kick-off document is a description of the research gap in the field of your study. Your project is meant to reduce this gap with	

scientific methods. A small literature review and a month spent with the problem owner help to identify this research gap. The practical relevance for a problem owner is a second order effect. Take care that you balance the aims of your commissioner and your TUD supervisors.

*The next milestone is your Mid-term Meeting. In a mid-term meeting the content and status of your project will be on the agenda. As a benchmark, you should have finished the analysis part of your work and have rewritten your initial problem definition in terms of solutions. An outlook into the solution space could be discussed, if the time allows. An updated planning scheme should always be discussed. If you experience problems while working at the commissioner, discuss this also before or in this meeting.

*The Green light meeting is very important. A green light report should contain:

- Summary (and Executive summary if a commissioner wants this) in English (and also in Dutch in case of a Dutch government agency);
- Analysis in full;
- Design/solution space in full (in case of a design project);
- The conclusions in draft;
- Tools verified, validated, experiments with results;
- (Where appropriate:) a user manual for the tool can be written later.
- Appendices with details about data sets, model structure (not code), statistics, evaluation etc.

Work to be done: Research paper, final conclusions, final summary, minor editing and language check, finish appendices, printing and binding of hard copies.

*The Final Thesis Defense is the Grand Finale of your project.

The final thesis report = the improved green light report plus

- Acknowledgement;

- Research Paper as Appendix A.

The structure and length of the Final Thesis Defense meeting may differ per department and chair(wo)man. Contact the chair of your Committee or his or her secretary.

A TIL5060 Grading rubric is used to assess your thesis project in a break-out meeting. It is used to grade the

A. Research quality;

B. Research competences;

C. Quality of final report;

D. Quality of oral presentation & defense.

At most one learning outcome (A - D) may receive a mark below 6. The four sub marks are weighted and then summed. The final mark is determined by the assessment committee and ultimately by its chair.

Confidentiality/non-disclosure (CA/NDA) agreement as part of a graduation/apprenticeship contract.

Students graduating at a company usually sign a contract with that company. In many cases this contract includes CA/NDA clauses. Such contracts are bilateral agreements. Neither the TUD nor your supervisory team are contract partners, which means that we are not allowed to co-sign. Even if we would, then our signatures would not have legal meaning.

CA/NDA restrict the publication of your work. Clauses can deal with protection of company data, documents and practices until an obligation to hand-over authorship. The first are acceptable, but the last surely is not, because it violates your fundamental legal rights as well as the academic freedom and progress of science.

In no way should the contract restrict your or our academic freedom as TUD supervisors.

Given these concerns, we are of the opinion that NDA/CA's should be avoided or at least be edited in order to arrange a proper balance of interests. The confidentiality period should not be longer than 2 years.

Don't automatically sign any document, because of pressure or fear that you will loose the contract. Ask the Graduation coordinators to check the clauses of the contract and either remove or edit unacceptable clauses. This helps in the subsequent contract negotiation with the company. Experience shows that there is usually some space to negotiate, but it may delay the start of your project a bit.

Confidentiality may influence your final report in several ways. You could decide to keep the full document, but not upload it to the repository. Another option is to wipe-out or generalize confidential data and then upload the report to the repository. In either case, your supervisors should be able to read and assess the full unedited final version of your thesis document.

A new situation arises in case of exceptional quality of your work, giving way to a future journal publication. This needs further discussion with all parties involved.

Finally, a student is always the owner of his or her thesis report and any material used to write it, with the exception of data or tools owned or licensed by the TUD or an external commissioner.

A neutral NDA form can be found in the TIL Graduation Portal: <https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/master/transport-infrastructure-logistics/programme-2020-2021/5-graduation-portal-thesis-project/3-proposal-writing/>.

Always read the information in the Graduation portal of the TIL website first.

During preparation the Graduation coordinators may help you with your choices and questions:
Dr. Stefano Fazi TPM B3.200 - s.fazi@tudelft.nl

The GC's can also be contacted in case of

- Coaching/matching problems with TUD supervisors / change of supervisors;
- Serious content issues (change of topic or scope, promised data set not available on time);
- Insufficient coaching by (external) supervisors (non-response on request for meetings, no follow-up on agreements, not reading documents, very general comments, not connecting with internal or external colleagues);
- Structural health issues or other private matters;
- Delays due to dyslexia or other health issues;
- Serious problems with planning and structuring of the project;
- Foreign students with specific questions.

During the project, contact your

1 - Daily supervisors or chair;

2 - The (secretary of the) professor in charge.

Put these issues on the table when they happen and after you tried to solve them yourself. Waiting does not solve serious problems.

Start: Information from earlier BSc and MSc courses that matches with the scope, topic, research questions, methods and available data for your thesis. This also helps to digest the literature and carry out the necessary field work.

Special Information

Contact

Expected prior Knowledge

Academic Skills	Defense: The thesis Assessment Committee usually restricts its questions to the content of your final thesis report. However, sometimes there are arguments to relate your work to (generic) theories or methods from earlier courses. In other cases questions about more general applications of your findings can be asked. Use and refer to your thesis report when answering questions. The thesis student -has the ability to gather, integrate and interpret relevant base material (literature, datasets, interview results etc.) and translate this into information and knowledge. - understands the complexities in the TIL domain and is able to reason about and reflect on possible social, scientific, policy and ethical dimensions, implications and restrictions linked to the application of this data, information and knowledge. - is able to form judgements, formulate conclusions and recommendations based on these academic activities; - is able to reflect in an academic way on his or her own work (including learning points), literature and specialists.
Literature & Study Materials	Topic specific information. Read the website and the 'tips' sections. https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/master/transport-infrastructure-logistics/programme-2022-2023/5-thesis-and-graduation.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/onderwijsinformatie/educational-rules-and-regulations
Permitted Materials during Exam	Audiovisual equipment, final thesis report, thesis input material, slide presentation and handout, (simulation) model, movie, notes.
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Electives: between 19 and 21 (depending on the chosen specialization)

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Transport and Logistics (T&L)
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SEN115B	Law and Institutions	5
Module Manager	mr. J.M. Kooijman	
Instructor	Dr. R.S. van Wegberg	
Instructor	Dr. A.F. Correlje	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	0/0/2x45 min/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Suppose we want to switch to a hydrogen based economy in which fuel cell cars are used as power plants. Or suppose we want a fully automated inland terminal. And what about the construction of multifunctional dikes? You cannot just realize that but you have to be aware of the relevant legal and institutional framework. That framework will enable you to make things happen, but it can also contain constraints. This course focuses on the legal and institutional framework to take into account with that kind of projects. Are there for instance legal instruments to force people to switch to fuel cell cars? And to what institutional layer do those instruments belong? Are they easy to change? Or: can the government subsidize the transition to a fully automated inland terminal? What are the constraints? And can we work around those constraints? Or: what exactly has to be studied for the environmental impact assessment regarding the dike? Does an environmental impact assessment have to be conducted at all? Students form groups and pick a case. Then, the group will present its case in an elevator pitch. On the basis of those pitches, we will dive into the world of law in order to see what defines the design space for those projects, what areas of law are relevant, whether the law forms a barrier or not and how the law can be used to realize those projects. Next to the group assignments there will be an individual exam.	
Study Goals	At the end of the course, students are able to: - Relate legal instruments to the institutional framework - Work with the law as a system - Relate the legal instruments to the institutional framework - Identify the legal borders of the design space regarding a socio technical system - Identify relevant law for a socio technical system - Apply relevant law to a socio technical system and identify barriers and opportunities - Deal with legal language	
Education Method	Interactive lectures.	
Assessment	In order to pass the course, the students write a group paper and take an individual exam. The paper is based on the institutional and legal framework, the legal barriers and the legal opportunities regarding a specific case. The individual exam will be (if possible) conducted in ANS. The final grade is based on: - a group elevator pitch (10%) - two peer reviews (P/F) - a final paper (40%) - an individual exam (50%) There will be a resit possibility for the pitch and final paper when the obtained grade is lower than 6. There will be a general resit possibility for the individual exam. Every partial result will be registered in Osiris and can therefore be taken to the next academic year.	

SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5
Module Manager	N. Mouter	
Instructor	Dr. J.A. Annema	
Instructor	J. Rezaei	
Instructor	N. Mouter	
Instructor	Dr. R.M. Shortall	
Co-responsible for assignments	J. Rezaei	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Transport policies are almost always controversial. Several methods are applied worldwide to support decision-making in transport. Nowadays, societal cost-benefit analysis (CBA) and Multi Criteria Decision Methods (MCDM) are the most prominent and well-known transport policy appraisal methods. In this course you will learn about the newest scientific developments in these appraisal methods. Additionally, you will learn about new appraisal methods such as Participatory Value Evaluation.	
Study Goals	Describe and explain traditional evaluation methods (CBA and MCDM) and non-traditional evaluation methods. Explain the pros and cons of different methods for different evaluation circumstances Describe the ethical critiques for different types of evaluation methods Define/pick a research gap (to be used in the paper) Write an academic paper, based on completed case study exercise, where: A research gap that the student can fill/contribute to is identified ; Pros and cons of ONE chosen method are critically reflected upon; The suitability of the method is convincingly argued; The results are analysed and interpreted, criticised with particular attention to ethical issues; Policy recommendations are made on the use of the results	
Education Method	In formal lectures the methods and theories will be explained. By two exercises students will apply the methods and theories. Finally, by analysing a pre-defined real-world controversial transport policy case and at least ten scientific papers (which they have selected themselves) students will learn to critically reflect on appraisal outcomes. They have to write a concise essay to report their findings and reflections.	
Assessment	The exam is an individual paper of 3500 words (excluding appendices and list of references). (70%) A CBA assignment (15%) A MCDM assignment (15%)	

SEN1721	Travel Behaviour Research	5
Module Manager	M. Kroesen	
Instructor	B. Pudane	
Instructor	Dr. E.J.E. Molin	
Instructor	M. Kroesen	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. E.J.E. Molin	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Knowledge about basic statistics and data-analysis (probability density functions, hypothesis testing, regression) is absolutely required as prior knowledge. Without this knowledge it will be very difficult to follow this course.	
Course Contents	In this course students learn how to perform travel behaviour research and how to translate the findings from such research into recommendations for transport policy design. The students will acquire knowledge about six modelling paradigms in travel behaviour research, namely the econometric, psychological, geographical, biographical, mobility style and sociological paradigm. The strengths and weaknesses of each paradigm will be discussed. In addition, three state-of-the-art methods to analyse travel behaviour (structural equation models, latent class choice models and latent class clusters models) will be treated. These methods figure prominently within the six modelling paradigms. For each method there will be a group assignment (so 3 in total). These group assignments will count for 50% of the grade.	
Study Goals	After this course the students will be able: (1) to carry out travel behavior research using three state-of-the-art statistical methods, namely structural equation modeling, latent class cluster analysis and latent class choice models. (2) to translate the study findings to recommendations for transport policy design.	
Education Method	Lectures Computer practicals Group project (4 persons)	
Computer Use	In the course the following statistical softwarepackages will be used: SPSS, Latent Gold, Apollo and AMOS	
Assessment	Knowledge about the modelling paradigms and three methods will be tested in an individual written exam in week 10, which accounts for 50% of the final grade. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be an online "open book" Mobius exam. The group project will consist of three assignments which together account for 50% of the final grade. Both grades should be >5.75 to pass the course. Both parts can be separately passed.	

SEN1741	Innovations in Transport and Logistics	5
Module Manager	Dr. M. Comes	
Instructor	Dr. S. Balakrishnan	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The transport and logistics systems are continuously changing due to the introduction of innovative concepts in vehicles, infrastructure, and services, examples being the introduction of the car, e-bike, drones, or high speed train (vehicles), ride-sharing applications, GPS tracking, or satellite navigation (services), or charging facilities (infrastructure). But many innovations fail. Others may undermine the resilience of the system or create unwanted cascading effects for other infrastructures, social or environmental systems or services. Therefore, policy-makers need to understand the implications of such innovations.	
Study Goals	In this course you will learn about factors important to understand the impact of transport and logistics innovations. By the end of this course, you should be able to LO1: Explain the main frameworks that describe the impact of innovations. LO2: Explain the main pathways that link transport and logistics innovations and their societal impacts within the system (e.g., accessibility/equality, resilience) and cascading effects. LO3: Analyse the success and failure factors of candidate transport and logistics innovations using a combination of frameworks from the literature. LO4: Design and develop a conceptual model that illustrates the impacts of candidate transport and logistics innovations.	
Education Method	Interactive Lectures (frameworks; impacts), Group projects, resulting in a presentation and a short essay Interactive group presentations	
Assessment	1. Group essay and group presentation 2. Individual exam (open book, take home) Each part has to be graded with at least 5.75. The final mark is composed as follows: part 1 40%; part 2 60%. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed individual assessment will be a timed open-book assignment, and it will be written at home.	

SEN9110	Simulation Masterclass	5
Module Manager	Prof.dr.ir. A. Verbraeck	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Experience with a simulation environment such as Arena, Simio, Tomas, NetLogo, GAMA or MESA. Basic knowledge about probability theory and statistics.	
Course Contents	System Theory, Object Orientation, Discrete Event System Specification, Multi-Formalism Simulation, Distributed Simulation, Real-Time Simulation, Multi-Paradigm Simulation, Multi-Resolution Simulation, and Simulation Language selection will be the core topics of the course. After an introduction to system theory, the inner working of simulation environments will be illustrated on the basis of the DEVS, DESS, and DTSS formalisms. Then, possible integrating of the different formalisms will be shown. Several special topics will be taught, based on the latest research in simulation. This material will be illustrated in intensive and interactive courses. In addition to the lecture topics, several other simulation topics will be studied by groups of students, who will write a scientific paper, and present their findings in class. These topics can be focused on the MSc program that the students participate in; special topics to study are available for TIL, CoSEM, EPA, Computer Science, and other students. Finally, groups of students will study a simulation package in-depth and discuss the commonalities and differences with other packages. Again, the package chosen can be targeted at the MSc program of the students. TIL students can, e.g. study a package that is more aimed at logistics and transport, CoSEM students can focus on a package that is used in systems design, whereas EPA students can focus on a package used in policy analysis. Most packages can be made available for use from home. This course requires active participation and contains group work.	
Study Goals	After taking this course the student will have knowledge about: - systemic principles behind simulation, and clear understanding of key terms, such as state, time, transition, event, process, and activity; - internal working of different kinds of discrete event simulation languages and environments, with a focus on state transitions over time; - underlying theories and formalisms of discrete event simulation, such as DEVS, DTSS and DESS; - the DEVS paradigm, including state trajectories and transition functions; quantized DEVS (QDEVS); discretized time; - Dynamic Structure DEVS (DSDEVS), and other DEVS variants; - transformation between simulation formalisms; the formalism-transformation graph; - object-oriented simulation environments; service-based simulation; - structure and abilities of distributed simulation; the concept of HLA to manage time and state in distributed simulations; classes of functions in a distributed simulation environment; - latest research activities in the field of simulation, with research topics like web-based simulation, real-time control using simulation, agent based simulation, interactive simulation and gaming, and simulation in special domains; - examples of successful and less successful simulation studies and the learning experiences of those studies; - important similarities and differences between simulation languages, and selection methods for simulation packages. Practical skills will have been developed with respect to: - usage of at least one simulation package that was previously unknown; - writing of a scientific paper on the topic of simulation. Insight will have been gained on: - comparing and contrasting simulation languages; - the importance of time and state relationships for any type of simulation modeling; - the formal foundations of the simulation domain.	
Education Method	Lectures about state-of-the-art simulation theory Computer use for studying one discrete-event or agent-based simulation package in-depth per group Paper writing by student groups on a special topic Lectures by student groups on a special topic.	
Computer Use	Each group of students will study and evaluate a discrete-event or agent-based simulation package. Examples are Simio, Arena, Enterprise Dynamics, Plant Simulation, AnyLogic, Extendsim, Flexsim, Jaamsim, DSOL, Salabim, MESA, GAMA, Netlogo, and Repast Symphony. Also here, the simulation package to study can be specifically adapted to the group of students. Most packages can be made available for use from home.	
Course Relations	The course contains very little overlap with other simulation courses such as EPA1324, EPA1352, TB231/TB233, etc.	
Literature and Study Materials	Set of scientific papers and book chapters that will be made available through Brightspace. Typically, every class will have on average 2 papers to study for the exam. The set of papers is also available as a Reader that can be obtained on paper.	
Reader	A paper-based or PDF version of the Reader is available. The content of the Reader is also electronically available in Brightspace.	
Assessment	The mark for this course will be based on the result of two summative group assignments (paper with presentation, lab study of simulation package consisting of 4 sub-assignments that all count equally heavy) and on a summative individual written exam. The individual written exam will count for 50%, the group paper (including the class presentation) for 30% and the study of the simulation package for 20%. Each of the three marks has to be 5.75 or higher to be included in the final mark. Partial marks will stay valid for the study year in which they were obtained and the study year thereafter. The written exam takes place at the end of Q1 with a retake at the end of Q2. When a group assignment is insufficient or missing, the group will be given the opportunity to repair the insufficient or missing assignment.	
Permitted Materials during Tests	All written materials are allowed during the exam: papers, class notes, slides.	
Elective	Yes	
Tags	Challenging Intensive Modelling Programming concepts Software Stochastics	
Targetgroup	Students in EPA, CoSEM, TIL, TEL, PEL, Computer Science, and any other MSc student with some prior knowledge about	

Extra Skills Trained	computer simulation. Writing a high-quality scientific paper; exploring a new simulation language / package.
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SEN9720	Logistics and Supply Chain Innovations	5
Module Manager	Ir. M.W. Ludema	
Instructor	Ir. M.W. Ludema	
Responsible for assignments	Ir. M.W. Ludema	
Co-responsible for assignments	Dr. J.H.R. van Duin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	Supply chain management is about the exchange of product, information and money between companies or organizational entities that are part of a supply chains or demand/supply network. Supply chain management contributes to the competitive strength of companies in the market place. The supply chain management function co-ordinates and controls operational business value adding processes in and between companies. For a large extent the supply chain management function determines the flexibility of companies to respond to market changes and determines the efficiency of the internal processes within companies, as well as the inter-company processes. Logistical and supply chain decision-making frequently relates to situations in which conflicting wishes and desires from various parts of companies or organizations have to be brought together cohesively. Gaining insight into the performance indicators of a company/organization, its logistical costs, customer expectations and the opportunities to improve these aspects is of vital importance for survival of companies/organizations. All engineers will encounter to a larger or lesser extent logistics and supply chain management in the business and institutional environment in where they will be working in, or in where they will be carrying out their assignments.	
Course Contents	This course contains logistical, supply chain and innovations in logistics and supply chain topics. Logistics Topics: - The significance of the logistics system of companies in supply chains; - Methods for analyzing logistics systems of companies; - The role of inventories in logistics systems of companies; - The role of transport between logistics systems of companies; - Connecting purchasing and customer service within companies; - Value added logistics, outsourcing and third party logistics; - Performance Measurement in logistics systems of companies; - Logistics control and information systems; - Design of logistics networks and locations for distribution centers. Supply Chain Topics: - Global trade lanes, industrial clusters and supply chains; - Strategy issues and the supply chain; - Organizational issues and supply chains; - Supply Chain Mapping, Benchmarking and the SCOR-reference model; - Physical and Virtual Value Systems (MRP-I, MRP-II, ERP and APS); - Supply Chain Value Adding Processes; - Lean and Agile Supply Chains and the strategic role of decoupling points; - Supply Chain Collaboration (Supply Chain Partnerships & Alliances); - Supply Chain Analysis; - Supply Chain Design. Innovations in Logistics and Supply Chain Management: - The Last-Mile of Supply Chains; - Multi-modal and Synchro-modal Logistics and Supply Chains; - Customs and Trade Compliance; - Supply Chain Integration and the use of Cross Chain Control Centers (4C); - Service Logistics (Asset Management, Maintenance Management and spare parts logistics); - Sustainable, Green and Closed-Loop Supply Chains within a sustainable economy, - Smart Supply Chains and the Internet of Things; - Supply Chain Finance; - Smart Contracts & Block Chains Technology and the Supply Chain.	
Study Goals	This course covers the supply chain management function in and between companies. This course will provide the student insight in the theoretical background of the supply chain, its current use and expected developments and a wide set of technical and informational innovations that companies consider to implement. The student learns and puts into practice the basic theoretical skills of a supply chain analyst and engineer and can show the contribution of relevant logistics and supply chain innovations. Theoretical knowledge and practical skills will be explained and exercised by means of a set of relevant cases.	
Education Method	The course contains 17-24 lectures (2 or 3 time 2 hours each week), in-class discussions about presented topics and distributed articles and cases. During the course assignment groups of 4-5 students will work on a logistics and supply chain analysis/engineering assignment. In the final weeks of the course the assignment groups will compile their results in a final report. Furthermore the course contains self-study and some group assignment feedback sessions. The draft report of the course assignment groups will discussed with the groups prior to final hand-in.	
Literature and Study Materials	This course works with a virtual reader. Suggested Course Materials is continuously under construction (see manual on Bright Space), relevant articles will be related to weekly lectures.	
Practical Guide	See Bright Space for the updated course and assignment manual and course assessment issues.	
Prerequisites	This course builds on the knowledge gained during courses like operations management, production logistics, logistics management or equivalent courses. Students without (sufficient) knowledge and background are advised to spend additional study time during the first two to three weeks of this course. Please consult the module manager / lecturer during the first lecture or earlier. During the first two lectures an overview of relevant operational and logistics management issues will be provided by means of intensive lectures and a few in-class assignments.	
Assessment	Final deliverable and group assessment: Students will work in groups on a large group assignment in one of the following example domains:	

Transportation Equipment Supply Chain;
Construction Supply Chain;
After-sales service /spare parts supply chain for the maintenance of durable goods;
Supply chain of conditioned goods for the retail market;
Supply chain of the inter city last mile;
Care supply chain of the Medical Equipment;
Other issues in respect to the interest of the student.

A final report including analysis of the current supply chain, impact assessment of a chosen logistics and supply chain innovation and a design of a feasible future supply chain.

SEN9725	Supply Chain Gaming	5
Module Manager	Ir. M.W. Ludema	
Instructor	Ir. M.W. Ludema	
Responsible for assignments	Ir. M.W. Ludema	
Co-responsible for assignments	Dr. J.H.R. van Duin	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Required (as a specialization elective) for : MOT Specialization Supply Chain Management (to be taken after SEN9720 (Q1) or after sufficient other logistics (fundamental) courses.	
Expected prior knowledge	This course builds on the knowledge gained during courses like operations management, production logistics, logistics management, supply chain management or equivalent courses. Students have ideally followed the course SEN9720 Logistics and Supply Chain Innovations (first quarter). Understanding of supply chain management trade-offs is necessary to be able to understand, analyze and develop supply chain game sessions. Students without this prior knowledge have to spend additional study time prior to this course. Please consult the module manager before the start of the course. During the first two lectures an overview of relevant operational, logistics and supply chain management issues and trade-offs will be discussed. By means of small in-class assignments students discuss possible serious (board) game development projects and themes for their group assignment.	
Summary	Supply chain management considers product and information exchange between companies in supply chains. It determines a company's flexibility to respond to market changes and determines the efficiency of the internal processes within that company, as well as the inter-company processes. Logistical and supply chain decision-making frequently relates to situations in which conflicting wishes and desires from various parts of supply chain organizations have to be brought together cohesively. By means of analyzing existing games and developing supply chain game sessions more awareness and insight in these conflicts can be gained. If supply chain stakeholders play games they will improve their awareness about supply chain mechanisms and possible means of actions to improve the alignment of supply chain processes. A serious game is a suitable tool to create awareness about relevant supply chain issues. It levels the experience of participants (either novice or experienced from the field). After playing a serious logistics or supply chain game it contributes to discussions and facilitates the exchange of knowledge and personal experience. Short introduction to some relevant supply chain concepts, playing a game, and a clear introduction and participating in relevant discussions together are part of what is called a game sessions. Playing games is a good learning exercise, but designing and developing helps the game developer to understand the essence of the trade-off from the real world. One of the approaches that can be applied for design and development of a serious design session is the Triadic Game Design (TGD)-approach of Hartevelde (2011). Games generally include three worlds, namely reality, meaning and play. These worlds must be in balance to provide a serious game which will be fun to play and also to learn from. The world of reality is needed so the players of the game can relate to elements of the real world, this will make experience in the game intuitive and understandable. Meaning is needed, because the game should have a learning purpose. At last the world of play is needed to make a game playable. The game should have clear goals and rules. The play factor will guide players in the right direction of the learning goals of the game. The three worlds are equally important, therefore they should be balanced. Balancing of these three worlds, is what makes designing a game session complex. Other approaches and techniques will be added to this TGD-approach. During the design, development and execution of a game session the players have to gain knowledge about a certain logistics or supply chain issues and/or trade-offs or the effect of changes in their part of the supply chain relative to changes in other parts of the supply chain. This could for example be the implementation of technological innovations to improve traceability in the logistic supply chain, the optimization of warehouses, the best collaboration approach, implementing IoT within the supply chain, the understanding of the benefits of synchromodality. In general playing and designing games helps to create a better awareness about logistics and the supply chain in general and certain addressed problem areas in particular. In some cases games can help to gather data for analysis and research purposes.	
Course Contents	Common trade-offs in Logistics and Supply Chain Systems: 1) Fundamentals of logistics and supply chain systems; 2) Methods for analyzing logistics and supply chain systems on multiple levels; 3) Strategic, Tactical and Operational Trade-Offs in logistics and supply chain systems; 4) Game Concepts to facilitate improvement of the awareness of trade-offs in logistics and supply chain systems. 5) Discussing trade-offs and ideas of game concepts. Game Essentials: - Sharing (board, card and role-playing and computer) game experiences and discussing fundamentals; - Flow experiences (Mihaly Csikszentmihalyi); - Game Concept Decompositions and Construction; - Systems and Process modeling and analysis for game design; - Triadic Game Design approach; - Design Methods (from established needs to workable results); - Playing and Learning: - Lemonade Tycoon; - Rise of Industry; - Playing MSc-Students developed games. Game Design Assignment: - Towards understanding a (chosen) supply chain problem situations; - Developing Game Session Objectives; - Setting Boundaries; - In Depth Analysis of the problem situation and value of possible solutions; - Setting Game Design Requirements; - Developing the Game (Session) Concept; - Generating Solutions;	

Study Goals	- Evaluating Solutions; - Game Session and Play (Delivery, Evaluation); - Reporting about the problem situation, game design, game development, game demonstration, game evaluation and improvements and final game deliverable. This course gives students knowledge and understanding in theoretical and practical issues concerning the design, development and execution of supply chain gaming sessions. The fundamentals of game session design and its application area logistics and supply chain analysis and management will be discussed and practiced. Important and problematic logistics and supply chain issues and opportunities for innovations will be explained and related to game sessions design alternatives. Upon completion of this course the student should be able to: - Elaborate on how companies can develop and maintain their position as an adequate supply chain member. - Elaborate on the inter-relationship between processes in the silos of business logistics systems and the need to align them. - Understand important logistics and supply chain trade-offs that are suitable to incorporate in supply chain games. - Suggested and develop supply chain (re)design directions for the future of the companies in a supply chain that need awareness between supply chain partners that can be facilitated by means of game sessions. - Demonstrate that they can decompose logistics and supply chain activities in meaningful game elements. - Understand the fundamentals of strategic game design approaches and elaborate on reality, meaning and play aspects and their relationships. - Develop and apply a practical descriptive and prescriptive design method for game session design. - Design, development, demonstrate and evaluate a functional supply chain game session including an playable logistics / supply chain game.
Education Method	The course contains 14 lectures (1 time 2 hours each week and once 3-4 hours each week), in-class discussions about presented topics and of distributed articles, cases and reviewing and playing games. The first two course weeks will be class room lectures, the third, fourth and fifth course weeks will be a combination of class room lectures and working on the group assignment, and the sixth and seventh week will be working on the group assignment. The last lecture will be playing and assessing each others game session. At the the end of the educational period students have to hand in the developed game and accompanying report and game session manual.
Assessment	During the course assignment groups will work on the design, the development and the execution of a supply chain game sessions on a relevant supply chain issue. Assessment: Game design consisting of the design project progress and the final delivered supply chain game and accompanying report (including a manual), [90% of 100%]. The game demonstration and deliverable will be assessed by the students playing the demonstrated game, [10% of 100%].

TPM004A	Transport Safety	4
Module Manager	Dr. E. Papadimitriou	
Instructor	Dr. E. Papadimitriou	
Instructor	Dr. A.P. Afghari	
Responsible for assignments	Dr. E. Papadimitriou	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Acceptance to a TUD Msc program	
Course Contents	The course presents a systematic approach towards safety in different modes of transportation (rail, road, maritime and aviation), integrated in system design and management. The focus is on the role of humans in causing safety related problems and in controlling them. The course deals with the structure and development of the notions of safety and risk by focusing on: Practice, with a focus on the (historical and future) developments, perceptions and acceptance of safety and risk in transport systems; Control, focusing on various strategies available in dealing with risk at the level of policy and management, including ethical issues; Technology, focusing on the way safety and risk are involved in emerging technological systems and applications; A micro systems level, dealing with accident analysis, the complexity of problem modelling, causality and explanatory models for failure at the operator level; A meso systems level, in developing scenarios and applications of quantitative risk analysis; A macro systems level, focusing on approaches in dealing with and deciding on risk by safety impact assessment procedures. The course provides students with strengths and weaknesses of methods and techniques in problem analysis and problem solving at various systems levels and from different perspectives based on the notion of safe systems. It focuses on applying basic skills in accident analysis, quantitative risk assessment and the evaluation of control strategies. Keywords: Safety (design) principles & approaches, human factors, research methodology.	
Study Goals	After taking the course the students should be able to: 1. Appraise individual and societal risk of transport systems 2. Discuss the role of human factors in control of transportation systems 3. Explain the principles of accident investigation and casuistic analysis 4. Apply conceptual and quantitative models of safety management within a safe system approach 5. Analyse data with statistical analysis techniques for risk estimation 6. Assess the societal, ethical and policy implications of transport safety 7. Explain the challenges of automation in all transport modes	
Education Method	Blended learning: pre-lecture work, lectures, post-lecture work and 3 group assignments.	
Literature and Study Materials	Literature and other material will be made available through Bright Space.	
Assessment	A final exam and three group assignments. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: A remote written exam and three group assignments.	
Elective	Yes	
Tags	Transport & Logistics	
Category	MSc level	

TPM023B	Cost-Benefit Analysis: Theory and Application	4
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	N. Mouter	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	The course TPM023B focuses not only on private (e.g. project, firm, industry) costs and benefits but also on social costs and benefits. While the private CBA only considers financial aspects, the social CBA considers both financial and non-financial aspects. For instance, in the construction of new highway, besides the financial aspects such as toll revenues or construction costs, we also assess the non-financial aspects such as safety benefits (lives saved, accidents avoided). Hence, the course actually has a much broader view than the one of the private CBA. This offers students a higher level of practical applications.	
Course Contents	Cost-benefit analysis (CBA) has become a widely used (often mandatory) tool for public policy decision making on infrastructural projects or environmental and climate policies. Through CBA, the potential current and future gains and losses of a proposed policy are identified, and then converted into monetary units in order to make them comparable with policy alternatives; certain decision rules of CBA help determine whether the project is desirable from society's standpoint. Because CBA evaluates policy and/or project proposals from the point of view of the public interest, the market prices used in CBA are corrected for possible market distortions or failure (for example, un-priced environmental damage). Nevertheless, the application of CBA is fraught with difficulties and limitations that need to be taken into account. What rates of interest are appropriate for determination of present and future costs and benefits of a policy proposal? Why and how do the private benefits and costs of a policy proposal differ from the social benefits and costs? How can we express e.g. environmental damage, or human health and life risks, in monetary units? How can we adjust market prices to account for external effects? How can we use alternative measures of welfare to identify the benefits and costs of a policy proposal? How can we ensure that politicians who aspire to use the results of CBA in their decisions can actually do so?	
Study Goals	At the end of this course, students will be able to: understand why CBA can be used to appraise projects with an objective to improve social welfare understand the strengths and limitations of CBA in project appraisal criticize and defend the CBA methodology from an ethical perspective explain the advantages and disadvantages of applying a CBA methodology in a political process with multi-stakeholders gain an understanding of the fundamentals of CBA perform steps or approaches to evaluation financial and economic values of a project interpret the results of a cost-benefit analysis	
Education Method	The lectures are on campus. In case of any unforeseen circumstances (e.g. the previous COVID-19 measures), further arrangement will be announced.	
Literature and Study Materials	* Textbook: Anthony E. Boardman, David H. Greenberg, Aidan R. Vining & David L. Weimer: Cost-Benefit Analysis: Concepts and Practice, 4th International Edition, Pearson, 2013. * Selected articles (available in Brightspace)	
Assessment	There are two types of the course assessment: 1. An individual essay (30%) 2. A written exam (70%) Both types of assessment are mandatory and the passing grade for each assessment is 5.5. Only the final grade will be entered in OSIRIS. Please register for the exam in OSIRIS on time (see: https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations). For both types of assessment, students who do not pass in the first attempt are given the second attempt. Students may review their works for each type of assessment. This will be held physically according to the time scheduled by the lecturer(s). Rules for the individual essay only: For the individual essay, please be well informed that the maximum grade for the second attempt for the essay is a 6.0. Note that the second attempt for the essay is only meant for students who do not pass and NOT intended for students who have passed nor new students who would like to enrol in the course after the first attempt. Further details about the individual essay will be provided later in Brightspace.	
Category	MSc level	

TPM028A	Decision Making in Multimodal Transport Systems	5
Module Manager	M.Y. Maknoon	
Instructor	M.Y. Maknoon	
Responsible for assignments	M. Farhani	
Co-responsible for assignments	J. Rezaei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Knowledge in operations research (mathematical modelling, integer programming) - Basic knowledge of optimization solver (e.g. CPLEX) and programming In case of deficits, students need to invest additional time to acquire these knowledges.	
Course Contents	Coordinating physical, information and decision flow is a focal point in managing multi-modal freight transportation systems. By having a global view on the entire supply chain - from raw material acquisition to the end customer delivery multimodal systems aim to provide a rapid, efficient and low-cost solution for freight transportation. On the other hand, social and environmental concerns have a central role in political and economic decision making. These concerns result in laws and regulations that affect the production of goods and services. Therefore, in addition to provide efficient, reliable and green services multimodal systems should be flexible enough to respond to short and long-term environmental, social and political changes. To address these challenges, effective, powerful and intelligent methods and techniques are required to analyse, plan and manage multi-modal systems. Operations research is at the heart of this intelligence. Operational research models and methods, integrated with enterprise information management systems, analyse and transform data into useful information for planning and management. This course focuses on the multimodal freight transportation. In particular design, planning and operation of multimodal network. The models and techniques taught in this course deal with a wide range of problems. From the long-term physical design and operating structure to the transport route selection, service mode and timetabling.	
Study Goals	The course focuses on multimodal transportation network. It presents the main models and methods to analyse and solve problems arises in this context. The course aims at knowledge acquisition and development of problem solving skills. The goal is to enhance students analytical and modeling capacities in research and practice.	
Education Method	The course is based on active participation. In case of unforeseen circumstances or measures resulting from COVID-19, student participation will be facilitated by on-line platforms (e.g. Microsoft Teams, Skype). Readings will be given for each session and presentations will be supplemented by discussions with the students.	
Assessment	The course is assessed based on two mid-term take-home exams (15% each adding up to 30% of the total grade) and one closed-book final exam (70% of the total grade). In case of unforeseen circumstances or measures resulting from COVID-19, the final exam will be take-home and open-book. To pass this course, students need to get more than 5.5/10 for the final exam. The results of mid-term exams are valid only for the current year and are not transferable to the next academic year. In addition, re-take is not allowed for the mid-term exams.	
Elective	Yes	
Tags	Design Mathematics Modelling Optimisation Programming Transport & Logistics	
Category	MSc level	

TPM032A	Multi-criteria Decision Analysis	5
Module Manager	J. Rezaei	
Instructor	J. Rezaei	
Responsible for assignments	J. Rezaei	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic mathematics and programming	
Course Contents	During this course, students will learn how a multi-criteria decision problem can be structured, solved, and analyzed. A multi-criteria decision problem is a problem where a number of alternatives (options) need to be evaluated with respect to a number of criteria (attributes) in order to (i) select the best alternative, (ii) rank all the alternatives, or (iii) sort the alternatives into a number of classes. A multi-criteria decision analysis (MCDA) method is a method to find a solution for (one or more parts of) an MCDA problem, where the main phases are: - Formulating the problem (identifying the goal(s); alternatives; and criteria), - Evaluating the alternatives with respect to the criteria, - Finding the relative importance of (or tradeoff among) the criteria, - Synthesizing the data collected in the previous phases to find a solution, - Checking the reliability and validity of the outcomes. These phases are executed with close interaction between decision-maker(s)(DMs) and analysts. This course covers the main phases of MCDA problems and some of the main MCDA methods. An overview of the contents of the course is as follows. - Introduction - Structuring an MCDA problem - Multi-Attribute Value methods - Multi-Attribute Value Theory (MAVT) - Analytic Hierarchy Process (AHP) - Best Worst Method (BWM) - Outranking Methods - ELECTRE methods - PROMETHEE methods - Group Decision-Making - Behavioral considerations	
Study Goals	On successful completion of this course, the students should be able to: - Structure multi-criteria decision problems; - Analyze decision problems using different MCDA methods, including multi-attribute value methods and outranking methods; - Structure and analyze decision problems considering different views of a group of decision-makers/stakeholders; - Explain and analyze behavioral aspects of decision problems (e.g., cognitive biases); - Apply an MCDA method to a real-world decision problem and analyze the outcomes.	
Education Method	Lectures Assignments Experimental analysis Real-world applications	
Assessment	The course has two partial assessments as follows: - Assignments (students in groups of size 2 apply MCDA methods to different decision-making problems): 40% - Application (students in groups of size 2 apply an MCDA method to a real-world decision-making problem and write a report of 8000 words): 60% The review of the Assignments and Application is done following a rubric. In order to pass the course, the grade of each part (Assignments, Application) should be at least 5.0, and the average of the parts $(0.4 * \text{Assignments} + 0.6 * \text{Application})$ should be at least 5.8. Resit: Students who fail the course have a possibility to repair the Assignments and Application to a maximum of 6.0. If other group members have already passed the course, the students may do this individually. The deadline for the resit is the end of the fifth week of Q2. If a student fails the course, they take separate partial grades the following year (partial grades are registered separately in Osiris).	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Multi Machine Engineering (MME)

ME44101	Dynamics and Interaction of Material and Equipment	4
Responsible Instructor	Prof. Dr.ir. D.L. Schott	
Instructor	Dr.ir. Y. Pang	
Co-responsible for assignments	H. Shi	
Co-responsible for assignments	E.F.L. Stok	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ME44115	
Course Contents	This course focuses on modelling and designing machine-cargo interactions and using those interactions to conceptually design equipment taking into account the transport demands and logistics processes. The course starts with the logistic context with emphasis on (bulk and container) terminal level, for which the functionality of equipment types will be addressed with regard to the cargo type (dry bulk, container, piece goods) and its characteristics. Specific equipment types will be discussed, analysed and designed, such as a belt conveying systems used for bulk materials (e.g. iron ore and coal) as well as for baggage and parcel handling. The interaction between cargo and equipment will be explicitly taken into account. Specifically for the bulk solid cargo type, an experimental assignment to determine the properties of a particular bulk solid material is part of the course. Bulk solid materials include granular materials such as coal, sand, limestone, iron ore, grain. These materials can be free flowing through bunkers and chutes as well as stored in silos, handled by stackers and reclaimers or transported by conveyors. With the experimentally determined properties conceptual design of a silo and belt conveyor will be undertaken. State-of-the-art particle based simulation with Discrete Element Method (DEM) will be introduced. The parameters, algorithms, and applications will be addressed, as well as the procedure for validation of DEM simulations to obtain realistic bulk material behaviour in a virtual environment. It will be shown how DEM has led to new breakthroughs and developments in the field of port handling equipment.	
Study Goals	The student will be able to: - Differentiate among various types of cargo terminals based on the specific characteristics of cargo and their corresponding logistical requirements. - Conduct experimental analysis to determine key properties of bulk solid materials, and utilize the findings to inform effective equipment design. - Design equipment solutions that are tailored to specific logistical requirements, taking the performance of the equipment within the logistics context into account, supported by thorough calculations. - Assess the effectiveness of equipment designs by considering the relationship between material characteristics, material behavior, and material equipment interaction parameters on equipment performance. - Evaluate the role of Discrete Element Method (DEM) in the design of equipment, including understanding the theory and algorithm used for DEM calculations and analyzing the impact of input parameters (particle level, simulation level) on the output, specifically on bulk behavior and erratic behavior.	
Education Method	Lectures, including guest lectures from industry experts. Practical for testing and characterising bulk solid materials and its material equipment interaction properties by experiments.	
Course Relations	ME44115 DEM simulations	
Literature and Study Materials	Schulze, D., Powders and Bulk Solids - Behavior, Characterization, Storage and Flow, Springer, 2008 ISBN 978-3-540-73767-4 (print), 978-3-540-73768-1 (online), DOI: 10.1007/978-3-540-73768-1 http://www.springerlink.com/content/155416/?p=fbeb6748815f4e4c92f56519a15f8837&pi=0 Rhodes, M., Introduction to Particle Technology by Martin Rhodes, John Wiley & Sons, 2008. ISBN 978-0-470-01427-1 (print), 9780470727102 (online), DOI: 10.1002/9780470727102 http://www3.interscience.wiley.com/cgi-bin/bookhome/117932420?CRETRY=1&SRETRY=0	
Assessment	Slides and supplementary material provided on Brightspace and/or during lectures Because of measures resulting from COVID-19, the prescribed assessment will be as follows: 1. practical assignment - remote (25% of the mark) 2. written exam - remote open book with several fraud prevention measures (75% of the mark) If the opportunity arises for examinations on campus, the method of assessment will be: 1. practical assignment (25% of the mark) 2. written exam - closed book (75% of the mark).	
Enrolment / Application	Please enroll yourself as soon as possible. It helps us to schedule and setup enrollment for practical (in pairs or threesomes)	
Remarks	MSc ME track Multi Machine Engineering MSc TIL Engineering specialisation	
Percentage of Design	20%	
Design Content	Conceptual design of material handling equipment considering cargo characteristics and interaction parameters between cargo and equipment	
Department	Dynamics of Material and Equipment Interaction 3mE Department Maritime & Transport Technology	

ME44106	Structural Design with FEM	4
Responsible Instructor	Ir. W. van den Bos	
Instructor	X. Jiang	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic mechanical engineering knowledge is expected. Experience with FEA programs is not required.	
Summary	Understand fundamental theory of finite element method and identify various failure mechanisms of metallic structure and its connection with relevant design standards. Design a ship-shore crane according to Dutch design Standards, check and evaluate the structural integrity applying a finite element model.	
Course Contents	For complete information see bright space! Understand fundamental theory of finite element method and identify various failure mechanisms of metallic structure and its connection with relevant design standards. Design a ship-shore crane according to the Dutch Standards (NEN 2017 to 2023) and control your design with the use of a finite element model (Femap and SDC Verifier) for detailed description and or examples see brightspace . This course is a continuation of the original course ME44105 and wb3416 Design with FEM Software used is: SDC Verifier (www.sdeverifier.com) and FEMAP (https://www.plm.automation.siemens.com/store/en-se/femap/index.html)	
Study Goals	The general objective of this course is to apply the finite element method in a proper way to design and evaluate the structural strength and durability of transport equipment as a crane. After the course the student is able to: 1. Understand the fundamental basis of FEM including the underlining linear and non-linear theory 2. Develop a reasonable FEM model and Judge the FEM results correctly 3. Apply the Finite Element method as a Design Tool 4. Design according to standards 5. Understand the structural failure mechanisms, buckling, plate buckling, crippling, fatigue and static overload (yielding) and how these failure modes influence the checks implemented in design standards	
Education Method	Weekly 2 hours theory lectures and 4 hours computer room time to work on the assignment.	
Assessment	The assessment consists of two parts- practical part and theoretical part. The practical part will be assessed through an oral exam of 1/2 hour on the final structural design report. The report contains the complete description of the structural design, the design specification and the translation of these specification into loads the design, and the final result of the FEA calculation. The FEA results should prove that the crane is structural save by complying with the design standard to be used. The theoretical part will be assessed based on a regular, closed book written exam. Detail information will be announced on Bright space in due time. The practical part and the theoretical part account for 75% and 25% of final grade respectively. Both parts are compulsory: a grade at least 5.0 is demanded per part in order to complete the course and obtain the final grade.	
Tags	Design Modelling Numeric Methods Structural Mechanics Transport & Logistics	

ME44110	Integrated Design Project for Multi-Machine Systems	5
Responsible Instructor	Ir. W. van den Bos	
Instructor	Dr. J. Jovanova	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Course Contents	The course Integration Project Multi-Machine Systems brings together the skills students develop throughout their studies. The students groups work together to develop designs of specific port equipment (ex: cranes; autonomous vehicles) and complex multi-machine systems (ex: floating terminals, bulk material terminals). Students are encouraged to use acquired knowledge in their previous studies and apply it to solve real engineering problems. They need to make multiple decisions in the design process and justify it based on their previous knowledge and experience. The students are expected to raise their technical confidence and learn how to combine the skills gained from other courses to design complex multi-machine systems. A graduate in Mechanical Engineering MME track is able to model the dynamics of the interaction between equipment and materials design, control and automate transport equipment and facilities develop, monitor and control components for transport and logistic systems and networks. This course Integration Project Multi-Machine Systems brings them a step closer to a real working scenario. Exit qualification Competent in designing. A graduate in Mechanical Engineering MME track is able to model the dynamics of the interaction between equipment and materials design, control and automate transport equipment and facilities develop, monitor and control components for transport and logistic systems and networks. The course Integration Project Multi-Machine Systems teaches to students of how to apply their knowledge in design of complex systems. The course is in Q3 and Q4 (the second semester) so the students are expected to already have high level of accumulated knowledge. As the course is project based, groups of students (6 -7) work on different tasks within a project. This allows for students from different entry levels to work together and support each other in the process. Another exit qualification is considering the temporal and social context. In this course the importance of transport systems and logistics in society is stressed as well as designs that are safer, more efficient and sustainable.	
Study Goals	The learning goals are: LO1: Review on design principles for multi-machines. LO2: Apply different design conceptual approaches (unit level, system level, integration level). LO3: Use standards for equipment design. LO4: Asses and recommend system integration (market availability, custom designs) LO5: Design a project for multi-machine system	
Education Method	Lectures, feedback sessions	
Assessment	Report	
Department	3mE Department Maritime & Transport Technology	

ME44115	Discrete Element Method (DEM) simulation	4
Responsible Instructor	Prof. Dr.ir. D.L. Schott	
Instructor	H. Shi	
Instructor	R.N. Roeplal	
Instructor	A.H. Hadi	
Instructor	Ir. M.C. van Bente	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	ME44101	
Course Contents	This hands on course teaches the use of particle based simulations using Discrete Element Method (DEM). DEM can be applied to any system that handles or consists out of (groups of) particles. DEM models individual particles (microscopic level) with the aim to model realistic behaviour of a the system of particles (macroscopic level). DEM can be used as an engineering tool to provide insight in influencing factors (e.g. particle size, shape and density) in particle processes, such as segregation and mixing. It can help to analyse and solve bottlenecks in industrial systems, but then validation of the model is required. After validation DEM can be used as a virtual prototyping environment for design of equipment that handles particulate materials. The application range is wide: from powder handling (small particle sizes, cohesive materials in food or pharmaceutical industry) to large scale systems such as rock dumping (e.g. 300 kg stones, 500 mm in size), and anything in between including bulk handling. Next to the theoretical background of DEM, the major part of this hands on course is to perform an assignment using the software EDEM. In this assignment you will model and redesign or optimise a (part of a) handling system with a given particulate material. This might also require laboratory work to measure real material properties on a macroscopic level such as angle of repose and wall friction angle. Your DEM model has to represent realistic material behaviour and realistic interaction between material and equipment. With the properly working DEM model you will analyse influential parameters and where possible improve performance of equipment. Typical application areas for DEM modelling are: bulk handling of materials coal and iron ore, also including solid biomass such as wood pellets or biocoal mining offshore food industry, handling of powders pharmaceutical industry, powders and tablets agricultural industry, for planting and soil preparation, post harvesting, fruit handling and sorting machines pavement engineering, creating mixtures and more.	
Study Goals	During the course the student will learn to: 1. explain the Discrete Element Method (DEM) theory 2. define the performance criteria of the system to be modeled 3. choose an appropriate contact model for material-material and material-equipment interaction in a DEM simulation 4. choose and implement a simplified but appropriate representation of the system in the simulation environment 5. evaluate the stability of the model by investigating a range of timesteps 6. calibrate and verify the model with experiments, experimental data or analytical solutions 7. test and evaluate the influence of different simulation parameters on the performance of the system 8. analyse datasets obtained from simulations linked to but not limited to the defined performance criteria 9. define confidence intervals for simulation and experimental results due to numerical and measurement errors	
Education Method	Lectures and assignment (computer and possibly lab)	
Computer Use	60%	
Course Relations	ME44101	
Literature and Study Materials	Study materials will be provided throughout the course	
Assessment	The examination will be as follows: 1. presentation on model development (20%) 2. report and presentation on model and results (80%) In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: 1. remote or recorded presentation on model development (20%) 2. report and remote or recorded presentation on model and results (80%).	
Enrolment / Application	As soon as possible via Brightspace	
Special Information	Design Lab Research Mechanics Modelling Practicals Prototyping Software Transport & Logistics Contact Models Offshore Mining Ballast Railway Scour protection Penetration Particle Interaction Equipment Machine Cargo Interaction Bulk material	

	Design of Experiments Experimental plan Bulk
Department	3mE Department Maritime & Transport Technology

ME44125	Reliability and Maintenance of Transport Equipment	3
Responsible Instructor	X. Jiang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Technical academic BSc	
Course Contents	Transport equipment is in general a mechanical system, growing in size, capacity and complexity in order to accommodate continuous development of transport. It is essential to ensure sufficient safety, serviceability and durability of the equipment at the reduced cost in order to maintain the effectiveness of the transport system. This belongs to the research context of system reliability. Typically, a transport equipment consists of multiple subsystems or components with different function, and they would fail in different failure modes with different mechanism behind. Contributed by various non-destructive techniques, online sensor system and data analytics, potential failure of the equipment (fault, defect, damage) could be detected, predicted and mitigated in such a way that the occurrence of a catastrophic accident would be prevented, although with some extent of uncertainty caused by, i.e., the sensitivity of NDTs & sensors, environmental noise, etc. Upon the detection of a fault(or damage), a follow-up maintenance strategy shall be made in order to determine whether, when, where and how to repair/maintain the equipment, constrained by available budget, facilities and resources, etc. The overall objective of the course is to prepare engineering graduates with an in-depth knowledge of system reliability, maintainability and availability. A series of modules will be given covering the fundamental theory embedded in the reliability and maintenance of transport equipment. Application of the theory in practice will be exemplified and practiced through in-class exercises and discussion, weekly assignments and final written report. One or two guest lectures will be introduced to demonstrate the implication of system reliability in the operation and maintenance of the transport equipment and the infrastructure.	
Study Goals	After successfully completing the course, students will be able to: -Define system Reliability, Maintainability and Availability(RAM)and describe associated basic measures using statistic method. -Carry out system reliability analysis to identify critical component / subsystem using statistic models, like FMECA, FTA, ETA and Markov process among others. -Apply the acquired data to predict Remaining Useful Life (RUL) using statistic, data driven or hybrid models. -Analyze common maintenance strategies at both component and system levels based on statistic, data driven or hybrid models. -Develop and reflect a maintenance strategy for a simplified transport equipment/system based on statistic, data driven or hybrid models.	
Education Method	Lectures and supplementary material provided during lectures, in class exercises and discussions, weekly assignment, group work on final written report and presentation.	
Assessment	The final grade is based on 30% of the grade for weekly assignments plus 70% of the grade for the final report and associated presentation(50/50). Four to five students are anticipated to work as a group on the final report. Weekly assignments are compulsory and they can be done either individually or as a group work (the same group on the final report). The grade for assignments will be the average grade of all weekly assignments. To pass the course, the grade for each weekly assignment is not allowed to be lower than a 5.0. The final presentation will take place on campus. In case of unforeseen circumstances or measures resulting from i.e., COVID-19 etc., the final presentation will then be given online. Detail information will be announced on Brightspace in due time. The final grade must be a 5.75 or higher in order to pass the course.	
Remarks	Reference books: 1. Marvin Rausand, Arnljot Høyland, System Reliability Theory: Models, Statistical Methods, and Applications. 2. Kumar, U. D., Crocker, J., Knezevic, J., & El-Haram (2000), M. Reliability, Maintainability and Logistic Support - A Lifecycle Approach. 3. Charles E. Ebeling (2010), An Introduction to Reliability and Maintainability Engineering.	
Department	3mE Department Maritime & Transport Technology	

ME44200	Operations and Maintenance	3
Responsible Instructor	Dr.ir. Y. Pang	
Instructor	Dr. V. Reppa	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course focuses on intelligent control to optimize the operational performance and to maintain the reliability of multi-machine systems in large-scale transport and logistics domain. Along with the understanding of diverse transport systems and logistic processes, including the dynamic interactions of the equipment and facilities involved, the control for multi-machine systems contains two folds, the integrated control for efficient operations and the intelligent decision-making for reliability control. To ensure efficient and reliable operations at both equipment level and system level, technologies and methodologies of automated data acquisition, remote communication, intelligent data analysis and integrated operational decision-making will be studied. Condition-based maintenance of systems and equipment will be analysed and discussed in combination with on-line techniques for monitoring the health of the sensors used for maintenance. During the course industrial case studies will be given to present the state-of-the-art development and foreseen challenges. The course is concluded by practical assignments where such intelligent control systems will be researched and conceptually designed.	
Study Goals	After successfully completing this course, you will be able to: - explain the concepts of intelligent control for operational support systems and advanced maintenance strategies; - identify the properties that determine efficient operation and reliable performance; - analyse the complexity of the operations and interactions of multi-machine systems; - derive models to represent the interactions and operational processes of multiple equipment and facilities; - design intelligent operational control systems in term of mathematical models and integrated control methods; - apply the methodologies and technologies to improve the efficiency and reliability for both individual machine and overall multi-machine system; - describe the effects of sensor faults in operations and maintenance of transport systems; - explain the design of model-based sensor fault detection methods; - explain the design of multiple sensor fault isolation methods; - design a sensor fault detection and isolation method for the operation and/or maintenance of a transport system.	
Education Method	Lectures (2 hours per week), case study, practical assignments	
Literature and Study Materials	Handouts and references to relevant literature and media. Additional teaching materials to be provided during lectures and available via Brightspace . Expected prior knowledge: ME44101 (not obligatory). Prior knowledge in system engineering (dynamical systems) is recommended.	
Assessment	Group assignment: Assignment Reports & Oral Exam Final grade is calculated by weighted average of Assignment Part I (75%) and Assignment Part II (25%). In case of a final grade less than 6, the group can have one opportunity to resubmit or repair the insufficient work via one of two modalities: 1. Repair: to improve the work of the same assignment towards a maximum retake grade of 6; 2. Resubmit: to choose a new assignment to be assessed by general assessment criteria towards a retake grade higher than 6. After successful delivery of the group assignment reports, the Oral Exam takes place in the exam week of the education period.	
Department	3mE Department Maritime & Transport Technology	

ME44210	Drive & Energy Systems	3
Responsible Instructor	Dr.ir. H. Polinder	
Contact Hours / Week x/x/x/x	3/0/0/0 (1st part Q1 2x2 hours, 2nd part Q1 1x2 hours)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This course is designed for students without a BSc in Electrical Engineering. Proper knowledge of mathematics (differential equations, complex numbers) and mechanics is required.	
Course Contents	- Introduction to drive and energy systems - Electricity and electrical circuits - Magnetism and magnetic circuits - Power electronics - DC machines - AC voltages, currents, powers and phasors - Transformers - Synchronous machines (including PM machines) - Induction machines - Choices in electrical drive and energy systems	
Study Goals	After following this course, students should be able to - Calculate DC and AC (single phase and three-phase) voltages and currents in simple electrical circuits (such as equivalent circuits of electrical machines) with sources, resistive, capacitive and inductive loads using Kirchhoffs laws. - Sketch phasor diagrams with voltage and current phasors. - Calculate active, reactive and apparent power in AC circuits. - Calculate energy stored in capacitors and inductors. - Describe the most important power semiconductor (diodes, IGBTs, MOSFETs) and their switching behaviour. - Describe how power electronic converters use these switches to convert DC from one voltage level to another (choppers or phase legs using IGBTs or MOSFETs), AC into DC (passive rectifiers using diodes or active rectifiers using IGBTs or MOSFETs) or DC into AC (phase legs using IGBTs or MOSFETs). - Derive an expression for the output voltage of a single-phase or three-phase diode bridge rectifier as a function of input voltage in continuous conduction mode (neglecting commutation). - Derive an expression for the output voltage of a DC-DC converter as a function of duty cycle and input voltage in continuous conduction mode. - Calculate induced voltage using Faradays law. - Calculate electromagnetic forces and torques using Lorentz law or the power balance. - Derive voltage equations and equivalent circuits of transformers, DC machines, synchronous machines, induction machines and permanent magnet machines. - Derive performance characteristics based on these equivalent circuits (torque-speed characteristics). - Describe advantages and disadvantages of electrification of drives. - Select suitable machine and drives for different applications.	
Education Method	Lectures	
Books	T. Wildi, "Electrical Machines, Drives and Power Systems"	
Prerequisites	Mathematics, Mechanics	
Assessment	The method of assessment will be: written exam, closed book, partly multiple choice. An equations form will be handed out with the exam. If on campus assessment is not possible (for example, because of measures resulting from COVID-19) the assessment will be as follows: written exam, open book, online, partly multiple choice.	
Permitted Materials during Tests	An equations form will be handed out.	
Department	3mE Department Maritime & Transport Technology	
Contact	h.polinder@tudelft.nl	

ME44300	Multi-Machine Coordination for Logistics	3
Responsible Instructor	Dr. V. Reppa	
Instructor	Prof.dr. R.R. Negenborn	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this course students will get familiar with automatic control techniques and their benefits and applications for real-time coordination of actions of large-scale transport systems. Theoretical concepts mostly related to model predictive control are discussed, as well as their application to real-world transport systems. Illustrative examples of large-scale systems include among others intermodal / synchromodal (i.e. road, waterborne, rail) transport networks. The operation of these networks relies on the interoperability of various machines such as cargo ships, trains, road tracks, automated guided vehicles, quay/yard cranes and many more.	
Study Goals	Students are able to: LO1. apply system theory to describe large-scale machines and transport systems LO2. explain the design of an automatically controlled system and of the automatic coordination of several controlled systems LO3. explain the design of control architectures for large-scale interconnected machines and transport systems (centralized, distributed, single-agent, multi-agent, single-level, multi-level) LO4. evaluate the pros and cons of different control architectures and their impact on multi-machine coordination for transport LO5. design an optimal control strategy for a single machine used in the transport network (truck, ship, crane, etc) LO6. develop a multi-agent framework based on model predictive control for real-time logistics of large-scale transport systems	
Education Method	Study material will be provided to support the lectures during which new concepts will be discussed and explained. Small groups of students will be created to conduct a project that will help them to develop deeper understanding of the theoretical (optimal) control and coordination concepts applied to a specific machine that is part of a larger transport network. Throughout the course illustrative examples will be used for showing the application of new concepts. Feedback sessions will be organized to discuss questions about the project.	
Course Relations	The course is related to the exit qualifications 1.b, 1.c, 3.b, 3.c and 7.a of the track MME of the Master Programme Mechanical Engineering, and to the specialization Engineering of the Master Programme Transport, Infrastructure and Logistics. Prior knowledge in automation of transport systems, dynamical systems, optimization and simulation is recommended.	
Literature and Study Materials	We will discuss chapters from the book "Intelligent Infrastructures", Negenborn et al., Springer, Dordrecht, The Netherlands, 2010. ISBN: 978-90-481-3597-4. Additional handouts and references to relevant scientific literature and media will also be provided.	
Assessment	Assessment will be based on a written group project report (50%) and on a written individual (closed-book) exam (50%). In principle, the written exam will take place on campus. Resit options are offered for both the project and the exam. Note that the grade of the exam and the grade of the project are only valid for the academic year 2023-2024.	
Tags	Information & Communication Modelling Optimalisation Project Transport & Logistics	
Department	3mE Department Maritime & Transport Technology	
Contact	Dr. Vasso Reppa (v.reppa@tudelft.nl)	

ME44305	System Analysis and Simulation	5
Responsible Instructor	Ir. M.B. Duinkerken	
Instructor	F. Schulte	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Required for	Multi-Machine Engineering	
Expected prior knowledge	Basic programming	
Parts	Systems and Data Analysis Discrete-Event and Agent-Based Simulation	
Course Contents	The course is an introduction to "Problem Engineering"; how to formulate a well-defined problem in a complex situation. The course introduces a systems approach to define the elements and structure of a problem situation and the process approach of discrete simulation to understand and quantify the time-dependent relations (behavior)	
Study Goals	Learning objectives: - Describe all relevant elements and their relations for at least one complex industrial or transportation system (e.g., cyber-physical production systems) using a combination of at least two conceptual models: process models, system models, steady-state models; - Collect data for the described system and analyse the system using the collected data and introduced data analysis methods; - Design industrial and transportation systems and processes applying an established innovation model; - Transform the developed conceptual models into a process-oriented simulation model, using a suitable language (e.g., Delphi/Tomas/Lazarus or Python/Salabim/Simpy).	
Education Method	lectures + group assignments	
Computer Use	Windows laptop with Python	
Literature and Study Materials	"The Delft Systems Approach, Analysis and Design of Industrial Systems", 2008, H.P.M.Veeke, J.A.Ottjes, G. Lodewijks, Springer, ISBN 978-1-84800-176-3 Lecture notes	
Assessment	Report of group assignment	
Exam Hours	80 for assignment	
Tags	Abstract Broad Challenging Design Industry Logics Modelling Process Programming Programming concepts Prototyping Stochastics Transport & Logistics	
Percentage of Design	50%	
Design Content	Logistic systems	
Department	3mE Department Maritime & Transport Technology	

ME44311	Advanced Operations and Production Management	5
Responsible Instructor	A. Napoleone	
Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	To enroll in the course, students must have a BSc in Engineering. The course is especially for students following TIL, MME and SDPO.	
Course Contents	The course requires high engagement and commitment, and includes both self-study and group work. Students are expected to be willing to work in teams, to collaborate and use social and project management skills. The course revolves around different topics of operations and production management, including: system analysis and process mapping, modularity in product design and process design, smart manufacturing and Industry 4.0, reconfigurability and adaptability in operations management.	
Study Goals	With this course, the student is expected to develop a structured approach to address operational problems using the methods provided in the course and/or methods identified with individual study. After successfully completing this course, students will be able to: - Describe the fundamental concepts of operations and production management - Search for and use methods to solve operational problems in different applications - Have a structured approach to manage familiar as well as unknown operational problems	
Education Method	Problem-based and interest-driven approach. PART ONE in Q3: (i) Plenary lectures / Cases by industrial speakers AND (ii) Individual study and deepening of one/more operational topics of interest. Attending lectures is required to acquire the fundamentals of production and operations management. Active participation during plenary lectures is strongly advised. At the end of Q3, students submit individual reports (one report per student). PART TWO in Q4: Group projects (up to 6 students per group) with case companies AND Group supervision (on demand). In Q4, students work in groups at different projects with case companies. The course leader can assist students in finding a company. Groups work autonomously and ask for supervision when needed. At the end of Q4, students submit group reports that include multiple analyses of operational problem at the case company and the solution to the problem. Q5 is available groups if preferable for time constraints.	
Assessment	The assessment is based on the evaluation of the individual report (Q3) and group report (Q4).	
Department	3mE Department Maritime & Transport Technology	
Judgement	The report is judged by Dr A. Napoleone The final grading considers the individual report (50%) and the group report (50%)The grading is based on a Rubric. Results can be discussed in feedback sessions (on demand)	
Contact	a.napoleone@tudelft.nl	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Traffic & Transport Engineering (TTE)
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CIEM6301	Railway Traffic Management	5
Responsible Instructor	E. Quaglietta	
Module Manager	E. Quaglietta	
Instructor	Prof.dr. R.M.P. Goverde	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Railway timetabling: Microscopic and macroscopic rail traffic simulation models Timetable design techniques and optimization models Timetable assessment using microscopic simulation Energy-efficient train operation Real-time railway traffic management: Driver Advisory Systems Disruption management Train delay prediction and analysis using a statistical train delay analysis tool Railway traffic management systems Conflict detection and resolution Simulation-based assessment of traffic management strategies This elective will include discussions about societal impacts of advanced tools and techniques for train service planning and management. Students will be aware of responsibilities and ethical implications in using automated decision support systems in rail traffic planning/management practice. Roles and responsibilities of railway stakeholders in planning / managing rail traffic will be tackled together with interaction processes during nominal and degraded operational conditions.	
Study Goals	On completion of this module, the student will be able to: LO1: Design railway timetables (CE ILO1) LO2: Assess rail traffic management methods (CE ILO4) LO3: Analyse train delays (CE ILO2) LO4: Apply rail traffic simulation (CE ILO2) LO5: Evaluate energy-efficient train operation (CE ILO3)	
Education Method	Blended learning mixing on-campus lectures, reading, videos, guest lectures and formative assignments	
Assessment	Formative group assignments on rail traffic simulation, scheduling and traffic data analysis with supervised practicum sessions. Formative assessment also includes pre- and/or post-lecture tests, on-line self-assessment, training with model exams with answers. Final digital written exam. The exam will include questions on lectures and formative group assignments. A minimum grade of 5.8 is required to successfully pass the exam.	
Expected prior Knowledge	Prior knowledge about planning and design of public transport systems is recommended, such as units of Module B3 of the MSc programme in Transport and Traffic Engineering (i.e. Public Transport Planning and Operations, Public Transport networks and modelling, Railway operations and control). In particular, blocking time theory is assumed working knowledge. Additional reading (one chapter, 3 hours) is required for those not familiar with blocking time theory. In addition, an introduction course on discrete optimization is recommended.	
Academic Skills	Problem formulation and problem solving Critical and analytical thinking Interpretation Writing technical reports Applying professionally-relevant tools and methods	
Literature & Study Materials	I.A. Hansen & J. Pachl (eds.), Railway Timetabling & Operations, Eurailpress, Hamburg, 2nd edition, 2014. Lecture slides of the course available on Brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Open book - digital exam	
Collegerama	No	

CIEM6303	Transitions, Sustainability & Innovation	5
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Module Manager	Dr.ir. A.J. van Binsbergen	
Contact Hours / Week x/x/x/x	4hrs lectures per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Transport and mobility is changing rapidly - partially as a consequence of technological and societal developments, partially also because our demands towards the transport system change. This course is about understanding this changing context and being able to develop and/or assess new ideas and concepts for transportation. This is done along the lines of sustainability (requirements), innovation (introducing new ideas) and transitions (radically changing the perspective on transport and mobility). Sustainability: on basis of (changing) demands in the fields of wellbeing ('people'), economic activities ('prosperity') and the environment ('planet') as well as anticipated developments in demography, economy and spatial development, we analyse the requirements and needs for future transport systems. The concepts will be discussed during lectures, but you'll work on these during the first part of an assignment as well: identification of main developments and future aims and ambitions for a specific urbanised area, to be described in a base case, Business-as-usual (BAU) and two alternative (near or far) future scenarios. Innovation: introduction of innovation theory (steps, roles of actors, role of technologies, product life cycle, technology readiness levels, pilots, living labs, upscaling); tools for analysing and assessing 'new' ideas, what is really new, is a new concept really useful, what about interoperability and interconnectivity? Some concrete examples will be discussed, for instance MaaS, mobility hubs and synchro-modality. The topics will be introduced and discussed in class, and in part two of the assignment you're asked to identify, analyse and assess 'new' transport and mobility concepts. Transitions: introduction of transition theory and transition management (wicked problems, regimes and regime change; actors, co-evolution, self-organisation, adaptation), illustrated with some concrete examples. The approach will be discussed in class and in the third part of the assignment you're asked to identify or a potentially 'transitional' development yourself, to identify current regimes and actors, needed changes, barriers and ways to overcome these. During the fourth part of the assignment you're asked to develop a futuristic transport system based on (assumptions of) spatial and demographic development, complying to sustainability requirements. Analysing potential future use and effects under different future scenarios, developing innovation or transition pathways, identifying relevant regimes, key actors, practices and needed changes therein, identifying the necessity of co-evolution and the potential of self-organisation/co-makership. Overall assessment of the validity of the envisaged transport system.	
Study Goals	On completion of this module, the student will be able to: LO1. Create innovation pathways for passengers and logistics concepts answering to societal challenges: ILOs 1, 5, 8, 9, 10. LO2. Evaluate performance and (sustainability) impacts of innovative transport concepts in future contexts: ILOs 2, 3, 9, 10. LO3. Assess suitability of tools for evaluating innovative transport concepts in future contexts: ILOs 4, 6, 10. LO4. Analyse the expected functionality and (sustainability) performance of future transport concepts: ILOs 2, 5, 9, 10. LO5. Apply different standard methods and approaches for developing innovation pathways: ILOs 4, 7, 8, 10. LO6. Operationalise performance and (sustainability) effects indicators for (future) transport systems, including estimation methods and approaches: ILOs 5, 6, 10. LO7. Explain the complexity of sustainable transport system transitions and innovation pathways: ILOs 6, 7, 9, 10. LO8. Define (current and future) transport systems, their (envisaged, expected) characteristics, and main development challenges: ILOs 6, 9, 10. LO9. Effectively communicate results and opinions with stakeholders of diverse backgrounds: ILOs 6, 9	
Education Method	Mix of on-campus lectures, reading, videos, guest lectures and a summative assignment with intermediate feed-back	
Assessment	Formative in-class exercises / interactive sessions with direct feed-back. Group assignment (via report and presentation) Summative group assignment with supervisor assessment resulting in a group grade based for 85% on the report and 15% on the presentation, with the possibility of tuning individual grades via peer-assessment (varying max. 20% of the group grade). The minimum passing grade is 5.8. In case an individual grade is below 5.8 but higher than 4.5, an additional task can be given; when done satisfactory, the final grade can be improved to max. 5.8.	
Expected prior Knowledge	1. Basics of (transport) economics: basics of economic theory, economic rationale and behaviour (utility theory), costs & benefits (fixed, variable; direct, indirect), economic assessment (cost/benefit ratio), value of time. 2. Basic concepts of Transport Modelling: including both agent based modelling (as modelling technique) and activity based modelling (as modelling approach next to the traditional trip-based modelling). 3. Basic insights in societal goals such as (preconditions for) economic developments including accessibility, sustainability and social equity/inclusiveness, and terms such as reliability, robustness, resilience. 4. Insight in societal challenges and social, economic and ecological sustainability objectives. For participants lacking background knowledge in these fields, self-study materials sourced from the MSc Civil Engineering Transport Technology and Engineering (CE-TTE) track will be available.	
Academic Skills	Critical, interdisciplinary and ethical thinking, problem aware and solution oriented learning, information collection and synthesis, communication and responsiveness, scientific writing and presentation, application of theory to new situations.	
Literature & Study Materials	Lecture slides available on Brightspace, and: Book chapter: René Kemp, Jan Rotmans (2004) Managing the transition to sustainable mobility (Chapter 7 in System Innovation and the Transition to Sustainability, Edgar Elgar) Kemp R., Avelino, F., Bressers, N. (2011) Transition management as a model for sustainable mobility in European Transport / Trasporti Europei No. 47, April 2011 Snelder, M., Wilmink, I., van der Gun, J., Bergveld, H.J., Hoseini, P., van Arem, B. (2019). Mobility impacts of automated driving and shared mobility: explorative model and case study of the province of North-Holland. European Journal of Transport and Infrastructure Research, vol. 19, n. 4. https://doi.org/10.18757/ejtr.2019.19.4.4282 Milakis, D., Snelder, M., Arem, B. van, Wee, B. van, & Homem de Almeida Correia, G. (2017). Development and transport implications of automated vehicles in the Netherlands: scenarios for 2030 and 2050. European Journal of Transport and Infrastructure Research, 17(1).	

Judgement	Lucas, K., Maat, K. & van Wee, GP., (2016) A method to evaluate equitable accessibility: combining ethical theories and accessibility-based approaches, in: Transportation. March, p. 1-18 18 p. See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

CIEQ6003	Traffic Modelling and Management	6
Responsible Instructor	Dr. V.L. Knoop	
Instructor	Dr. M. Wang	
Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Prof.dr.ir. S.P. Hoogendoorn	
Contact Hours / Week x/x/x/x	x/x/6/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Roadway capacity and cumulative curves Intersection control Car-following behaviour, micro-macro relations, trajectories Railway capacity Timetable stability and max-plus algebra Speed contour plots & capacity drop Fundamental diagram & shock waves Traffic flow properties for buses, trams and metro, incl. macroscopic fundamental diagrams and dwell time functions. Transit signal priority. Pedestrians: expand traffic flow theory to two dimensional planes Safety, environment; control objectives Systems and control Network management - hierarchical and multi-level (Application of) traffic simulation tools (reliability, seeds, ...)	
Study Goals	On completion of this course, the student will be able to: TBU3-LO1 Describe traffic flows in space and time at microscopic and macroscopic level (TTE-TB-LO1) TBU3-LO2 Discuss and apply theories and models at microscopic and macroscopic level for different transport modes (TTE-TB-LO3). TBU3-ILO3 Develop and assess control strategies at local and network level (TTE-TB-LO2). TBU3-ILO4 Set-up, perform, and report a traffic flow simulation experiment (TTE-TB-LO4).	
Education Method	Lectures, feedback on group work	
Assessment	Written exam (100%)	
Tags	Analysis Modelling Transport & Logistics Transport phenomena	
Expected prior Knowledge	Calculus; basic insights into modelling and insights into simulation setup.	
Academic Skills	Critical thinking	
Literature & Study Materials	Lecture notes provided by the teachers	
Judgement	100% exam For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Self-made notes (limited in size; see brightspace)	
Collegerama	Yes	

CIEQ6211	Transport Modelling: Traffic Assignment	3
Responsible Instructor	Ir. A.J. Pel	
Instructor	Dr. O. Cats	
Instructor	N. van Oort	
Instructor	Dr. M. Snelder	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we study a number of more advanced modelling techniques related to travel demand, traffic assignment and network loading, and how these are used (and needed!) for various model applications. We will approach advanced transport modelling in this course in two ways. On one hand, we will discuss modelling techniques such that you will become aware of a wider variety of models and that you understand their characteristics and (dis)advantages. On the other hand, we will discuss aspects to address when selecting and applying a transport model such that you will become skilled in how to execute a model study. The course continues on basic concepts of transport modelling (as studied in the TTE track base) and explores a variety of more advanced transport modelling, both in research and practice. Main themes relate to activity-based models, non-equilibrium route choice models, multi-modal travel, capacity-restricted static models, crowding in transit assignment, and dynamic traffic assignment. These topics are illustrated by case study applications on a variety of topics related to network performance and analysis.	
Study Goals	At the end of this course you will be able to design, apply, and assess advanced transport models (beyond the state-of-the-practice and the modelling examples taught in the TTE track base). You will be able to: I. explain the conceptual framework and operations of transport models II. describe advanced traffic assignment and choice modelling techniques, and assess their key notions, strengths, and weaknesses III. assess the realism of transport models with respect to their intended application IV. apply these advanced techniques within a case study: design the model requirements, model specifications, and model verification and application	
Education Method	Weekly lecture recordings, selected literature, in-class Q&A, and discussion of example (exam) questions	
Assessment	Digital exam (100%)	
Expected prior Knowledge	Basic concepts of transport modelling (four step transport models, static traffic assignment)	
Academic Skills	Critical thinking, analytical thinking, scientific reasoning.	
Literature & Study Materials	Weekly list of key concepts, selected literature, lecture recordings and slides, and additional literature (optional, for reference) are made available through Brightspace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	None	
Collegerama	Yes	

CIEQ6212	Regions, transport and networks	4
Responsible Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr.ir. M. Saeednia	
Instructor	M. Rinaldi	
Instructor	Dr. C. Maat	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The concept of accessibility and its relevance for user behaviour and transport network and spatial design; Spatial and social external effects of transport infrastructures and use, and its relevance for transport network and spatial design; Characteristics, data and data analysis of population, economic activity and land use; Interactions in regional, land use and transport network developments and their dynamics in time; Urban and regional transport demand models; Transport networks and regional development optimisation and design problem; Base introduction and application of Transport Economics, System Dynamics, and Agent-Based Modelling.	
Study Goals	On completion of this course, the student will be able to: LO1. To explain the concept of accessibility and its indicators. LO2. To analyse population, economic activity and land-use data and to 342 evaluate the accessibility and spatial, social and economic impact of existing or new land use transport plans. LO3: To explain the conceptual, dynamic relationships between land use, transport system quality and transport behaviour, both of passengers and freight transport. LO4: To apply location theory and network design optimization methods on land use and transport challenges. LO5: To create transport network land use design on basis of formulated societal requirements. LO6: To model the complex, dynamic relationship between land use and transportation development.	
Education Method	Lectures, practical lecture/workshop	
Assessment	Written exam (100%)	
Expected prior Knowledge	Transport Demand Modeling and optimization techniques.	
Academic Skills	Mathematical modeling	
Literature & Study Materials	- Lecture slides; - Book: Jean-Paul Rodrigue (2020) 'The geography of Transport Systems'. Fifth Edition; New York: Routledge, 456 pages. ISBN 978-0-367-36463-2	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Closed book.	
Collegerama	No	

CIEQ6213	Freight transport networks and systems	4
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Prof.dr.ir. L.A. Tavasszy	
Instructor	Dr. S. Fazi	
Instructor	Dr.ir. A.J. van Binsbergen	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Economic activities, trade, logistics and freight flows; Logistics networks with consolidation and transshipment points; Modelling of freight transport demand and supply; Unimodal and intermodal freight transport networks; Freight transport and logistics optimisation from the user perspective; Terminal and port operations; Urban freight transport, urban distribution; Sustainability effects of freight transport, sustainability policy consequences for freight transport Transport Economics (applied)	
Study Goals	On completion of this course, the student will be able to: LO1. To explain the current state and future developments of freight transportation. 343 LO2. To explain logistic processes. LO3. To design, model and analyse freight flows. LO4. To explain and analyse the interaction between freight demand and supply. LO5. To design, model and analyse freight traffic flows for multimodal networks in global, national and urban settings.	
Education Method	Lectures, guest lectures, formative group assignment (multiple parts) with instruction	
Assessment	Written exam (100%)	
Expected prior Knowledge	1. Basic insights performance, impacts, ethical and social implications and their indicators - of traffic and transport networks; 2. Basic insights in transport economics (choice preferences, business perspectives); 3. Basic insights in Transport Modelling and Analysis, including aggregated, disaggregated models, gravity models and discrete choice models for freight networks; 4. Basic knowledge on distribution (OD-estimation), assignment and equilibrium models, Basic knowledge on mixed-integer linear optimisation; 5. Insight in characteristics of unimodal and intermodal networks, physical and service networks for freight transport; 6. Insight in data-driven modelling approaches for verification, calibration and validation of models, input and parameters in the context of transport infrastructure, network, operations and behaviour.	
Academic Skills	analytic skills; data analysis; academic writing (for the exam)	
Literature & Study Materials	- Lecture slides; - Book: Tavasszy, L., G. de Jong (eds) (2014), Modelling Freight Transport, Elsevier (digital edition) - Report: CE Delft (2020) 'STREAM Freight Transport' (digital edition; annual update) - Paper: Jin Y. Yen (2015) Finding the K shortest Loopless Paths in a Network, Management Science 17(11)	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	handheld calculator	
Collegerama	Yes	

CIEQ6221	Active Modes: Traffic and Transport	3
Responsible Instructor	Dr.ir. D.C. Duives	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Active mode operations Empirical relations in active mode traffic operations Phenomena in active mode traffic Active mode models Active mode travel behaviour Sensing and monitoring of active mode travel behaviour Active mode route and activity choice modelling Crowd management theory and practice	
Study Goals	On completion of this course, the student will be able to: B2U1-LO1 List and select data collection methods for active modes, and design a data collection plan B2U1-LO2 Identify and demonstrate the elements of active mode monitoring systems B2U1-LO3 Identify, predict and analyse active mode route choice behaviour B2U1-LO4 Formulate and apply various simulation approaches for active mode traffic in both normal and evacuation conditions and discuss the results B2U1-LO5 Explain and perform calibration and validation of active mode simulation models B2U1-LO6 Design active mode facilities B2U1-LO7 Identify, apply and design various approaches for crowd management B2U1-LO8 Explain, identify and evaluate shared space	
Education Method	Lectures (2x 2 hours per week), formative group assignments	
Assessment	Written exam (100%)	
Special Information	This course is the TIL equivalent of the CEG course CIEM6220 - Unit 1: Active Modes. To enroll as a NON-CEG student, enroll in CIEQ6221 AND CIEM6220. The material of the course will be published on the Brightspace page of CIEM6220.	
Expected prior Knowledge	Preferably a module A from the T&P track of the CEG master. For universities of other faculties, no prior Knowledge is expected, but you might need to put in some work to catch up with the CEG students.	
Academic Skills	Technical literacy Data analysis Time management Critical thinking	
Literature & Study Materials	Presentation slides and reading material will be provided after each lecture	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	A simple calculator	
Collegerama	No	

CIEQ6222	Traffic Safety	3
Responsible Instructor	Dr.ir. H. Farah	
Instructor	Dr. M.P. Hagenzieker	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Road safety paradigms & road safety research Overview research methods in road safety; History of road safety research; Models and theories in road safety: road user behaviour, road design, safe systems approaches; Traffic safety (accident) modelling; Traffic safety of and road design for vulnerable road users (cyclists, pedestrians); Surrogate safety measures New technology, automated driving and traffic safety; Perceived safety and user acceptance of automated vehicles; Writing a research proposal.	
Study Goals	On completion of this course, the student will be able to: B2U2-LO1. Explain the different behavioural theories and models in the field of traffic safety; B2U2-LO2. Explain the different research and evaluation methods applied in the field of road user behaviour and traffic safety; B2U2-LO3. Analyse the contributing factors, characteristics and mechanisms of a selected traffic safety problem; B2U2-LO4. Evaluate the strengths and weaknesses of different data collection and research methodologies and their ethical implications; B2U2-LO5. Apply the basic concepts, behavioural theories and models, and the research methods to the selected traffic safety problem; B2U2-LO6. Develop a research proposal by conducting a literature review on the selected traffic safety problem, defining research gaps, defining research objectives, and research method; B2U2-LO7. Write a clear and reproducible scientific research proposal; B2U2-LO8. Evaluate strengths and weaknesses of a peer group research proposal; B2U2-LO9. Present concisely (pitch-like) to your peers the traffic safety problem and the developed research proposal.	
Education Method	Lectures (2 x 2 hours per week), in-class small groups practice problems, in-class discussions, self-assessment quizzes, peer review, guest speakers.	
Assessment	100% group assignment (the assessment rubric will include: report, presentation in a pitch format, interview including individual questions to each member of the group, peer review)	
Expected prior Knowledge	TTE Track Base (TB-1: Planning and design of Traffic and Transportation Networks and Infrastructure)	
Academic Skills	Knowledge and understanding of key elements, theoretical frameworks and how they are applied in the road safety field; How road safety research is conducted; Critical thinking and analytic skills; Use logic and reasoning as well as analytic thinking in assessing strengths and weaknesses of road safety research and applied road safety measures; Writing skills: write a research proposal including problem definition and argumentation, appropriate use of scientific literature, propose methodology, discuss implications of possible outcomes.	
Literature & Study Materials	Lecture notes and reading material will be provided in each lecture	
Judgement	See Assessment.	
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	The assignment (i.e., research proposal).	
Collegerama	No	

CIEQ6223	Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment	4
Responsible Instructor	M. Rinaldi	
Instructor	Dr. M.P. Hagenzieker	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	Dr.ir. W.J. Schakel	
Instructor	Dr. V.L. Knoop	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Introduction, classification and functional description of intelligent vehicles; System architecture, decision and control methods for intelligent vehicle systems; Technologies for intelligent vehicles, sensors, communication, and actuators; Behavioural adaptation to intelligent vehicles; Impact assessment of intelligent vehicles on traffic safety; Impact assessment of intelligent vehicles on traffic efficiency; Simulation experimental research of impacts of intelligent vehicles on traffic flow using traffic flow/driving simulation	
Study Goals	On completion of this course, the student will be able to: B2U3-LO1. describe and analyse the essential components of intelligent vehicle systems; B2U3-LO2. analyse and design the tactical and/or operational level decision-making systems of your intelligent vehicle system; B2U3-LO3. compare the characteristics of different sensing and communication technologies for intelligent vehicles; B2U3-LO4. identify and assess behavioural adaptation effects of intelligent vehicles; B2U3-LO5. (a) analyse the main methods of traffic safety impact assessment for intelligent vehicle systems; (b) assess the traffic safety impacts of intelligent vehicle systems; B2U3-LO6. (a) analyse the main methods of traffic flow impact assessment for intelligent vehicle systems; (b) assess the traffic flow impact of intelligent vehicle systems; B2U3-LO7. apply traffic simulation to assess the traffic flow impacts of a selected intelligent vehicle system.	
Education Method	Lectures (2x 2 hours per week)	
Assessment	100% group assignment.	
Expected prior Knowledge	The course is closely related to CIEM6220-U4 Traffic Flow Modelling and Control and CIEM6220-U1 Traffic Safety.	
Academic Skills	In addition to the technical materials, students will also practice scientific writing, literature research, presenting, giving feedback and working in teams	
Literature & Study Materials	Reader, recommended papers and slides	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	None	
Collegerama	No	

CIEQ6224	Urban and motorway Traffic Flow Modelling and Control	5
Responsible Instructor	Dr.ir. A. Hegyi	
Instructor	Dr.ir. A.M. Salomons	
Instructor	Dr. V.L. Knoop	
Contact Hours / Week x/x/x/x	x/x/x/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Modelling: Macroscopic traffic flow models, node models Shock wave theory - moving bottlenecks Network fundamental diagram Control: Traffic state estimation Traffic control design methodology Applications: Advanced vehicle-actuated signal control Urban network control Freeway traffic control (coordinated ramp metering, route guidance, variable speed limits, cooperative speed control)	
Study Goals	On completion of this course, the student will be able to: B2U4-LO1. Describe and apply macroscopic traffic flow models, such as the Cell Transmission Model, node models, to determine the evolution of traffic over time; B2U4-LO2. Discuss the shape of the macroscopic fundamental diagram and explain perimeter control; B2U4-LO3. Apply and explain Kalman filtering for traffic state estimation problems; B2U4-LO4. Explain the different purposes and designs of (1) coordinated ramp metering, (2) route guidance, (3) variable speed limit systems. Explain and apply the discussed designs for such systems to a traffic situation. B2U4-LO5. Design and analyse a vehicle-actuated intersection controller. B2U4-LO6. Describe and discuss the main features of network-oriented urban signal control approaches. B2U4-LO7. Explain the steps of the design methodology for traffic control systems. Apply the steps to design problems.	
Education Method	Lectures (2x2 hours per week) Practical by supervised work on exercises for assignments (1x4 hours per week)	
Assessment	Written exam (100%)	
Expected prior Knowledge	Road traffic flow modeling and control related parts of CIEM6000.	
Academic Skills	Application of theory to new situations, conceptual thinking, critical thinking, problem-based learning and solving.	
Literature & Study Materials	Lecture notes provided by the lecturers.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Equation sheet will be handed out at (and uploaded to Brightspace). Simple calculator. One A4 sheet (two-sided) with notes in own(!) handwriting, max 30 lines, 80 characters per page.	
Collegerama	No	

CIEQ6231	Public Transport System and Supply Planning and Operations	5
Responsible Instructor	N. van Oort	
Instructor	Dr. O. Cats	
Contact Hours / Week x/x/x/x	x/x/x/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Service analysis Transit data sources and performance analytics Data fusion and analysis Technology design and properties (BRT, light rail,...) Access and egress, Micromobility Emerging modes and concepts (autonomous shuttles, MaaS, Hubs) Service design Frequency setting Fleet size determination Resource allocation and vehicle scheduling Control Reliability analysis Real-time control measures Societal aspects of the entire public transport planning process and operations are discussed. Also, the (expected) impacts of new technologies and emerging modes on sustainability and equity will be evaluated. The part of the unit dealing with first and last-mile solutions will discuss bicycle-related and micromobility solutions.	
Study Goals	LO1. Explain and relate the functions of strategic, tactical and real-time operation and control of public transport systems, first and last-mile solutions and emerging modes and concepts. LO2. Devise a data collection and analysis scheme directed at gaining knowledge on travel patterns and system performance of public transport. LO3. Compute and analyse the capacity and reliability of public transport services. LO4. Construct timetables and vehicle schedules for public transport services. LO5. Identify and evaluate potential control measures and to learn when and where to apply them, considering trade-offs of impacts on passengers, crew, vehicles and costs.	
Education Method	On-campus lectures from instructors Multiple industry lectures from operators and consultancy companies Active learning activities such as a data challenge, a serious game and small class exercises	
Assessment	Written exam (100%)	
Literature & Study Materials	Lecture slides and multiple papers per week provided during the course	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

CIEQ6232	Public Transport Demand and Network Planning and Operations	5
Responsible Instructor	Dr. O. Cats	
Instructor	N. van Oort	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Modelling PT planning problems. The course consists of three blocks: network and demand analysis, network assignment and design, and service design. The course covers the following topics: - Planning process - Network analysis - Demand and behavioral modelling - Network assignment - Network design - System resilience - Cost-benefit analysis - Data and modelling in planning practices - Demand and supply real-time management - Mobility-on-Demand - Service optimisation in operations and planning practices - Long-distance travel The course includes lectures, lab tutorials and talks highlighting insights from practice. Societal aspects of the entire public transport planning process are discussed. The unit will discuss the relations between public transport service design and societal/urban development goals such as accessibility, coverage, sustainability, equity and resilience. Attention is also given to communicating modelling and analysis results to decision makers and implications of mobility service developments for different user groups. This unit is the official substitute of the course CIE4811 "Planning and Operations of Public Transport Systems".	
Study Goals	LO1. Analyse key dependencies among design variables, parameters, and criteria in public transport service planning and management and their dynamics. LO2. Design and examine public transport networks while considering coverage, accessibility, travel and operational costs. LO3. Characterize the structure of public transport networks by applying graph theory techniques. LO4. Compare alternative network models, simulation, optimization and choice modelling techniques in the realm of public transport, and their key notions, strengths, and weaknesses. LO5. Select the modelling approach adequate for a given application concerning network modelling, optimization and assignment models, and interpret the results of these models	
Education Method	The course involves lectures, in-class discussions, feedback sessions, pre and post-lecture material such as reading scientific articles and watching videos.	
Assessment	Group assignment including an individual component: 100%.	
Tags	Algorithmics Analysis Business Modelling Programming Sustainability Transport & Logistics Transport phenomena	
Expected prior Knowledge	This courses provides knowledge on planning and operations of public transport systems and does not have specific prior knowledge requirements. Notions related to network science, transport fundamentals and mathematical programming will be used in several classes.	
Academic Skills	The main academic transferable skills that will be developed in this unit are critical thinking, analytical thinking and problem solving, specifically applied to the planning and operations of public transport networks and demand analysis. Students are expected to exercise the following academic skills: - Problem formulation - Data analysis - Statistical interpretation - Comparing alternatives quantitatively and qualitatively - Working in diverse groups - Critically reflecting on their own work - Report structuring and writing - Reading and summarizing scientific papers The main personal skills that the student will develop are organizational, team-building, and interpersonal and presentation abilities by means of concrete cooperation with colleagues in assignments having the scope of emulating projects and interactions in a professional environment.	
Literature & Study Materials	Lecture slides and pre- and post-lecture material including 3-5 reference papers are made available through Brightspace.	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable (no exam)	

Collegerama	No
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CIEQ6233	Railway Operations and Control	5
Responsible Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Dr. D.M. van de Velde	
Contact Hours / Week x/x/x/x	x/x/x/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Railway safety and signalling Railway safety principles Train separation, Interlocking, Automatic train protection Blocking time theory Railway capacity assessment Next generation signalling (ERTMS, CBTC, Virtual Coupling) Railway automation Automatic Train Operation Grades of Automation ATO-over-ETCS Railway traffic Control - Metro control Migration to ERTMS and ATO Coordination in the railway value chain Railway governance Railway policy Railway value chains Short-term, Mid-term and Long-term coordination This unit will underline the impacts of railway signalling and automation on railway performance and public safety, making the students aware of the responsibilities and ethical implications related to the introduction of fully automated railway operations. Roles and responsibilities of the different railway system components and stakeholders are introduced, alongside with potential conflicts of interest in the railway business. Regulations and policies over the entire railway value chain are discussed in line with international objectives on sustainable mobility.	
Study Goals	LO1: Assess railway safety and signalling systems LO2: Assess railway automation LO3: Analyse railway value chains and coordination misalignments LO4: Evaluate railway governance and main policy options LO5: Evaluate migration issues, including ERTMS and ATO	
Education Method	This course uses a mixture of lectures, reading, guest lectures, and formative exercises.	
Assessment	Written exam (100%)	
Expected prior Knowledge	None	
Academic Skills	Modelling, design, critical thinking, reasoning	
Literature & Study Materials	J. Pachl, Railway Signalling Principles, 2nd ed., 2021. And a set of papers or book chapters made available on BrightSpace. Each lecture will require reading one paper/book chapter.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Simple calculator	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Control and Operations (C&O)

AE4321-15	Air Traffic Management	4
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Dr.ir. J. Ellerbroek	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.	
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.	
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.	
Reader	Hand-outs of the presentations are available on Brightspace.	
Assessment	Written, on site exam (50%) and the assignment (essay+presentation, also 50%). Both parts need at least a 5.0.	

AE4423-20	Airline Planning and Optimisation	4
Responsible Instructor	Dr.ir. B.F. Lopes Dos Santos	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	It is assumed that students are familiar with modelling and optimization techniques. It is assumed that students have experience in code development in Python or MatLab (or any other programming language).	
Course Contents	This course provides students with knowledge to analyse planning problems related to airline operations and to develop modeling approaches to solve these problems. The focus is on the relationship between planning models and their operational implications. It starts with a general overview of the airline demand analysis, followed by the study of the planning framework which airlines operate in. This planning framework includes strategic decisions, namely fleet planning and network development, tactical decisions, such as scheduling and maintenance planning.	
Study Goals	At the end of the course, the students should be able to: Obj1: identify the main strategic, tactical, and operational problems of an airline; Obj2: get familiarized with the development and implementation of some modeling techniques: (a) mix-integer linear programming, (b) multi-commodity flow networks, (c) time-space networks, (d) Markov decision programming. Obj3: get familiarized with the development and implementation of some solution techniques: (a) branch-and-bound, (b) column generation, (c) dynamic programming. Obj4: identify, formulate and solve airline strategic and tactical planning problems: (a) airline networks development, (b) fleet planning, (c) frequency planning, (d) aircraft assignment and routing planning, (e) crew scheduling, and (f) maintenance planning. Obj5: identify an airline operations problem, analyse and solve it; Obj6: explain the implications of airline planning decisions and report them in an academic manner.	
Education Method	The course will work according to a blended learning approach. Online lectures will be prepared for 9 content sessions. Discussion and Q&A sessions of 1 hour will be planned per week and there will be tutorial lectures to discuss the assignment(s). The students are assessed according to the grades of a group assignment (60%) and an individual online exam (40%).	
Literature and Study Materials	References literature: - Belobaba, P., Odoni, A., Barnhart, C. (2009), "The Global Airline Industry", John Wiley & Sons Ltd, West Sussex, UK. [ISBN: 978-0-470-74077-4] - Lectures notes and additional material posted on Brightspace during the course. Additional Literature: - Doganis, R. (2010), "Flying Off Course: Airline economics and Marketing", 4th Edition, Routhledge: London, UK. [ISBN: 0-415-44737-2] - Bazargan, M. (2010), "Airline Operations and Scheduling", 2nd Edition, Ashgate Publishing Company, Surrey, UK. [ISBN: 978-0-7546-7900-4] - Clark, P. (2007), "Buying the Big Jets: Fleet Planning for Airlines", Ashgate Publishing Company, Hampshire, UK. [ISBN: 978-0-7546-7090-2]	
Prerequisites	The students should have completed the AE4441 course, or the ME44206 course, or an equivalent operations research course. The students should also be familiar with programming. For python courses, students can follow TIL6010, AE1205, or TI3105TU.	
Assessment	Group assignments - divided into two parts (60%) Final exam (40%) The students need to have a grade equal to or higher than 5.5 in both examination elements. The group assignments are done by groups of (maximum) 3 students.	
Permitted Materials during Tests	None	

AE4446	Airport and Cargo Operations	4
Responsible Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of the following lectures: Lecture 1 Introduction Lecture 2 Landside operations Lecture 3 Airside operations Lecture 4 The Gate assignment problem Lecture 5 The Aircraft taxiing problem Lecture 6 Air cargo supply chain Lecture 7 Bin packing problems in air cargo operations Lecture 8 Airport environmental consideration Lecture 9-14 Guest lectures A short quiz is given after lectures 1 to 8. Bonus points for the test can be earned by participating in these quizzes. Note that the sequence of the lectures might change pending the availability of the invited speakers. In addition, if not enough invited speakers are found, such lectures will be replaced with lectures covering additional topics (e.g., uncertainty in airport operations) or a Q&A related to the assignment. Knowledge of linear optimization and familiarity with Mixed Integer Linear Programming (MILP) formulations is a strongly suggested pre-requisite because many models presented in the lecture and the assignment are MILPs that the students must understand and (potentially) reproduce. Additional resources, examples, and tutorials will be provided to help students.	
Study Goals	At the end of the course, students will be able to: - Understand and critically assess key airport characteristics - Understand the air cargo supply chain in terms of stakeholders and operations - Understand, reproduce, and modify mathematical models describing different operations occurring inside or around airports	
Education Method	The course consist of about 8 base lectures and 6 guest lectures (based on availability of guests). Students work in a group of up to 4 students to assess and reproduce an airport/cargo related optimization model (MILP model) and write a report. As such, some knowledge about linear programming is a strongly suggested pre-requisite for the course (e.g., courses such as Operations Optimization or Quantitative Methods for Logistics are recommended courses). A closed book computer test is used to test and grade the individual knowledge of the students.	
Literature and Study Materials	Slides and other information and supplementary material are posted on Brightspace. Some lectures and the assignment are based on academic papers describing MILP models. Such papers are also uploaded on Brightspace.	
Books	Airport Systems: Planning, Design, and Management, Richard de Neufville and Amadeo Odoni, McGraw-Hill	
Assessment	50% Individual computer test (must be 5.0 or higher) 40% Group report (must be 5.0 or higher) 10% Homework quizzes	
Permitted Materials during Tests	Non graphical calculator	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Urbanism (U)

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'inteken-en-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168.
Period of Education	The maximum marking period is 15 working days.
Concept Schedule	Quarter 4. The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Course Coordinator	T. Kuzniecowa Bacchin	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge. The course programme consists of lectures, workshops and individual mentoring.	
Study Goals	Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical- - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3CS021	Seminar Cross Domain City of the Future	5
Course Coordinator	Dr.ir. R. Cavallo	
Course Coordinator	Dr.ir. T.A. Daamen	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. J.H. Baggen	
Instructor	Ir. J.A. Kuijper	
Instructor	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	4 hours/week (1.1-1.8 & 2.1-2.5)	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Given the framework of the project, the aim of the seminar is to: - analyse and understand which factors, tendencies, (design research) approaches, technological innovations, development and governance models will be playing a role in the future developments of our cities. - study relevant examples and/or best practices around the world and compare them with the city-study areas of the graduation studio. - collect data into research books to be shared among the participating students. - argument and develop the investigations (research positions) into a written article. Multiple aspects / disciplinary perspectives regarding the (research design) project will be presented by the involved researchers in interactive sessions. A.o., following topics will be discussed: transportation, connectivity, new mobility concepts, housing challenges, climate change mitigation, cultural planning, health and private, public and semi-public spaces, circular economy, politics, resiliency, adaptation strategies. The seminar challenges students to develop critical and comparative investigations focusing on specific insights and positions to be determined within the (research) framework of the studio.	
Study Goals	- Become aware of different (design) research methods in order to inform own graduation process. - Acquire necessary research skills in order to set up and develop a research article - Explore different inquiry methodologies and research approaches. - Work in a collaborative way within a multidisciplinary group and probe different constraints to define own (design) research approach.	
Education Method	Workshops/tutorials given by the involved tutors and/or invited guests/experts	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	- Writing Assignment - Analytical Assignment Next to the participation in the seminar, the assessment of the course will enclose a writing assignment: Option 1, a position paper Option 2, a report of the thematic seminar sessions related to a personal research standpoint. The assessment criteria will be: -Is the paper or report coherent/concise, regarding structure and clearness of style? -Is the language adequate? -How did the paper or report follow the review comments? Problem field and research question: -Has a research question / standpoint been clearly defined? -Has the question / standpoint been developed beyond its initial formulation? -Does the paper / report acknowledge a state of the art, regarding this question? Societal and scientific relevance -Is the paper /report relevant to the studio theme City of the Future from a societal point of view? -Is the paper / report relevant to the studio theme City of the Future from a scientific point of view? -Has a position been taken, in relation to this state of the art? Theoretical and operational frameworks -How is the paper / report connected with the graduation topic? -Are the sources pertinent, and well used and is the use of reference and own words balanced? -Does the paper / report follow a clear approach / methodology and shows scientific quality? On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Period of Education	1-2	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Other TIL fields

CME2300	Financial Engineering	4
Responsible Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Compulsory course for master CME - Construction Management & Engineering (4TU)	
Summary	Short overview of the theory of the firm Sources of finance, cost of finance, capital markets Cash flow models, decision rules like payback period, C/B ratio, NPV Financial accounting Financing (international) projects, project finance Political and social factors Portfolio management Financial risk management Influence of future trends on sustainability and circularity on finance	
Course Contents	This course deals with the finance issues related to the implementation of civil engineering projects. It introduces economic engineering concepts and finance-related topics such as project financing and financial accounting. This course requires the student to study in detail: # Finance and the firm, covering topics such as sources of finance, cost of finance, financial structures, working capital management and financial accounting # Capital Budgeting Decisions and Risk, covering topics such as Capital Budgeting, Political and Social factors, Portfolio Management and Risk Considerations # Project finance, covering topics such as international capital markets, stakeholder/actors viewpoints and cash flow modelling of projects including characteristics of typical projects like oil wells, open mining, infrastructure and office buildings # Future trend topics for finance, especially implications of sustainability and circularity	
Study Goals	The final attainments of this course that students are able to: # Understand the financing and financial implications of civil engineering projects from both a firm and a project perspective, including perspectives from financial involved actors # Create a cash flow overview from cost engineering information # Apply investment and financing analytical techniques # Assess a project proposal and investment strategies # Apply risk analytical techniques on financial data of a project or proposal # Understand the project life-cycle and its impact on and relationship with project finance # Apply and understand the valuation of (financial) assets and projects under uncertainty. # The ability of students to deal with uncertain political and social factors and financial risks # Understand future financing and business trends with regards to sustainability and circularity	
Education Method	This course will include lectures, case studies and an individual assignment.	
Books	Principles of Corporate Finance, Brealey, Myers and Allen, Eleventh version or later, Global Edition	
Assessment	The final assessment is a written examination (40 multiple choice questions). The use of a calculator is allowed. No cell phones or tablets are allowed. Continued pandemic risks may imply an alternative examination for 2022/23. - The result of the multiple choice exam should be 6.0 or higher.	
Contact	CME Secretary's Office, CiTG room 6.49, mrs. Sandra Schuchmann - Hagman, s.c.m.schuchmann@tudelft.nl tel: +31 (0)15 27 84774	
Expected prior Knowledge	No prior knowledge is required.	
Academic Skills	To prepare for the future interdisciplinary world of work (engineering and finance, projects), to broaden academic skills with respect to optimization and to obtain interdisciplinary competences.	
Literature & Study Materials	Principles of Corporate Finance, Brealey, Myers and Allen, Eleventh version or later, Global Edition Related academic papers	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/.	
Permitted Materials during Exam	* The use of a calculator is allowed, but no mobile phones or tablets / iPads. * No books or dictionaries are allowed	
Collegerama	No	

ME41106	Intelligent Vehicles 3ME	5
Responsible Instructor	Prof.dr. D. Gavrila	
Practical Coordinator	R.M. Ensing	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	ME41106 Intelligent Vehicles is an introduction course on the (on-board) AI technology of automated driving. Whereas the application setting is that of vehicles, most concepts apply to the broader mobile robotics setting. Students seeking a one-course overview of AI for mobile robotics are thus encouraged to attend, also from outside 3ME (e.g. EEMCS, Civil Engineering and Aerospace Engineering faculties). Introduction Course organization Intelligent Vehicles (IV) domain Motivation for automated driving SAE levels of automation Main components of IV technology Remaining challenges Sensor Overview (camera, radar, LiDAR) 3D Machine vision Perspective camera model Extrinsic and intrinsic camera transformations Stereo vision Object Detection and Classification Detection vs. classification Object proposals Handcrafted features (e.g. HOG) & classification (e.g. linear SVM) End-to-end learning: neural networks, deep learning Performance metrics: confusion matrices, precision vs. recall, ROC curves State Estimation Bayesian Filtering Kalman Filtering Particle Filtering Object Tracking Data Association Track Management Self-Localization & Sensor Fusion Absolute vs. relative localization Ego-motion compensation (e.g. odometry) Extrinsic sensor calibration Environment representations (grids, voxels) Motion Planning Planning as graph-search (the A* algorithm) Trajectory generation and optimization The Road Ahead IV technology currently on the market Future developments The course has a substantial practicum component (about 65% of course work load), where learned concepts are put into practice by means of programming assignments (Python, using Jupyter notebooks).	
Study Goals	At the end of the course students will understand the main methodical components of a intelligent vehicle (mobile robot). They will be able to program basic algorithms and test these on simulated and real-world data. Students will be able to express an educated opinion on the benefits and risks of automated driving, the current developments from driver assistance to driver-less cars, and the forces driving this transformation.	
Education Method	Lectures (2-4 hours per week) Practicum (2 hours per week) Formative feedback: Practica - Pairwise collaboration (alternating random groups) - Pass/Fail (counts for knock-out criteria) - No late submissions or resit (except due to motivated medical conditions) Students can in principle run the software locally on their own computers (laptop/PC). However, we only provide limited support for setting up the necessary software environment (i.e. Anaconda). If the software environment cannot be installed properly (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms.	
Course Relations	This course, ME41106 Intelligent Vehicles <i>*cannot*</i> be selected in combination with RO47004 Machine Perception, due to the content overlap between the courses. This means, for example, that this course ME41106 cannot be selected by students following the MS Robotics program (for which RO47004 is required). This course mainly addresses the on-board AI technology of an intelligent vehicle. Complementary to this course is CIE 5805 Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment which focuses more on automated vehicles as part of a larger intelligent transportation system (i.e. including smart infrastructure, connected/cooperative driving) and on associated issues at the macro level (e.g. traffic flow efficiency, fuel consumption, behavior adaptation).	
Literature and Study Materials	Slides Hand-outs recommended text book: Artificial Intelligence: A Modern Approach (Russel and Norvig) 3rd edition freely available as PDF Jupyter notebooks	
Prerequisites	Basic linear algebra and probability theory Intermediate Python programming skills, e.g. Numpy, object oriented programming (classes, inheritance), modules, namespaces and scope. Recommended: Pattern Recognition / Machine Learning (RO47002, CSE2510, or equivalent) It is possible to start the course with beginner Python skills if your aim is to take advantage of this course to also improve your programming expertise (a laudable aim, given its importance for both industry and academia). Be aware, however, that this	

Assessment	catching up will likely result in an increased study load (e.g. 1-2 EC). Final practicum assignment (50%) - Individual assignment - Late submission: max. one day late for -1.0 grade point - Students having an initial grade of < 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the Q3 quarter. Final written exam (50%) - Written exam on campus. Knock-out criteria - At most one of the Pass/Fail practicum assignments is Fail - The minimum final practicum assignment grade is 5 - The minimum final written exam grade is 5 - The minimum final course grade is 6 The final practicum assignment will be evaluated on the computers in the designated lab rooms. If a student chooses to work using own computing equipment, the student needs to test the software on the designated computers before handing-in. Both final practicum assignment and final written exam results remain valid for one year. Each result can separately be transferred to the next academic year, upon request.
Special Information	For more information about this course, see the slides of the Introduction lecture from past academic year at http://intelligent-vehicles.org/teaching/courses/ . TU Delft students might also be able to view the lectures from past academic year on Collegerama.
Department	3mE Department Cognitive Robotics

MT44070	Shipping Management	5
Responsible Instructor	Prof.dr.ir. E.B.H.J. van Hassel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course offers a comprehensive insight into the transport economic aspects of a maritime transport chain. This includes: - Short history of maritime trade and transport - Fundamental of transport economics - Maritime (sub) markets - Risk and Risk Management in shipping - Ports - Inland navigation - Hinterland transportation - Emission reductions (legislation and current developments)	
Study Goals	The objective of this course is to provide the students the insights into how ships are used to earn money. In order to do so, the functioning of the maritime markets needs to be understood. Also, it is the aim to explain that ships are part of a larger logistics chain and that the interaction in the port and with the other transport modes is vital. Next to that, also the working principles of a port are explained even as hinterland transport. Finally, also the current developments regarding emission reductions in the total transport chain are dealt with.	
Education Method	Physical lectures with audiovisual presentations.	
Assessment	Final exam in written form (75%). Group coursework (2 to 3 students): a short paper based on an assignment for which a model will be provided to the students (25%).	
Remarks	Old course code: MTM313-15	
Department	3mE Department Maritime & Transport Technology	

RO47016	Human Factors of Automated Driving	5
Responsible Instructor	Prof.dr.ir. R. Happee	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	Master Robotics	
	Track Vehicle Engineering (as replacement of ME41000)	
	Recommended for Transport Infrastructure & Logistics	
Summary	Human Factors of Automated Driving	
	Today's cars & trucks already support drivers with cruise control, lane keeping assist and automated emergency braking. In the near future automation will allow drivers to take the eyes off the road and use the driving time for work or leisure. Driverless taxis and buses are already tested on public roads and remote driving is explored to handle conditions which the automation cannot handle. This course will address key challenges in the human interaction with automated vehicles: Can we safely delegate control to automated vehicles? Can we safely resume manual control after prolonged automation use? Can remote operators effectively control driverless vehicles? Will we feel safe? Will we be comfortable, and not become motion sick? How to redesign the vehicle interior such that we can use the driving time for work or leisure? Through lectures and a creative team assignment you will learn about the current state of the art in human factors of automated driving in terms of research findings and methods to design automation matching human abilities and preferences.	
Course Contents	Content in detail Introduction From manual to supervisory control (monitoring), delegated control and driverless operation Road safety The evolution from manual to automated driving Longitudinal Control: Human behaviour in car following versus Adaptive Cruise Control Lateral Control: Human path following versus Lane Keeping Assistance & Automated Steering Transitions of Control and safety Teleoperator state in remote driving Steering feedback controllers in normal and remote driving Situational awareness in remote drivers Occupants comfort in remotely operated vehicles Motion Comfort & Acceptance Trust in automation and perceived safety Comfort and motion sickness Ability to perform non-driving tasks Seating, vision and reach of instruments Vehicle cockpits for automated driving Several lectures will be used for students to present their team assignments.	
Study Goals	After completing this course, students will be able to: describe and analyse the interaction of drivers and users of automated vehicles, with their vehicles contribute to the design of vehicle automation with a focus on human factors express your own vision on the role of the driver/user in future vehicles	
Education Method	Contact hours Integrated plenary sessions, including lectures, discussions, and team presentations of the design assignment (6 hours per week) Group assignment designing an innovative automated driving system. Specify the operation modes and range of the automation. 1) Review the human factors challenges. 2) Design the human machine interface. 3) Propose experiments to evaluate safe and acceptable user interaction. Optionally teams may design 1) A market introduction strategy 2) Redesign the vehicle interior for non-driving tasks 3) Integrate the system into (public) transport	
Course Relations	Replaces the course ME41000 (3ECTS) which was last given in April-June 2020	
Literature and Study Materials	Literature Lecture slides & recordings	
Assessment	Written exam Individual assignments Team assignment	
	In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: online.	
Department	3mE Department Cognitive Robotics	

TIL6020	TIL Scientific Assignment	7
Responsible Instructor	Dr. J. Jovanova	
Instructor	Dr. J. Jovanova	
Contact Hours / Week x/x/x/x	-	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course gives the student the opportunity to conduct and participate in a concrete scientific research project in the field of Transport, Infrastructures and Logistics (TIL). Students will experience what it is to perform socially, industrially, and/or environmentally relevant TIL research within an academic setting. A student will complete a 'mini' project dedicated to a scientific topic with all typical project stages including: problem definition, definition of research questions, problem analysis, research set-up & time plan, working out of the research plan, evaluating results and eventually coming to findings and recommendations. Literature research and, when appropriate, (small scale) empirical research, model development, case study, and analysis will be part of the approach. Writing a draft scientific paper as well as writing a process report are essential parts of this course. A scientific researcher (PhD, researcher, professor) will supervise the student. The mini projects either: - relate to ongoing PhD research, research by a senior scientific staff member: here, the student works only with the PhD or senior scientific staff member active on the ongoing research project; - or - fall within the scope of a coherent research program defined in relation with external partners within the domain of Transport, Infrastructure, and Logistics. In this case, the student will also interact with other students working on related research questions of the research programme and a program coordinator.	
Study Goals	To learn the basics of performing (applied) scientific research and associated activities like literature research, empirical or case analysis, writing a scientific paper, and planning and managing such a project set-up.	
Education Method	The course consists of a (mini-)project. During a period of about three months (part time) students will participate closely with scientific researchers and possibly other students in a research project.	
Literature and Study Materials	Day, R. 1998. How to write and publish a scientific paper, 5th edition. Orynx Press. Alley, M. 1996. The craft of scientific writing, 3rd edition. Prentice Hall, NJ. Goben, G., and J. Swan. 1990. The science of scientific writing. Am. Scientist 78: 550-558. Course information on BrightSpace.	
Practical Guide	Guidelines will be provided by the course coordinator and via BrightSpace.	
Assessment	The assessment is based on two products: - process report, including a self-evaluation - (concept) scientific (conference or journal) paper The products will be reviewed and assessed by the scientific researcher and the course coordinator. Both products need to be sufficient in order to pass the course.	
Enrolment / Application	This course can be started anytime by contacting the course coordinator.	
Special Information	The 'course' is to be seen as a working project; the project is formed in cooperation between student and supervisor, tuned to the requirements and needs of the associated research project and tuned to the interests of the student. A proactive and curious attitude of the student is expected.	
Elective	Yes	
Tags	Adventurous Challenging Personal Project Research Methods Specific Transport & Logistics	
Contact	Dr.ir. Arjan van Binsbergen E: a.j.vanbinsbergen@tudelft.nl	
Expected prior Knowledge	TIL4030-16 (TIL Research and Design Methods)	
Academic Skills	Setting up an academic research project (plan); literature research (source selection, critical reflection, effective use); empirical test set-up (optional); theoretical/model approach (optional); scientific analysis; deriving conclusions on basis of the research approach; scientific reporting.	
Literature & Study Materials	The literature and study materials are to be determined together with the supervisor - literature and study materials should relate to and be relevant for the chosen research topic. Guidelines for the course (describing the aim, the procedure and the table of contents of the process report) will be handed to the student when starting the course.	
Judgement	The assessment is based on two products and the work process: - process report, including a self-evaluation - (concept) scientific (conference or journal) paper The performance will be assessed on basis of the criteria: 'Focus and initiative', 'Method/Approach', 'Analysis', 'Argumentation', 'Reference list', 'Feedback process (including self-evaluation)', 'Inventiveness', 'English', 'Time (incl. timescheduling and updates thereof)' and the 'General presentation'. These criterions have an equal weight. On basis of the criterion scores, the overall mark	

	is set.
Permitted Materials during Exam	Not applicable.
Collegerama	No

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	- The student is able to recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - The student is able to solve the shortest path problem and the transportation problem as well as some small flow shop problems. - The student is able to solve the travelling salesman problem by the Branch and Bound algorithm. - The student knows several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - The student knows some basic theorems concerning the mentioned problems and is able to prove some of these theorems. - The student has knowledge about methods to solve large scale problems, especially shortest path and vehicle routing problems. - The student has basic knowledge about scheduling problems and complexity theory - The student is able to implement several of the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WM1301TU	Ethics of Transportation	3
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems - can see these problems within the broader background of moral, legal and political philosophy - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way	
Education Method	A series of 6 lectures and some working group sessions, focused on the discussion and application of the theories presented in the course	
Assessment	Written exam and assessment participation in working groups	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Transport, Infrastructure & Logistics

Convergence Program TIL (9 EC)

IFEEMCS010500	Probability and Statistics	3
Responsible Instructor	Dr. R.J. Fokkink	
Responsible Instructor	Dr. C. Kraaikamp	
Contact Hours / Week x/x/x/x	x/x/4/x or x/x/x/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch English	
Expected prior knowledge	Linear Algebra, Calculus for Engineering part 1 and part 2 or Calculus for Science part 1 and part 2.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Interactive lectures, pre-lecture videos, exercises	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one endterm (week 10) on Part 2: Statistics. Final grade is the average of both tests. A maximum of 1 bonus point can be earned for the computer assignments. The grades of the midterm and the endterm need to be at least 4.	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Simple (non-graphical and non-programmable) calculator	
Remarks	Please note that this course is available in Q3 and in Q4.	
Co-Instructor	Dr.ir. W.T. van Horssen	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Ir. D. den Ouden-van der Horst	

WI1807TH1-21	Linear Algebra	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programme students are applying for.	
Course Contents	- Systems of linear equations, elementary row-operations, (reduced) echelon form - Vector equations, spans - Solution sets of (non)homogeneous systems of linear equations - Linear independence - Linear transformations and standard matrices - Matrix algebra (sum, product), inverse of a matrix - Subspaces in $\mathbb{R}^n$, column space and null space - Basis and Dimension - Determinants - Eigenvalues and eigenvectors - Diagonalization	
Study Goals	After successfully completing the course you will be able to: - solve systems of linear equations. - identify if a system of linear equations is solvable, and if the solution is unique. - show the equivalence between a vector equation, a matrix equation and a system of linear equations. - give a geometric description of solution sets of vector equations. - determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - identify if a transformation is linear and determine the standard matrix of a linear transformation. - perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - determine the determinant of a matrix by applying the properties of a determinant. - apply the properties of determinants (product, transpose, inverse). - determine the eigenvalues and eigenvectors of a matrix. - determine the algebraic and geometric multiplicity of the eigenvalues of a matrix. - determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - analyze the case of complex eigenvalues and eigenvectors.	
Education Method	Lectures, 4 hours per week (2x2)	
Literature and Study Materials	- Book: David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216. - Slides on Brightspace. - Prelecture videos. - Online homework assignments (Grasple). - Old exams on Brightspace.	
Books	David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216.	
Assessment	Digital exam (in Grasple) on campus.	
Remarks	Lectures will be in Dutch!	
	Exams will be in offered in Dutch and also in English.	
Co-Instructor	Dr. E. Emsiz	

Ir. K.P.M. Aalbers

Unit	Bouwkunde
Department	Environmental Technology and Design
Telephone	+31 15 27 88560
Room	08.BG.West.600

Unit	Bouwkunde
Department	Environmental Design
Telephone	+31 15 27 88560
Room	08.BG.West.600

Dr. A.P. Afghari

Department	Safety and Security Science
Telephone	+31 15 27 83026
Room	31.C1.020

A.S. Alkan

Unit	Bouwkunde
Department	Theory & Territories
Telephone	+31 15 27 84123
Room	08.01.Oost.430

Dr. J.A. Annema

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 88912
Room	31.a3.230

Prof.dr.ir. B. van Arem

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 86342

P. Ashkrof

Department	Transport and Planning
Room	23.HG 4.01

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Room	23.HG 4.01

Dr. B. Atasoy

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 82706
Room	34.B-4-290

Dr.ir. J.H. Baggen

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 84813
Room	23.HG 4.25

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 84813
Room	23.HG 4.25

Dr. S. Balakrishnan

Department	Transport en Logistiek
Telephone	+31 15 27 85831

Ir. M.C. van Bente

Department	Marine and Transport Technology
Room	34.B-3-210

Dr.ir. A.J. van Binsbergen

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 81067
Room	23.HG 4.07

A. Bombelli

Department	Air Transport & Operations
Telephone	+31 15 27 84303

Ir. W. van den Bos

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 86766
Room	34.B-3-310

Dr.ir. S.C. Calvert

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Room	23.HG 4.02.2

Dr. O. Cats

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 83621
Room	23.HG 4.22

Dr.ir. R. Cavallo

Unit	Bouwkunde
Department	Theory & Territories
Telephone	+31 15 27 85352

Dr. M. Comes

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 84459
Room	31.B3.040

Dr. A.F. Correlje

Unit	Techniek, Bestuur & Management
Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 87924

Ir. S. van Cranenburgh

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 86957
Room	31.A3.210

Dr.ir. T.A. Daamen

Unit	Bouwkunde
Department	Urban Development Management
Telephone	+31 15 27 87725
Room	08.01.West.620

Dr. J.H.R. van Duin

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek

Telephone Room	+31 15 27 81142 31.a3.210
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Ir. M.B. Duinkerken

Unit Department	Mech, Maritime & Materials Eng Transport Engineering and Logistics
Telephone Room	+31 15 27 81790 34.B-3-320

Dr.ir. D.C. Duives

Unit Department	Civiele Techniek & Geowetensch Transport and Planning
Telephone Room	+31 15 27 86304 23.HG 4.24

Dr.ir. J. Ellerbroek

Unit Department	Luchtvaart- & Ruimtevaarttechn Control & Simulation
Telephone Room	+31 15 27 89613 63.SIM 0.03

Dr. E. Emsiz

Unit Department	Elektrotechn., Wisk. & Inform. Applied Probability
Telephone Room	+31 15 27 84579 36.HB 07.060

R.M. Ensing

Unit Department	Mech, Maritime & Materials Eng Intelligent Vehicles
Telephone Room	+31 15 27 89767 34.E-0-200

Dr.ir. J.T. van Essen

Unit Department	Elektrotechn., Wisk. & Inform. Discrete Wiskunde en Optimalisering
Telephone Room	+31 15 27 86266 36.HB 04.280

Dr.ir. H. Farah

Unit Department	Civiele Techniek & Geowetensch Transport and Planning
Telephone Room	+31 15 27 83401 23.HG 4.20

M. Farhani

Department	Transport en Logistiek
-------------------	------------------------

Dr. S. Fazi

Department	Transport en Logistiek
-------------------	------------------------

Dr. R.J. Fokkink

Unit Department	Elektrotechn., Wisk. & Inform. Applied Probability
Telephone Room	+31 15 27 89215 28.1.E300

Prof.dr. D. Gavrilă

Unit Department	Mech, Maritime & Materials Eng Intelligent Vehicles
Telephone Room	+31 15 27 88504 34.E-0-330

Prof.dr.ir. P.H.A.J.M. van Gelder

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
Telephone	+31 15 27 86544
Room	31.c1.170

Prof.dr. R.M.P. Goverde

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 83178
Room	23.HG 4.03

Dr. J.B. van Grunsven

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 84715

A.H. Hadi

Department	Transport Engineering and Logistics
-------------------	-------------------------------------

Dr. M.P. Hagenzieker

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 86909
Room	23.HG 4.29

Prof.dr.ir. R. Happee

Unit	Mech, Maritime & Materials Eng
Department	Intelligent Vehicles
Telephone	+31 15 27 83213
Room	34.E-0-240

Dr.ir. M.G.A.D. Harteveld

Unit	Bouwkunde
Department	Urban Design
Telephone	+31 15 27 84164
Room	08.BG.West.150

Prof.dr.ir. E.B.H.J. van Hassel

Department	Ship Design, Production and Operations
-------------------	--

Dr.ir. A. Hegyi

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 89644
Room	23.HG 4.45

mr. F.A.M. Hobma

Unit	Bouwkunde
Department	Design & Construction Management
Telephone	+31 15 27 83170
Room	08.01+.West.650

Prof.dr.ir. J.M. Hoekstra

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Control & Simulation
Telephone	+31 15 27 87587
Room	62.LB 0.29

Dr.ir. G. Homem de Almeida Correia

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 81384

Prof.dr.ir. S.P. Hoogendoorn

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 85475
Room	23.HG 4.09

Dr. F.L. Hooimeijer

Unit	Bouwkunde
Department	Environmental Technology and Design
Telephone	+31 15 27 89638

Dr.ir. W.T. van Horssen

Unit	Elektrotechn., Wisk. & Inform.
Department	Mathematical Physics
Telephone	+31 15 27 83524
Room	36.HB 05.080

Y. Huang

Unit	Techniek, Bestuur & Management
Department	Systeemkunde
Telephone	+31 15 27 83822

Dr.ir. T.M.L. Janssen

Department	Discrete Wiskunde en Optimalisering
Telephone	+31 15 27 83882
Room	36.HB 04.070

X. Jiang

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 88511
Room	34.B-3-320

mr.dr. P. Jong

Unit	Bouwkunde
Department	Design & Construction Management
Telephone	+31 15 27 83405

Dr. J. Jovanova

Department	Transport Engineering and Logistics
Telephone	+31 15 27 83376
Room	34.B-4-220

Dr. V.L. Knoop

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 88413
Room	23.HG 4.37

mr. J.M. Kooijman

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 81173
Room	31.b2.220

Dr. C. Kraaikamp

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 81910
Room	36.HB 07.300

P.K. Krishnakumari

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Room	23.HG 4.02

M. Kroesen

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 87183
Room	31.A3.250

Ir. J.A. Kuijper

Unit	Bouwkunde
Department	Theory, Territories & Transitions
Telephone	+31 15 27 84857
Room	08.01.Oost.430

T. Kuzniecowa Bacchin

Unit	Bouwkunde
Department	Urban Design
Telephone	+31 15 27 87081

Dr.ir. B.F. Lopes Dos Santos

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Air Transport & Operations
Telephone	+31 15 27 82713
Room	62.4.08

Ir. M.W. Ludema

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 81885

Dr. C. Maat

Unit	Bouwkunde
Department	Transport and Planning
Telephone	+31 15 27 87640
Room	23.HG 4.25

M.Y. Maknoon

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 83479

Dr. E.J.E. Molin

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 88510
Room	31.a3.250

N. Mouter

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 89748
Room	31.A3.060

A. Napoleone

Department	Transport Engineering and Logistics
-------------------	-------------------------------------

Prof.dr. R.R. Negenborn

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 86718

Room	34.B-3-300
-------------	------------

N. van Oort

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Room	23.HG 4.10.2

Ir. D. den Ouden-van der Horst

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Room	36.HB 03.250

Dr.ir. Y. Pang

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 88685
Room	34.B-3-310

Dr. E. Papadimitriou

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
Telephone	+31 15 27 84913

G. Papaioannou

Department	Intelligent Vehicles
Telephone	+31 15 27 89329
Room	34.E-0-220

Department	Intelligent Vehicles
Telephone	+31 15 27 89329
Room	34.E-0-220

Ir. A.J. Pel

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 84874
Room	23.HG 4.35

Dr.ir. H. Polinder

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 81844
Room	34.B-3-290

B. Pudane

Department	Transport en Logistiek
Telephone	+31 15 27 86903

E. Quaglietta

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 82761
Room	23.HG 4.18

Dr. V. Reppa

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 88545

J. Rezaei

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 81716

Room	31.B3.100
-------------	-----------

M. Rinaldi

Department	Transport and Planning
-------------------	------------------------

R.N. Roeplal

Department	Transport Engineering and Logistics
Telephone	+31 15 27 82039

Dr.ir. Z. Roosenboom-Kwee

Unit	Techniek, Bestuur & Management
Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 84711

Dr.ir. M. Saeednia

Department	Transport and Planning
Telephone	+31 15 27 83309

Dr.ir. A.M. Salomons

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 88556
Room	23.HG 4.45

Dr. F. Santoni De Sio

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 85141

Dr.ir. W.J. Schakel

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 84427
Room	23.HG 4.02.2

Prof. Dr.ir. D.L. Schott

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 83130
Room	34.B-4-300

Dr. D.F.J. Schraven

Unit	Civiele Techniek & Geowetensch
Department	Real Estate Management
Telephone	+31 15 27 85967
Room	23.HG 3.30

F. Schulte

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 84414

Dr. S. Sharif Azadeh

Department	Transport and Planning
Telephone	+31 15 27 82307

H. Shi

Department	Transport Engineering and Logistics
Room	34.B-3-340

Dr. R.M. Shortall

Dr. M. Snelder

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 84981
Room	23.HG 4.18

E.F.L. Stok

Unit	Mech, Maritime & Materials Eng
Department	Support Marine and Transport Technology
Telephone	+31 15 27 86839
Room	34.B-2-240

Dr.ir. R.F. Swarttouw

Department	Analysis
Telephone	+31 15 27 83634
Room	36.HB 08.290

Prof.dr.ir. L.A. Tavasszy

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 83346
Room	23.HG 4.10.1

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 83346
Room	23.HG 4.10.1

Dr. D.M. van de Velde

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Room	31.B2.020

Prof.dr.ir. A. Verbraeck

Unit	Techniek, Bestuur & Management
Department	Beleidsanalyse
Telephone	+31 15 27 83805
Room	31.B2.270

Dr. J.M. Vleugel

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 86487
Room	23.HG 4.18

Dr. M. Wang

Department	Transport and Planning
Telephone	+31 15 27 83862
Room	23.HG 4.20

Dr. R.S. van Wegberg

Department	Organisatie en Governance
Telephone	+31 15 27 81320
Room	31.B2.210